

Theoretical Statistics

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1 Introduction

1.1 Confidence Interval For Means

Suppose we have $X_1, \dots, X_n$ are iid random variables with $\mathbb{E}X_i = \mu$ and $\text{Var}(X_i) = \sigma^2$. We can construct a confidence interval for the mean μ as

$$[\bar{X} - \frac{3\sigma}{\sqrt{n}}, \bar{X} + \frac{3\sigma}{\sqrt{n}}]$$

This is a CLT driven result. Note that $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$. Some potential issues with this are

1. In the CLT we assume $n \rightarrow \infty$. How many samples do we actually need to use this approximation. The answer is not clear.
2. What if σ is growing as the experiments go on? Then what can we do?

1.2 Linear Regression

Consider the model

$$Y_i = \langle x_i, \theta^* \rangle + \xi_i$$

where $x_i \in \mathbb{R}^d$ and $\xi_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ and $\theta^* \in \mathbb{R}^d$ is the target parameter, $i = 1, \dots, n$. We want to study the least squares estimator

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n (Y_i - \langle x_i, \theta \rangle)^2$$

We will show that

$$\mathbb{E} \left(\frac{1}{n} \sum_{i=1}^n \left(\langle x_i, \hat{\theta} \rangle - \langle x_i, \theta^* \rangle \right)^2 \right) \leq \frac{\sigma^2 d}{n}$$

This result is not asymptotic! It may be sub-optimal, but it works for any n and d . We can consider cases where d and σ are functions of n and then plug into this result.

2 Concentration Inequalities and Sub-Gaussian Tools

Proposition 2.1 (Markov's Inequality). If X is a non-negative random variable with finite first moment $\mathbb{E}X$ then for any $t > 0$,

$$P(X > t) \leq \frac{\mathbb{E}X}{t}$$

Good for answering bounds for $P(X > a\mathbb{E}X)$ type questions where $a > 0$.

Proof.

$$\begin{aligned}
\mathbb{P}\{\omega : X(\omega) > t\} &= \frac{1}{t} \int_{\{\omega: X(\omega) > t\}} t \, d\mathbb{P} \\
&\leq \frac{1}{t} \int_{\{\omega: X(\omega) > t\}} X \, d\mathbb{P} \\
&\leq \frac{1}{t} \int X \, d\mathbb{P} \\
&= \frac{1}{t} \mathbb{E}X
\end{aligned}$$

■

Proposition 2.2 (Chebyshev's inequality). Assume X is a random variable with finite $\mathbb{E}X$ and $\mathbb{E}X^2$. Then for any $t > 0$

$$P(|X - \mathbb{E}X| \geq t) \leq \frac{\text{Var}(X)}{t^2}$$

Answers a different question than Markov's inequality. Strictly looks at distance from the mean.

Proof. For any $p \geq 1$ and $t > 0$ we have by Markov's inequality

$$\mathbb{P}(|X - \mathbb{E}X| > t) = \mathbb{P}(|X - \mathbb{E}X|^p > t^p) \leq \frac{\mathbb{E}(|X - \mathbb{E}X|^p)}{t^p}$$

when $\mathbb{E}(|X - \mathbb{E}X|^p) < \infty$. If all p -th centered moments exists we can then say

$$\mathbb{P}(|X - \mathbb{E}X| > t) \leq \inf_{p \geq 1} \frac{\mathbb{E}(|X - \mathbb{E}X|^p)}{t^p}$$

■

2.1 Tails of Gaussian Distribution

Proposition 2.3 (Gaussian Tails). Let $Z \sim \mathcal{N}(0, 1)$, then for any $t > 0$,

$$\left(\frac{1}{t} - \frac{1}{t^3}\right) \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-t^2}{2}\right) \leq \mathbb{P}(Z > t) \leq \frac{1}{\sqrt{2\pi}t} \exp\left(\frac{-t^2}{2}\right)$$

Note that applying Chebyshev's inequality we can only obtain $\frac{1}{t^2}$ order decay.

Proof. (1) Fix $\lambda > 0$ then by Markov's inequality and the moment generating function equality

$$\mathbb{P}\{X > t\} = P\{e^{\lambda X} > e^{\lambda t}\} \leq \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda t}} = e^{\lambda^2/2 - \lambda t}$$

The RHS is minimized when $\lambda = t$. Thus the bound becomes $e^{-t^2/2}$. The desired result follows by symmetry. ■

Proof. (2) (Full result)

$$\begin{aligned} P(Z \geq t) &= \int_t^\infty \frac{1}{\sqrt{2\pi}} \exp(-x^2/2) dx \\ &= \int_0^\infty \frac{1}{\sqrt{2\pi}} \exp(-y^2/2 - yt - t^2/2) dy \\ &\leq \frac{\exp(-t^2)}{\sqrt{2\pi}} \int_0^\infty \exp(-yt) dy \\ &= \frac{\exp(-t^2)}{\sqrt{2\pi}t} \end{aligned}$$
■

Definition 2.4 (Moment Generating Function, MGF). If X is a RV define the real valued function $M_X(\cdot)$ as

$$M_X(\lambda) = \mathbb{E} \exp(\lambda X)$$

Note that this does not have to exist for all distributions.

Proposition 2.5 (Properties of the MGF). The following are properties of the Moment generating function:

1. If $X_1, \dots, X_n$ are iid random variables then

$$M_{X_1 + \dots + X_n}(\lambda) = \prod_i M_{X_i}(\lambda)$$

due to the fact that the expectations and multiplication commute when random variables are independent.

2. Only guaranteed to exist at $\lambda = 0$
3. For the class of random variables for which $\frac{d}{d\lambda}$ and $\mathbb{E}$ commute, we have

$$\frac{d}{d\lambda^k} \mathbb{E} \exp(\lambda X) = \mathbb{E} \frac{d}{d\lambda^k} \exp(\lambda X) = \mathbb{E} X^k \exp(\lambda X)$$

Thus $\frac{d}{d\lambda^k} M_X(\lambda) \big|_{\lambda=0} = \mathbb{E} X^k$.

Proposition 2.6 (Interchanging $d/d\lambda$ and $\mathbb{E}$). To rigorously justify the interchange

$$\frac{d}{d\lambda}\mathbb{E}[e^{\lambda X}] = \mathbb{E}\left[\frac{\partial}{\partial\lambda}e^{\lambda X}\right]$$

we use the **Dominated Convergence Theorem**.

The Rigorous Condition: The moment generating function $M_X(\lambda)$ must exist in an **open neighborhood** of zero. That is, there exists some $h > 0$ such that $\mathbb{E}[e^{\lambda X}] < \infty$ for all $\lambda \in (-h, h)$.

Why this condition is sufficient: If $M_X(\lambda)$ exists for $|\lambda| < h$, it implies that X has “exponentially decaying tails.” This allows us to construct a **dominating function** Y such that:

- (a) $|\frac{\partial}{\partial\lambda}e^{\lambda X}| = |Xe^{\lambda X}| \leq Y$ for all λ in a small interval around 0.
- (b) $\mathbb{E}[Y] < \infty$.

By the Dominated Convergence Theorem, this integrability condition permits the differentiation to be moved inside the expectation.

Proposition 2.7 (Higher Order Derivatives). The condition required for higher order derivatives is identical to the first derivative.

Theorem: If the moment generating function $M_X(\lambda)$ exists (is finite) in an open neighborhood $(-h, h)$ around zero, then:

- (a) $M_X(\lambda)$ is infinitely differentiable (C^∞) with respect to λ in $(-h, h)$.
- (b) We can validly interchange the derivative and expectation operator for any order $k \geq 1$:

$$\frac{d^k}{d\lambda^k}\mathbb{E}[e^{\lambda X}] = \mathbb{E}\left[\frac{\partial^k}{\partial\lambda^k}e^{\lambda X}\right] = \mathbb{E}[X^k e^{\lambda X}]$$

Justification: The existence of $\mathbb{E}[e^{\lambda X}]$ in a neighborhood of zero implies that the tails of the distribution decay exponentially fast. Since exponential growth dominates polynomial growth, the finiteness of $\mathbb{E}[e^{\lambda X}]$ guarantees the finiteness of $\mathbb{E}[|X|^k e^{\lambda X}]$ for all k . This allows us to apply the **Dominated Convergence Theorem** inductively for every k .

4. **Example:** If $X \sim \mathcal{N}(0, \sigma^2)$ then $M_X(\lambda) = \mathbb{E}\exp(\lambda X) = \exp(\frac{\lambda^2 \sigma^2}{2})$ because

$$\begin{aligned}\mathbb{E}\exp(\lambda X) &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(\lambda x) \exp(-x^2/(2\sigma^2)) dx \\ &= \exp(\frac{\lambda^2 \sigma^2}{2}) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(x - \lambda\sigma)^2 dx \\ &= \exp(\frac{\lambda^2 \sigma^2}{2})\end{aligned}$$

Remark 2.8 (Chernoff's method). If we have a random variable X with mgf $M_X(\lambda)$, fix $\lambda > 0$ and then we can say

$$\mathbb{P}(X \geq t) = \mathbb{P}(\lambda X \geq \lambda t) = \mathbb{P}(\exp(\lambda X) \geq \exp(\lambda t)) \leq \frac{\mathbb{E} \exp(\lambda X)}{\exp(\lambda t)} = \frac{M_X(\lambda)}{\exp(\lambda t)}$$

Thus,

$$\mathbb{P}(X \geq t) \leq \inf_{\lambda > 0} \frac{M_X(\lambda)}{\exp(\lambda t)}$$

Remark 2.9 (Chernoff's method for gaussian). Last class we saw that for $Z \sim \mathcal{N}(0, 1)$,

$$\mathbb{P}(Z \geq t) \leq \frac{\exp(-t^2/2)}{t\sqrt{2\pi}}$$

Applying Chernoff's method gives us

$$\mathbb{P}(Z \geq t) \leq \exp(\lambda^2/2 - \lambda t)$$

The bound is minimized when $\lambda = t$. Thus the bound is

$$\mathbb{P}(Z \geq t) \leq \exp(-t^2/2)$$

Proposition 2.10. If the Moment Generating Function (MGF) of a random variable X satisfies $M_X(\lambda) \leq \exp(\frac{\lambda^2 \sigma^2}{2})$ for all λ in a neighborhood of zero, then $\mathbb{E}[X] = 0$.

Definition 2.11 (Little-o Notation). We write $f(x) = o(g(x))$ as $x \rightarrow 0$ if $f(x)$ approaches zero strictly faster than $g(x)$. Formally:

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 0$$

In the context of Taylor series, $o(\lambda)$ represents higher-order error terms (like λ^2, λ^3) that become negligible much faster than the linear term λ .

Proof. We prove the claim by comparing the Taylor expansions of the MGF and the exponential bound around $\lambda = 0$.

First, recall the Taylor expansion of the MGF $M_X(\lambda)$ around 0. Since $M_X(0) = 1$ and $M'_X(0) = \mathbb{E}[X]$, we have:

$$M_X(\lambda) = 1 + \lambda \mathbb{E}[X] + o(\lambda)$$

Next, consider the Taylor expansion for the bound $f(\lambda) = \exp(\frac{\lambda^2 \sigma^2}{2})$. Since the exponent contains only λ^2 , the linear term is zero:

$$e^{\frac{\lambda^2 \sigma^2}{2}} = 1 + \frac{\lambda^2 \sigma^2}{2} + o(\lambda^2)$$

By hypothesis, $M_X(\lambda) \leq e^{\frac{\lambda^2 \sigma^2}{2}}$. Substituting the expansions into this inequality yields:

$$1 + \lambda \mathbb{E}[X] + o(\lambda) \leq 1 + \frac{\lambda^2 \sigma^2}{2} + o(\lambda^2)$$

Subtracting 1 from both sides:

$$\lambda \mathbb{E}[X] + o(\lambda) \leq \frac{\lambda^2 \sigma^2}{2} + o(\lambda^2)$$

We now analyze the limit as $\lambda \rightarrow 0$ from both sides.

Case 1: Right-hand limit ($\lambda > 0$)

Divide the inequality by λ . Since λ is positive, the inequality sign is preserved:

$$\mathbb{E}[X] + \frac{o(\lambda)}{\lambda} \leq \frac{\lambda \sigma^2}{2} + \frac{o(\lambda^2)}{\lambda}$$

Taking the limit as $\lambda \rightarrow 0^+$, the term $\frac{o(\lambda)}{\lambda}$ vanishes by definition, and the right side goes to 0:

$$\mathbb{E}[X] \leq 0$$

Case 2: Left-hand limit ($\lambda < 0$)

Divide the inequality by λ . Since λ is negative, the inequality sign **flips**:

$$\mathbb{E}[X] + \frac{o(\lambda)}{\lambda} \geq \frac{\lambda \sigma^2}{2} + \frac{o(\lambda^2)}{\lambda}$$

Taking the limit as $\lambda \rightarrow 0^-$, the higher order terms vanish as before:

$$\mathbb{E}[X] \geq 0$$

Since $\mathbb{E}[X] \leq 0$ and $\mathbb{E}[X] \geq 0$, we conclude that $\mathbb{E}[X] = 0$. ■

Remark 2.12. Note that for any random variable we can work with its centered version when doing MGF work by

$$\exp(\lambda X) = \exp(\lambda \mathbb{E}X) \exp(\lambda(X - \mathbb{E}X))$$

Definition 2.13 (Sub-Gaussian random variable). Zero mean random variable X is sub-gaussian with variance parameter $\sigma > 0$, if for all $\lambda \in \mathbb{R}$

$$\mathbb{E} \exp(\lambda X) \leq \exp(\lambda^2 \sigma^2 / 2)$$

This definition implies the following equivalent properties:

1. **Property 1:** $\mathbb{P}(X \geq t) \leq \exp(-\frac{t^2}{2\sigma^2})$. We can boost this property to $\mathbb{P}(|X| \geq t) \leq 2\exp(-\frac{t^2}{2\sigma^2})$ since

$$\mathbb{P}(|X| \geq t) \leq P(X \geq t) + P(X \leq -t) = \exp(-\frac{t^2}{2\sigma^2}) + \exp(-\frac{t^2}{2\sigma^2})$$

Please reference the equivalence of sub-gaussian properties Lemma from my non-asymptotic matrices notes. I go over the equivalence theorem and required Lemma's to prove it very carefully.

Lemma 2.14 (Hoeffding's lemma). If X is a random variable with $\mathbb{E}X = 0$ and $X \in [a, b]$ then

$$\mathbb{E} \exp(\lambda X) \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right)$$

Proof. Let X' be an independent copy of X . Since $\mathbb{E}[X'] = 0$, we can write $X = X - \mathbb{E}[X']$.

Step 1: Symmetrization via Jensen's Inequality Using the fact that the exponential function is convex, we apply Jensen's inequality to the conditional expectation with respect to X' :

$$\begin{aligned} \mathbb{E}[\exp(\lambda X)] &= \mathbb{E}_X[\exp(\lambda(X - \mathbb{E}_{X'}[X']))] \\ &= \mathbb{E}_X[\exp(\mathbb{E}_{X'}[\lambda(X - X')])] \\ &\leq \mathbb{E}_X[\mathbb{E}_{X'}[\exp(\lambda(X - X'))]] \quad (\text{Jensen's Inequality}) \\ &= \mathbb{E}[\exp(\lambda(X - X'))] \end{aligned}$$

The first line is true because $\mathbb{E}_{X'}X' = 0$.

Step 2: Utilizing Symmetry Let $Z = X - X'$. Since X and X' are i.i.d., the distribution of Z is symmetric about 0 (i.e., Z and $-Z$ have the same distribution). Therefore:

$$\mathbb{E}[\exp(\lambda Z)] = \mathbb{E}[\exp(-\lambda Z)]$$

We can rewrite the expectation as an average:

$$\begin{aligned} \mathbb{E}[\exp(\lambda Z)] &= \frac{1}{2}\mathbb{E}[\exp(\lambda Z)] + \frac{1}{2}\mathbb{E}[\exp(-\lambda Z)] \\ &= \mathbb{E}\left[\frac{e^{\lambda Z} + e^{-\lambda Z}}{2}\right] \\ &= \mathbb{E}[\cosh(\lambda Z)] \end{aligned}$$

Step 3: Bounding with Taylor Series We use the standard inequality $\cosh(u) \leq \exp(u^2/2)$ (derived from comparing Taylor series):

$$\mathbb{E}[\cosh(\lambda Z)] \leq \mathbb{E}\left[\exp\left(\frac{\lambda^2 Z^2}{2}\right)\right]$$

Step 4: Bounded Support Since $X, X' \in [a, b]$, the maximum difference is bounded by $|Z| = |X - X'| \leq b - a$. Consequently, $Z^2 \leq (b - a)^2$. Substitute this bound into the expectation:

$$\mathbb{E} \left[\exp \left(\frac{\lambda^2 Z^2}{2} \right) \right] \leq \exp \left(\frac{\lambda^2 (b - a)^2}{2} \right)$$

Combining all steps completes the proof. $\blacksquare$

Proposition 2.15 (Hoeffding Inequality). Let $X_1, \dots, X_n$ be independent random variables with means μ_i such that $X_i - \mu_i$ are sub-Gaussian with variance parameters σ_i^2 in the sense of the definition of the sub-gaussian random variable. Then for any $t \geq 0$,

$$\mathbb{P} \left(\sum_k (X_k - \mu_k) \geq t \right) \leq \exp \left(-\frac{t^2}{2 \sum_k \sigma_k^2} \right)$$

and

$$\mathbb{P} \left(\left| \sum_k (X_k - \mu_k) \right| \geq t \right) \leq 2 \exp \left(-\frac{t^2}{2 \sum_k \sigma_k^2} \right)$$

Proof. By independence and the sub-gaussian MGF bound

$$\mathbb{E} \exp \left(\lambda \sum_k (X_k - \mu_k) \right) = \prod_k \mathbb{E} e^{\lambda (X_k - \mu_k)} \leq \exp \left(\frac{\lambda^2}{2} \sum_k \sigma_k^2 \right)$$

If we apply the Markov inequality to $\exp(\lambda(\sum_k (X_k - \mu_k)))$ and then optimize over $\lambda > 0$ we obtain the desired upper tail. The lower tail follows by applying the same argument to $-\sum_k (X_k - \mu_k)$ and then using the union bound. $\blacksquare$

Proposition 2.16 (Hoeffding's Inequality for bounded random variable (with non optimal constant)). Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}X_i = \mu_i$ and $X_i \in [a_i, b_i]$ almost surely. Then for any $t \geq 0$,

$$\mathbb{P} \left(\left| \sum_{i=1}^n (X_i - \mu_i) \right| \geq t \right) \leq 2 \exp \left(-\frac{t^2}{2 \sum_{i=1}^n (b_i - a_i)^2} \right).$$

Proof. By the Hoeffding Lemma, each $X_i - \mu_i$ is sub-gaussian with variance parameter $\sigma_i^2 = (b_i - a_i)^2$. We just plug into the previous proposition to get the desired result. $\blacksquare$

Example 2.17 (Averages of random signs). Let $\epsilon_1, \dots, \epsilon_n$ be independent random signs taking values ± 1 with probability $1/2$. Since $\epsilon_i \in [-1, 1]$ and $\mathbb{E}\epsilon_i = 0$, the Hoeffding inequality implies for any $t \geq 0$

$$\mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n \epsilon_i \right| \geq t \right) \leq 2 \exp \left(-\frac{t^2}{2n(4/n^2)} \right) = 2 \exp \left(-\frac{nt^2}{8} \right)$$

Equivalently, for any $\delta \in (0, 1)$. with probability at least $1 - \delta$

$$\left| \frac{1}{n} \sum_{i=1}^n \epsilon_i \right| \leq \sqrt{\frac{8 \log(2/\delta)}{n}}$$

by setting $\delta = 2 \exp(-\frac{nt^2}{8})$ and then solving for $t = \sqrt{8 \log(2/\delta)/n}$.

For comparison, Chebyshev inequality only yields

$$\mathbb{P}(|\frac{1}{n} \sum_k \epsilon_k| \geq t) \leq 1/(nt^2)$$

so similarly with probability at least $1 - \delta$ we have

$$|\frac{1}{n} \sum_k \epsilon_k| \geq t$$

Note that the CLT implies $n^{-1/2} \sum_k \epsilon_k \xrightarrow{d} N(0, 1)$. The $\sqrt{8 \log(2/\delta)/n}$ corresponds to the convergence to sub-gaussian tails.

2.2 Sub-Gaussian Norm

The MGF definition is nice for sums but sometimes we want to work with a norm that measures the size of the tails. The standard choice is the Orlicz ψ_2 norm

Definition 2.18 (Sub-gaussian Norm). For a random variable X , define its sub-Gaussian norm by

$$\|X\|_{\psi_2} = \inf \left\{ t > 0 : \mathbb{E} \exp \left(\frac{X^2}{t^2} \right) \leq 2 \right\}$$

It is standard (and can be checked directly) that $X \mapsto \|X\|_{\psi_2}$ defines a norm on random variables modulo almost sure equality.

Proposition 2.19 (Equivalent characterizations). There exist absolute constants $c, C > 0$ such that for any random variable X :

1. for all $t \geq 0$, $\mathbb{P}(|X| \geq t) \leq 2 \exp(-ct^2/\|X\|_{\psi_2}^2)$;
2. $\mathbb{E} \exp(X^2/\|X\|_{\psi_2}^2) \leq 2$;
3. for all $p \geq 1$, $\|X\|_{L_p} \leq C\|X\|_{\psi_2} \sqrt{p}$;
4. if $\mathbb{E}X = 0$, then for all $\lambda \in \mathbf{R}$, $\mathbb{E}e^{\lambda X} \leq \exp(C\lambda^2\|X\|_{\psi_2}^2)$.

Remark 2.20 (Centering). Centering does not significantly change the ψ_2 size:

$$\|X - \mathbb{E}X\|_{\psi_2} \leq \|X\|_{\psi_2} + \|\mathbb{E}X\|_{\psi_2}$$

and for any $t > \|X\|_{\psi_2}$ we have

$$\mathbb{E} \exp(X^2/t^2) \leq 2$$

and then Jensen's Inequality gives

$$\exp((\mathbb{E}X)^2/t^2) \leq \mathbb{E} \exp(X^2/t^2) \leq 2$$

Hence

$$\|\mathbb{E}X\|_{\psi_2} = |\mathbb{E}X|/\sqrt{\log(2)} \leq t$$

Therefore

$$\|X - \mathbb{E}X\|_{\psi_2} \leq \|X\|_{\psi_2} + t$$

Letting $t \rightarrow \|X\|_{\psi_2}$ completes the proof.

Example 2.21 (Bounded Random Variables). Let $|X| \leq M$ almost surely, then $\|X\|_{\psi_2} \leq M/\sqrt{\log 2}$. Indeed $\exp(X^2/t^2) \leq \exp(M^2/t^2)$ almost surely so choosing $t = M/\sqrt{\log 2}$ makes $\mathbb{E} \exp(X^2/t^2) \leq 2$.

Proposition 2.22 (Sums of independent sub-gaussians). There exists an absolute constant $C > 0$ such that if $X_1, \dots, X_n$ are independent mean-zero random variables, then

$$\left\| \sum_{i=1}^n X_i \right\|_{\psi_2}^2 \leq C \sum_i \|X_i\|_{\psi_2}^2$$

Proof. By the equivalent characterizations theorem, for each i and all $\lambda \in \mathbb{R}$

$$\mathbb{E} e^{\lambda X_i} \leq \exp(C_0 \lambda^2 \|X_i\|_{\psi_2}^2)$$

for an absolute constant C_0 . Using independence and multiplying MGFs,

$$\mathbb{E} \exp \left(\lambda \sum_k X_k \right) = \prod_k \mathbb{E} e^{\lambda X_k} \leq \exp(C_0 \lambda^2 \sum_{i=1}^n \|X_i\|_{\psi_2}^2)$$

Apply the equivalent characterizations theorem again to identify the ψ_2 norm of the sum. ■

2.3 Khintchine-type bounds

For independent sub-Gaussian summands, the natural geometry is Euclidean: the size of $\sum_i a_i X_i$ is governed by $\|a\|_2$.

Proposition 2.23 (Khintchine inequality for sub-Gaussian variables). Let $X_1, \dots, X_n$ be independent mean-zero sub-Gaussian random variables such that $\text{Var}(X_i) = 1$ for all i . Let $a = (a_1, \dots, a_n) \in \mathbf{R}^n$ and $p \geq 2$. Then

$$\left(\sum_{i=1}^n a_i^2 \right)^{1/2} \leq \left\| \sum_{i=1}^n a_i X_i \right\|_{L_p} \leq CK \sqrt{p} \left(\sum_{i=1}^n a_i^2 \right)^{1/2},$$

where $K = \max_{1 \leq i \leq n} \|X_i\|_{\psi_2}$ and $C > 0$ is an absolute constant.

Proof. For the lower bound, recall $\|\cdot\|_p \geq \|\cdot\|_2$ (L^p norms) for $p \geq 2$ hence

$$\|\langle a, X \rangle\|_p \geq \|\langle a, X \rangle\|_2 = \sqrt{\text{Var} \langle a, X \rangle} = \sqrt{\sum_i a_i^2 \text{Var} X_i^2} = \|a\|_2$$

For the upper bound, the equivalence theorem gives

$$\|Y\|_p \leq C_1 \sqrt{p} \|Y\|_{\psi_2}$$

for an absolute constant C_1 . Apply this to $Y = \langle a, X \rangle$ and use the sums theorem to:

$$\begin{aligned} \|\langle a, X \rangle\|_p &\leq C_1 \sqrt{p} \|\langle a, X \rangle\|_{\psi_2} \\ &\leq C_2 \sqrt{p} \sqrt{\sum_k \|a_k X_k\|_{\psi_2}^2} \\ &= C_2 \sqrt{p} \sqrt{\sum_k a_k^2 \|X_k\|_{\psi_2}^2} \leq C_2 K \sqrt{p} \sqrt{\sum_k a_k^2} \end{aligned}$$

completing the proof. ■

2.4 Sub-Exponential Random Variables

Example 2.24 (Motivation). Suppose $X \sim N(0, 1)$. Then consider $Z = X^2$.

$$\mathbb{E} \exp(\lambda X^2) = \int_{\mathbf{R}} \frac{1}{\sqrt{2\pi}} \exp(\lambda x^2 - x^2/2) dx = \infty$$

when $\lambda \geq \frac{1}{2}$ thus X^2 cannot be sub-gaussian.

Definition 2.25 (ψ_1 -norm / sub-exponential). Define $\|X\|_{\psi_1} = \inf\{t > 0 : \mathbb{E} \exp(\frac{|X|}{t}) \leq 2\}$. If a random variable has bounded ψ_1 -norm then it is called sub-exponential.

Lemma 2.26 (sub-exponential is sub-gaussian squared). A random variable X is sub-gaussian if and only if X^2 is sub-exponential. Moreover,

$$\|X\|_{\psi_2}^2 \leq \|X^2\|_{\psi_1} \leq 2 \|X\|_{\psi_2}^2$$

Proof. Assume X^2 is sub-exponential. Then,

$$(\mathbb{E}|X|^p)^{1/p} = (\mathbb{E}|X^2|^{p/2})^{(1/2)/(p/2)} \leq (\|X^2\|_{\psi_1} p/2)^{1/2}$$

Thus X is sub-gaussian with

$$\|X\|_{\psi_2} \leq \|X^2\|_{\psi_1}^{1/2} / \sqrt{2} \leq \sqrt{\|X^2\|_{\psi_1}}$$

Suppose X is now sub-gaussian.

Then

$$(\mathbb{E}|X^2|^p)^{1/p} = (\mathbb{E}|X|^{2p})^{2/2p} \leq (\|X\|_{\psi_2} \sqrt{2p})^2$$

Thus X^2 is sub-exponential with

$$\|X^2\|_{\psi_1} \leq 2 \|X\|_{\psi_2}^2$$

■

Proposition 2.27 (More properties of sub-gauss and sub-exp norms). Props:

1. $\|X\|_{\psi_1} \leq C \|X\|_{\psi_2}$ where C is an absolute constant.
2. $\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2$
3. $\|XY\|_{\psi_1} \leq \|X\|_{\psi_2} \|Y\|_{\psi_2}$ [If we show this one then we get (1) as a special case!]

Proof. Starting with (3). Let $a = \|X\|_{\psi_2}$; $b = \|Y\|_{\psi_2}$. Then

$$\frac{XY}{ab} \leq \frac{X^2}{2a^2} + \frac{Y^2}{2b^2}$$

which implies

$$\begin{aligned} \mathbb{E} \exp\left(\frac{|XY|}{ab}\right) &\leq \mathbb{E} \exp(X^2/(2a^2)) \cdot \exp(Y^2/(2b^2)) \\ (\text{C.S. Inequality}) &\leq \sqrt{\mathbb{E} \exp(X^2/a^2)} \cdot \sqrt{\mathbb{E} \exp(Y^2/b^2)} \\ &\leq \sqrt{2}^2 = 2 \end{aligned}$$

Thus $\|XY\|_{\psi_1} \leq ab = \|X\|_{\psi_2} \|Y\|_{\psi_2}$.

■

Proposition 2.28 (Tails of sub-exp random variable). Assume that X is sub-exponential. There exists absolute constants $c_1, c_2, \dots, > 0$ such that

1. $\mathbb{P}(|X| > t) \leq 2 \exp(-\frac{c_1 t}{\|X\|_{\psi_1}})$
2. $\mathbb{E} \exp(\frac{|X|}{\|X\|_{\psi_1}}) \leq 2$ by def
3. $\|X\|_p = (\mathbb{E}|X|^p)^{1/p} \leq c_3 p \|X\|_{\psi_1}$
4. If additionally $\mathbb{E}X = 0$ then for $|\lambda| \leq \frac{c_4}{\|X\|_{\psi_1}}$ we have $\mathbb{E} \exp(\lambda X) \leq \exp(\lambda^2 c_5 \|X\|_{\psi_1}^2)$

2.5 Bernstein's Inequality

To some sense it is an analogue of Hoeffding's inequality.

Example 2.29 (Motivation). Suppose $X \sim \text{Exp}(\lambda)$. That is $\mathbb{P}(X > t) = \exp(-\lambda t)$ for all $t \geq 0$. We claim that

$$\|X\|_{\psi_1} = \frac{2}{\lambda}$$

We have

$$\mathbb{E} \exp(\frac{|X|}{t}) = \int_0^\infty \exp(x/t) \exp(-\lambda x) \lambda dx = \lambda \frac{1}{\frac{1}{t} - \lambda} \exp(x/t - \lambda x)|_0^\infty = \frac{\lambda}{\lambda - 1/t} \leq 2$$

when $\lambda - \frac{1}{t} > 0$. Setting equal to 2 gives $t = 2/\lambda$ as claimed.

Proposition 2.30 (Bernstein's inequality). Suppose $X_1, \dots, X_n$ are independent random variables with $\mathbb{E}X_i = 0$ and $\max_i \|X_i\|_{\psi_1} < \infty$. Then for any $t > 0$,

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp(-c \cdot \min\left\{\frac{t^2}{\sum_{i=1}^n \|X_i\|_{\psi_1}^2}, \frac{t}{\max_i \|X_i\|_{\psi_1}}\right\})$$

and

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \cdot \exp(-c \cdot \min\left\{\frac{t^2}{\sum_{i=1}^n \|X_i\|_{\psi_1}^2}, \frac{t}{\max_i \|X_i\|_{\psi_1}}\right\})$$

where $c > 0$ is an absolute constant.

Example 2.31 (χ^2 -distribution). If $Z = \sum_{i=1}^d X_i^2$ where $X_i \stackrel{\text{iid}}{\sim} N(0, 1)$. Then

$$\|X_i^2 - 1\|_{\psi_1} \leq \|X_i^2\|_{\psi_1} + \|1\|_{\psi_1} = 8/3 + \frac{1}{\log 2}$$

Then we apply Bernstein:

$$\mathbb{P}(|Z - d| \geq t) \leq 2 \exp(-c \cdot \min \left\{ \frac{t^2}{d}, t \right\})$$

where $c > 0$ was taken to be small enough. If we take the bound to be δ then we can say

$$|Z - d| \leq C \left(\overbrace{\sqrt{d \log(2/\delta)}}^{\text{Gaussian Tail}} + \overbrace{\log(2/\delta)}^{\text{Exponential Tail}} \right)$$

with probability at least $1 - \delta$.

Definition 2.32 $((\sigma, b)$ -sub-exponential). If $\mathbb{E}X = 0$, and for all $|\lambda| \leq 1/b$:

$$\mathbb{E} \exp(\lambda X) \leq \exp \left(\frac{X^2 \sigma^2 / 2}{1 - b|\lambda|} \right)$$

Proposition 2.33 (Bernstein's Assumption: A property that implies (σ, b) -sub-exponential). A random variable X with mean $\mathbb{E}X = \mu$ and variance $\text{Var}(X) = \sigma^2$ satisfies Bernstein's assumption with parameters (σ, B) if for all $k \in \mathbb{N}, k \geq 2$,

$$\mathbb{E}(X - \mu)^k \leq \frac{1}{2} k! \sigma^2 B^{k-2}$$

Example 2.34 (Bernstein's assumption). hi.

1. **Example 1: (Bounded Random Variable)** Consider X with $\mathbb{E}X = 0$, $\text{Var}(X) = \sigma^2$; $|X| \leq a$. Then

$$\mathbb{E}X^2 \leq \frac{1}{2} 2! \sigma^2 = \text{Var}(X)$$

$$\mathbb{E}X^k \leq \mathbb{E}X^2 |X|^{k-2} \leq \sigma^2 a^{k-2}$$

For $k = 3$, letting $B = a/3$ gives the correct bound

$$\mathbb{E}X^3 \leq \sigma^2 a = \frac{1}{2} 3! (a/3)$$

For $k \geq 4$,

$$\frac{1}{2} k! \sigma^2 (a/3)^{k-2} \geq \sigma^2 a^{k-2}$$

since $\frac{k!}{23^{k-2}} \geq 1$ for all $k \geq 4$. Thus X is $(\sigma, a/3)$ -Bernstein.

2. **Example 2:** If $X \sim \text{Bernoulli}(p)$ then $\text{Var}(X) = p(1 - p)$; $|X| \leq 1$. WLOG assume $\mathbb{E}X = 0$. Then

$$\begin{aligned}\mathbb{E} \exp(\lambda X) &= 1 + 0 + \frac{\mathbb{E}(\lambda^2 X^2)}{2} + \sum_{k=3}^{\infty} \frac{\lambda^k \mathbb{E}X^k}{k!} \\ &\leq 1 + \lambda^2 \sigma^2 / 2 + \lambda^2 \sigma^2 / 2 \sum_{k=3}^{\infty} B^{k-2} \lambda^{k-2} \\ &\leq 1 + \lambda^2 \sigma^2 / 2 + \lambda^2 \sigma^2 / 2 \left(\frac{B|\lambda|}{1-B|\lambda|} \right) \\ &\leq \exp\left(\frac{\lambda^2 \sigma^2 / 2}{1-B|\lambda|}\right)\end{aligned}$$

Proposition 2.35 (Bernstein's inequality (σ, b) -form). If $X_1, \dots, X_n$ are independent zero mean random variables with variances $(\sigma_1^2, \dots, \sigma_n^2)$ that satisfy Bernstein's assumption with (σ_i, b) then for all $t > 0$

$$\begin{aligned}\mathbb{P}\left(\sum_k X_k \geq t\right) &\leq \exp\left(\frac{-t^2/2}{\sum_k \sigma_k^2 + bt}\right) \\ \mathbb{P}\left(\left|\sum_k X_k\right| \geq t\right) &\leq 2 \exp\left(\frac{-t^2/2}{\sum_k \sigma_k^2 + bt}\right)\end{aligned}$$

The $+bt$ part creates the exponential tail part.

Proof. In home assignment. ■

Corollary 2.36. In particular when the X_i are zero mean; $\text{Var}(X_i) = \sigma^2$ and $|X_i| \leq B$ by the Bounded RV is Bernstein example and the last proposition we have

$$\mathbb{P}\left(\sum_k X_k \geq t\right) \leq \exp\left(-\frac{t^2/2}{n\sigma^2 + \frac{B}{3}t}\right)$$

Example 2.37. Suppose $X = \sum_{k=1}^n X_k$ and $X_k \stackrel{\text{iid}}{\sim} \text{Bern}(p)$.

1. Hoeffding's Inequality (prop 2.15):

$$|X - p| \leq 1 \implies \exp(\lambda(X_i - p)) \leq \exp\left(\frac{\lambda^2}{2.4}\right)$$

Then with probability $1 - \delta$ we have

$$\left|\frac{\sum_{k=1}^n (X_k - p)}{n}\right| \leq \sqrt{\frac{\log(2/\delta)}{2n}}$$

2. Bernstein's inequality: $\text{Var}(X_i) = p(1-p)$ and $|X_i - p| \leq 1 =: B$. Solving Bernstein's inequality with respect to δ we get

$$\left| \frac{1}{n} \sum_{i=1}^n X_i \right| \leq \sqrt{\frac{2 \log(2/\delta) p(1-p)}{n}} + \frac{2 \log(2/\delta)}{3n}$$

The $\frac{2 \log(2/\delta)}{3n}$ term corresponds to the $\frac{1}{3}t$ term. If p is very small then you are still not sub-gaussian and this term is the price to pay for it.

The $\sqrt{\frac{2 \log(2/\delta) p(1-p)}{n}}$ term corresponds to the $\sum \sigma_i^2$ term and is the CLT style term.

3. Recall Hoeffding's inequality. If X is a random variable with $\mathbb{E}X = 0$ and $\mathbb{E} \exp(\lambda X) \leq \exp(\frac{\lambda^2 \sigma^2}{2})$ where σ^2 is the sub-gaussian parameter. The Kearns-Saul bound is if X is $\text{Bern}(p)$ then

$$\mathbb{E}(\lambda(X - p)) \leq \exp\left(\frac{\lambda^2(1-2p)}{4 \log(\frac{1-p}{p})}\right)$$

Note

$$\mathbb{E}(\lambda(X - p)) = p \exp(\lambda(1-p)) + (1-p) \exp(-\lambda p)$$

For any value of $p \in [0, 1]$, Hoeffding's inequality gives a bound of $\frac{1}{4}$. The variance is the curve $p(1-p)$.

Claim: If σ^2 is a sub-gaussian parameter then $\text{Var}(X) \leq \sigma^2$.

Make a note about how different bounds behave for different values of p .

Proposition 2.38 (multiplicative bounds via Bernstein's inequality). Suppose $X \sim \text{Binom}(n, p)$. We would like a bound

$$\frac{1}{2} \mathbb{E}X + \dots \leq X \leq \frac{3}{2} \mathbb{E}X + \dots$$

We want to decide what the $\dots$ should be.

Let $X_i \stackrel{\text{iid}}{\sim} \text{Bern}(p)$ then

$$\begin{aligned} |X - \mathbb{E}X| &= \left| \sum_{i=1}^n (X_i - p) \right| \\ &\leq \sqrt{2np(1-p) \log(2/\delta)} + \frac{2}{3} \log(2/\delta) \\ &= \sqrt{2(\mathbb{E}X) \log(2/\delta)} + \frac{2}{3} \log(2/\delta) \\ &\leq \frac{1}{2} \gamma \mathbb{E}X + \frac{1}{2\gamma} 2 \log(2/\delta) + \frac{2}{3} \log(2/\delta) \end{aligned}$$

since $\mathbb{E}X = np$ and $1 - p \leq 1$.

Lemma: for all $a, b, \gamma > 0$ we have $\sqrt{ab} \leq \frac{1}{2}(a/\gamma + b\gamma)$.

Choosing $\gamma = 1$ in our case we get

$$\frac{1}{2}\mathbb{E}X - \frac{5}{3}\log(2/\delta) \leq X \leq \frac{3}{2}\mathbb{E}X + \frac{5}{3}\log(2/\delta)$$

Example 2.39 (Statistical Learning in the Realizable Setting). **Setup:** Let $\mathcal{X}$ be the instance space (e.g., the space of all possible emails) and let $\mathcal{F}$ be a family of binary classifiers (the hypothesis class). For any $f \in \mathcal{F}$, we have the mapping $f : \mathcal{X} \rightarrow \{0, 1\}$ (e.g., mapping an email to "spam" or "not spam").

We assume the *realizable case*, meaning there exists a true target function $f^* \in \mathcal{F}$ such that the labels are generated deterministically by f^* .

Data Generation: We observe a dataset $S_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$, where the inputs $X_i \stackrel{\text{i.i.d.}}{\sim} P$ are drawn from an unknown distribution P , and the labels are given by $Y_i = f^*(X_i)$. Note that while we see the values $f^*(X_i)$, the function f^* itself remains unknown.

Risk Definitions: We define the *true prediction risk* (generalization error) as the probability that f misclassifies a random sample drawn from P :

$$R(f) = \mathbb{P}_{X \sim P}(f(X) \neq f^*(X)) = \mathbb{E}[\mathbf{1}_{\{f(X) \neq f^*(X)\}}]$$

Since P is unknown, we cannot compute $R(f)$ directly. Instead, we compute the *empirical risk* based on the observed sample:

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{f(X_i) \neq f^*(X_i)\}}$$

Empirical Risk Minimization (ERM): Our goal is to construct an estimator $\hat{f}$ such that its true risk $R(\hat{f})$ is small with high probability. The Empirical Risk Minimizer (ERM) is defined as:

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} R_n(f)$$

Note that $\hat{f}$ is a random variable because it depends on the random sample S_n .

Concentration Bounds: Assume the hypothesis class is finite with cardinality $|\mathcal{F}| = M$.

Consider a fixed $f \in \mathcal{F}$. The loss on a single point, $Z_i = \mathbf{1}_{\{f(X_i) \neq f^*(X_i)\}}$, is a Bernoulli random variable. The variance of this variable is $\text{Var}(Z_i) = R(f)(1 - R(f)) \leq R(f)$.

Applying **Bernstein's Inequality** (which utilizes the variance information) to the

random variable $R(f) - R_n(f)$, we have that with probability at least $1 - \delta$:

$$\begin{aligned} R(f) - R_n(f) &\leq \sqrt{\frac{2\text{Var}(Z) \log(2/\delta)}{n}} + \frac{2}{3} \frac{\log(2/\delta)}{n} \\ &\leq \sqrt{\frac{2R(f) \log(2/\delta)}{n}} + \frac{2}{3} \frac{\log(2/\delta)}{n} \end{aligned}$$

Uniform Convergence (Union Bound): The previous bound holds for a *fixed* f . To guarantee the bound holds for $\hat{f}$ (which depends on the data), we must ensure the bound holds simultaneously for *all* $f \in \mathcal{F}$.

By applying the union bound over all M functions in $\mathcal{F}$, we distribute the failure probability δ across the class (replacing δ with δ/M). Thus, with probability at least $1 - \delta$, the following holds for all $f \in \mathcal{F}$:

$$R(f) - R_n(f) \leq \sqrt{\frac{2R(f) \log(2M/\delta)}{n}} + \frac{2}{3} \frac{\log(2M/\delta)}{n}$$

Since that bound holds for all $f \in \mathcal{F}$, we have that it also holds for all those $\hat{f}$ such that $R_n(\hat{f}) = 0$. For those $\hat{f}$ we get

$$\begin{aligned} R(\hat{f}) &\leq \sqrt{\frac{2R(\hat{f}) \log(2M/\delta)}{n}} + \frac{2}{3} \frac{\log(2M/\delta)}{n} \\ &\leq \frac{1}{2} R(\hat{f}) + \frac{\log(2M/\delta)}{n} \left(1 + \frac{2}{3}\right) \end{aligned}$$

Thus

$$R(\hat{f}) \leq \frac{10}{3} \frac{\log(2M/\delta)}{n}$$

Look into the exact step we applied Bernstein's inequality and make sure you understand that part :)

Alternate Argument Consider the bad event where $\exists f \in \mathcal{F}$ such that $R(f) \geq \epsilon$, $R_n(f) = 0$:

$$\begin{aligned} \mathbb{P}(\exists f \in \mathcal{F} : R(f) \geq \epsilon, R_n(f) = 0) &\leq \sum_{f \in \mathcal{F}} \mathbb{P}(R(f) \geq \epsilon, R_n(f) = 0) \\ &\leq M(1 - \epsilon)^n \\ &\leq M \exp(-n\epsilon) = \delta \end{aligned}$$

Then $\epsilon = \log(M/\delta)/n$. So with probability $1 - \delta$, $R(\hat{f}) < \log(M/\delta)/n$.

2.6 Maximal Inequalities

Suppose we have $X_1, \dots, X_n$ random variables not necessarily independent. We want to study $\mathbb{E} \max(X_1, \dots, X_n)$.

Proposition 2.40. Assume that the X_i are zero mean and sub-gaussian with σ . That is

$$\mathbb{E} \exp(\lambda X_i) \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right)$$

Then

$$\mathbb{E} \max(X_1, \dots, X_n) \leq \sigma \sqrt{2 \log n}$$

Alternatively, given $\|X_i\|_{\psi_2}$ we can say

$$\mathbb{E} \max(X_1, \dots, X_n) \leq C \max \|X_i\|_{\psi_2} \sqrt{\log n}$$

Proof. Suppose $\lambda > 0$. Then

$$\begin{aligned} \mathbb{E} \max(X_1, \dots, X_n) &= \frac{1}{\lambda} \mathbb{E} \lambda \max(X_1, \dots, X_n) \\ &\leq \frac{1}{\lambda} \log \mathbb{E} \exp(\lambda \max(X_1, \dots, X_n)) \\ &= \frac{1}{\lambda} \log \mathbb{E} \max \exp(\lambda(X_1, \dots, X_n)) \\ &\leq \frac{1}{\lambda} \log \left(\sum_{k=1}^n \mathbb{E} \exp(\lambda X_k) \right) \\ &\leq \frac{1}{\lambda} \log(n \exp(\lambda^2 \sigma^2 / 2)) = \frac{\log n}{\lambda} + \lambda^2 \sigma^2 / 2 \end{aligned}$$

Optimize wrt to λ to attain the answer.

To remember how to use Jensen's inequality use variance:

$$\mathbb{E}(X - \mathbb{E}X)^2 \geq 0 = (\mathbb{E}X - \mathbb{E}X)^2$$

■

Remark 2.41. Note the following

1. The $\sqrt{\log n}$ is achieved when X_i are independent Gaussians
2. $\mathbb{E} \max |X_i| \leq \mathbb{E} \max(X_1, \dots, X_n, -X_1, \dots, -X_n) \implies \mathbb{E} \max_i |X_i| \leq \sigma \sqrt{2 \log(2n)}$

Proposition 2.42. Assume that $X_1, \dots, X_n$ are zero-mean (σ, b) -sub-exponential random variables satisfying:

$$\mathbb{E}[\exp(\lambda X_i)] \leq \exp\left(\frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|}\right) \quad \text{for all } |\lambda| < \frac{1}{b}.$$

Then the expected maximum satisfies:

$$\mathbb{E} \left[\max_{1 \leq i \leq n} X_i \right] \leq \sigma \sqrt{2 \log n} + b \log n.$$

Equivalently, using the sub-exponential norm $\|\cdot\|_{\psi_1}$, there exists a universal constant C such that:

$$\mathbb{E} \left[\max_{1 \leq i \leq n} X_i \right] \leq C \max_{1 \leq i \leq n} \|X_i\|_{\psi_1} \log n.$$

Proposition 2.43. Generalization in terms of moments. For any $p \geq 1$,

$$\begin{aligned} \mathbb{E} \max(X_1, \dots, X_n) &\stackrel{\text{Jensen's}}{\leq} (\mathbb{E} \max(|X_1|^p, \dots, |X_n|^p))^{1/p} \\ &\leq \left(\mathbb{E} \sum_k |X_k|^p \right)^{1/p} \\ &\leq \left(n \max_k \mathbb{E} |X_k|^p \right)^{1/p} \\ &= n^{1/p} \max_k (\mathbb{E} |X_k|^p)^{1/p} \end{aligned}$$

We can rederive the sub-gaussian result by applying this result to $p = \log n$ and then also using the L_p bound which applied to sub-gaussian random variables.

Definition 2.44 (Bounded Difference Property). The function $g : \mathcal{X}^n \rightarrow \mathbb{R}$ has bounded differences property if for all $(x_1, \dots, x_n, x'_i)$ and for all $i = 1, \dots, n$

$$|g(x_1, \dots, x_n) - g(x_1, \dots, x'_i, \dots, x_n)| \leq c_i$$

with constants $c_1, \dots, c_n$.

Proposition 2.45 (Bounded Differences inequality (McDiarmid's Inequality)). If $X_1, \dots, X_n$ are independent random variables and g satisfies the bounded difference property for all $t \geq 0$,

$$\mathbb{P}(g(X_1, \dots, X_n) - \mathbb{E}g(X_1, \dots, X_n) \geq t) \leq \exp\left(\frac{-2t^2}{\sum_k c_k^2}\right)$$

and

$$\mathbb{P}(|g(X_1, \dots, X_n) - \mathbb{E}g(X_1, \dots, X_n)| \geq t) \leq 2 \exp\left(\frac{-2t^2}{\sum_k c_k^2}\right)$$

Proof. Define $Y_i = \mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_i] - \mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_{i-1}]$. We get the nice telescoping sum

$$g(X_1, \dots, X_n) - \mathbb{E}g(X_1, \dots, X_n) = \sum_{k=1}^n Y_k$$

Now

$$\begin{aligned}
\mathbb{E} \exp(\lambda \sum_{k=1}^n Y_k) &= \mathbb{E} \exp(\lambda \sum_{k=1}^{n-1} Y_k) \cdot \exp(\lambda Y_n) \\
(\text{tower property}) &= \mathbb{E} \left(\mathbb{E} \left(\exp(\lambda \sum_{i=1}^{n-1} Y_i) \cdot \exp(\lambda Y_n) \mid X_1, \dots, X_{n-1} \right) \right) \\
&= \mathbb{E} \left(\exp(\lambda \sum_{i=1}^{n-1} Y_i) \cdot \mathbb{E}(\exp(\lambda Y_n) \mid X_1, \dots, X_{n-1}) \right) \\
&\leq \mathbb{E} \left(\exp(\lambda \sum_{i=1}^{n-1} Y_i) \cdot \exp(\frac{\lambda^2 c_n^2}{8}) \right)
\end{aligned}$$

Note that given $(X_1, \dots, X_{n-1})$, Y_n is a function of X_n . We can see that by the bounded differences property $Y_n(X_n)$ belongs to an interval of length c_n .

$$\begin{aligned}
Y_n &= \mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_n] - \mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_{n-1}] \\
&= g - \mathbb{E}_X g(x_1, \dots, x_{n-1}, X) \\
&= \mathbb{E}_X [(g(x_1, \dots, x_n) - g(x_1, \dots, x_{n-1}, X))] \\
&\leq c_n
\end{aligned}$$

This implies $|Y_n| \leq c_n$.

The argument follows inductively and we get

$$\mathbb{E} \exp(\lambda \sum_{k=1}^n Y_k) \leq \exp(\frac{\lambda^2 \sum c_i^2}{8})$$

■

Example 2.46 (Kernel Density Estimation). We call $K(x)$ a kernel function if

1. $K(x) \geq 0$
2. $\int_{\mathbb{R}} K(x) dx = 1$

Typically, a kernel estimator is as: if f is the true density then

$$\hat{f} = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

Where $h > 0$ is a window size which needs to be tuned and $X_1, \dots, X_n$ is the observed sample. The natural question is

$$\int_{\mathbb{R}} |\hat{f}(x) - f(x)| dx$$

How far is $\int_{\mathbb{R}} |\hat{f}(x) - f(x)| dx$ from $\mathbb{E}_{X_1, \dots, X_n} \int_{\mathbb{R}} |\hat{f}(x) - f(x)| dx$? Take X'_i as in McDiarmid's inequality. We need to check the bounded differences property for the Kernel. By the triangle inequality

$$\begin{aligned} & \left| \int_{\mathbb{R}} \left| \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) - f(x) \right| dx - \int_{\mathbb{R}} \left| \frac{1}{nh} \sum_{i \neq i'}^n K\left(\frac{x-x_i}{h}\right) + \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-x'_i}{h}\right) - f(x) \right| dx \right| \\ & \leq \left| \int_{\mathbb{R}} \left| \frac{1}{nh} K\left(\frac{x-x_i}{h}\right) - \frac{1}{nh} K\left(\frac{x-x'_i}{h}\right) \right| dx \right| \leq \frac{1}{nh} \left(\int_{\mathbb{R}} K\left(\frac{x-x_i}{h}\right) + K\left(\frac{x-x'_i}{h}\right) dx \right) = \frac{2}{n} \end{aligned}$$

Thus with probability $1 - \delta$

$$\left| \int_{\mathbb{R}} |\hat{f}(x) - f(x)| dx - \mathbb{E} \int_{\mathbb{R}} |\hat{f}(x) - f(x)| dx \right| \leq \sqrt{\frac{2 \log(2/\delta)}{h}}$$

3 Concentration of the norm of random vectors

Suppose we start with vectors with independent, zero mean, variance coordinates. Such as $X \sim N(0, I_d)$. We know $\|X\|_2^2 \sim \chi_d^2$. By Bernstein's inequality

$$\mathbb{P}(|\|X\|^2 - d| \geq t) \leq 2 \exp(-c \min(t^2/d, t))$$

Sub-Gaussian Coordinates and Isotropic Distribution: We can generalize this with $X \in \mathbb{R}^d$; st. $\mathbb{E}X_i = 0$, $X_1, \dots, X_n$ are independent coordinates with $\text{Var}(X_i) = 1$; and $\max_i \|X_i\|_{\psi_2} \leq K$. In this case we have

$$\mathbb{E} \|X\|^2 = \mathbb{E} \sum_k X_k^2 = d$$

$$\max_i \|X_i^2\|_{\psi_1} = \max_i \|X_i\|_{\psi_2}^2 \leq K^2$$

This implies

$$\mathbb{P}(\|X\|_2^2 - d \geq t) \leq 2 \exp(-c \min(\frac{t^2}{dK^4}, \frac{t}{K^2}))$$

WLOG we may assume $K \geq 1$ since $\text{Var}X_i = 1$. Thus

$$\mathbb{P}(\|X\|_2^2 - d \geq t) \leq 2 \exp(-c'/K^4 \min(\frac{t^2}{d}, t))$$

What if we want to talk about $\|X\|_2$ instead of the squared norm? Then we can use this trick:

$$|z - 1| \geq u \implies |z^2 - 1| \geq \max\{u, u^2\}$$

We can use this to get

$$\mathbb{P}\left(\left|\frac{\|X\|_2}{\sqrt{d}} - 1\right| \geq t\right) \leq 2 \exp(-c'/K^4 \min\{\max(t^2, t^4), \max(t\sqrt{d}, t^2 d)\})$$

A few more notes on this: Note that by Cauchy schwarz

$$\mathbb{E} \|X\|_2 \leq \sqrt{\mathbb{E} \|X\|_2^2} = \sqrt{d}$$

Using the algebraic hint

$$\begin{aligned} \mathbb{P}\left(\left|\frac{\|X\|_2}{\sqrt{d}} - 1\right| \geq t\right) &\leq 2 \exp(-\frac{cd}{K^4} \min\{\max(t^2, t^4), \max\{t, t^2\}\}) \\ &= 2 \exp(-\frac{cd}{K^4} t^2) \end{aligned}$$

Thus

$$\mathbb{P}(|\|X\|_2 - \sqrt{d}| \geq t) \leq 2 \exp(-\frac{c}{K^4} t^2) \iff \left\| \|X\|_2 - \sqrt{d} \right\|_{\psi_2} \leq CK^2$$

3.1 Concentration of the norm for non-independent vectors

Definition 3.1 (second moment matrix & covariance matrix). The second moment matrix of a random vector X is defined as

$$\Sigma = \Sigma(X) = \mathbb{E}X \otimes X = \mathbb{E}XX^\top$$

where $\otimes$ denotes the outer product of vectors in $\mathbb{R}^n$.

The Covariance matrix of X is defined as

$$\text{Cov}(X) = \mathbb{E}(X - \mu) \otimes (X - \mu) = \mathbb{E}X \otimes X - \mu \otimes \mu$$

where $\mu = \mathbb{E}X$.

Definition 3.2 (Isotropic random vectors). A random vector X in $\mathbb{R}^n$ is called isotropic if $\Sigma(X) = I$. Equivalently, X is isotropic if

$$\mathbb{E} \langle X, x \rangle^2 = \|x\|_2^2 \text{ for all } x \in \mathbb{R}^n$$

Proof. of equivalence of definitions. Assume X is isotropic in the sense of the first definition. Let $x \in \mathbb{R}^n$. Then

$$\mathbb{E} \langle X, x \rangle^2 = \mathbb{E} x^\top X X^\top x = x^\top (\mathbb{E} X X^\top) x = x^\top x = \|x\|_2^2$$

Conversely, assume X is isotropic as in the second sense. Let $x \in \mathbb{R}^n$.

We have

$$x^\top (\Sigma(X) - I) x = x^\top \Sigma(X) x - x^\top x = x^\top x - x^\top x = 0$$

Since $\Sigma(X)$ is symmetric and I is symmetric, their difference is symmetric. Thus by the spectral theorem all the eigenvalues are real values. The above shows that all the eigenvalues are zero. Thus $\Sigma - I = 0$ so $\Sigma = I$. $\blacksquare$

Example 3.3 (Isotropic Random Variables). Some examples.

1. **Gaussian:** The standard Gaussian random vector X in $\mathbb{R}^n$ chosen according to the standard normal distribution $N(0, I)$. The coordinates of X are independent standard normal random variables. We know $\text{Cov}(X) = I$ and $\mu(X) = 0$ thus $\Sigma(X) = I$.
2. **Bernoulli:** A similar example of a discrete isotropic distribution is given by a Bernoulli random vector X in $\mathbb{R}^n$ whose coordinates are independent symmetric Bernoulli random variables.
3. **Product distributions:** More generally, consider a random vector X in $\mathbb{R}^n$ whose coordinates are independent random variables with zero mean and unit variance. Then clearly X is an isotropic random vector in $\mathbb{R}^n$.
4. **Coordinate:** Consider a coordinate random vector X , which is uniformly distributed in the set $\{\sqrt{n}e_i\}_{i=1}^n$ where $\{e_i\}$ forms a canonical basis in $\mathbb{R}^n$. Clearly X is an isotropic random vector in $\mathbb{R}^n$: $\mathbb{E} X_i X_j = 0$ for all $i \neq j$ since only one coordinate of X can be non-zero at a time. Then $\mathbb{E} X_i^2 = \frac{1}{n} \sqrt{n}^2 = 1$.
5. **Frame:** This is a more general version of the coordinate random vector. A frame is a set of vectors in $\mathbb{R}^n$ which obeys an approximate Parseval's identity, i.e. there exists numbers, $A, B > 0$ called frame bounds such that

$$A \|x\|_2^2 \leq \sum_{i=1}^M \langle u_i, x \rangle^2 \leq B \|x\|_2^2 \text{ for all } x \in \mathbb{R}^n$$

Put another way,

$$A \|x\|_2^2 \leq x^\top \left(\sum_{i=1}^M u_i u_i^\top \right) x \leq B \|x\|_2^2 \text{ for all } x \in \mathbb{R}^n$$

If $A = B$ the set is called a tight frame. Thus, tight frames are generalizations of orthogonal bases without linear independence. Given a tight frame $\{u_i\}_{i=1}^M$ with bounds $A = B = M$, the random vector X uniformly distributed in the set $\{u_i\}_{i=1}^M$ is clearly isotropic in $\mathbb{R}^n$.

$$\mathbb{E}XX^\top = \sum_{i=1}^M u_i u_i^\top \frac{1}{M}$$

Thus for any $x \in \mathbb{R}^n$,

$$x^\top (\mathbb{E}XX^\top) x = \frac{1}{M} x^\top \left(\sum_{i=1}^M u_i u_i^\top \right) x = \frac{1}{M} M \|x\|_2^2 = \|x\|_2^2$$

So X is isotropic.

6. **Spherical:** Consider a random vector X uniformly distributed on the unit Euclidean sphere in $\mathbb{R}^n$ with center at the origin and radius $\sqrt{n}$. Then X is isotropic. Indeed by rotation invariance $\mathbb{E} \langle X, x \rangle^2$ is proportional to $\|x\|_2^2$:

To expand on this, if U is a rotation matrix then UX has the same distribution as X . Thus $\Sigma(X) = \mathbb{E}(UX)(UX)^\top = U\Sigma(X)U^\top$. So $\Sigma(X)U = U\Sigma(X)$. Thus by Lemma 3.4 $\Sigma(X) = cI$ for $c \in \mathbb{R}$.

We have $\mathbb{E} \|X\|_2^2 = n$ since $\|X\|_2 = \sqrt{n}$ almost surely. Recall that $\mathbb{E} \|X\|_2^2 = \mathbb{E} \text{Tr}(XX^\top) = \text{Tr}(cI) = nc$. Thus $nc = n$ so $c = 1$. Thus X is isotropic.

Lemma 3.4 (Shur's Lemma applied to orthogonal group $O(n)$). If symmetric $A \in \mathbb{R}^{n \times n}$ commutes with all rotations matrices (orthogonal matrices) U , then $A = cI$ for $c \in \mathbb{R}$.

Proof. Since A is symmetric, it has at least one eigenvector v with real eigenvalue $\lambda \in \mathbb{R}$. So $Av = \lambda v$. Let w be any other unit vector in $\mathbb{R}^n$. Let U be a rotation matrix which maps v to w . Then $Aw = AUv = UAv = \lambda Uv = \lambda w$. Thus all unit vectors w are eigenvectors with the same eigenvalue λ . The only matrix that scales every vector by λ is λI . ■

Show that the coordinates of X are not independent!

7. **Uniform on a convex set:** In convex geometry, a convex set K in $\mathbb{R}^n$ is called isotropic if a random vector X chosen uniformly from K according to the volume is isotropic. As we noted, every full dimensional convex set can be made into an isotropic one by an affine transformation. Isotropic convex sets look “well conditioned”, which is advantageous in geometric algorithms (e.g. volume computations).

Remark 3.5. Assume that $X \in \mathbb{R}^d$ is such that $\mathbb{E}X = \mu$ and $\text{Cov}(X) = \Sigma$; Σ is invertible. Then $Z = \Sigma^{-1/2}(X - \mu)$ is isotropic since $\mathbb{E}Z = 0$ and

$$\begin{aligned}\mathbb{E}ZZ^\top &= \Sigma^{-1/2}\mathbb{E}(X - \mu)^\top(X - \mu)\Sigma^{-1/2} \\ &= \Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I_d\end{aligned}$$

Note that if Σ is not full rank then there is a lower dimensional basis of eigenvectors $v_1, \dots, v_k$ with $k < d$. Then in the orthogonal complement of the span of these vectors, the random vector has no variance (it must be constant a.s.).

Definition 3.6 (Sub-Gaussian Random Vectors). Assume $\mathbb{E}X = 0$. There are a few definitions which are not equivalent

1. Def 1: For all $s \in S^{d-1}$, $\|\langle X, s \rangle\|_{\psi_2} \leq K$
2. Def 2: For all $v \in S^{d-1}$, $\|\langle X, v \rangle\|_{\psi_2} \leq C\sqrt{v^\top \Sigma v}$ where $\Sigma = \text{Cov}(X)$. In some sense, this is a stronger assumption since the variance can get a lot smaller than the gaussian norm.
3. Def 3: If $v \in S^{d-1}$ then $\mathbb{E}\exp(\lambda \langle v, X \rangle) \leq \exp(\frac{\lambda^2}{2}v^\top \Sigma v)$. We can either think of Σ as the covariance or as a covariance proxy.

Proposition 3.7 (isotropic special case). If X is isotropic then $\Sigma = I_d$, so $v^\top \Sigma v = 1$ and then definitions 1 and 2 become the same.

Theorem 3.8. Assume that $X \in \mathbb{R}^d$ is a random vector with $\mathbb{E}X = 0$ such that $\mathbb{E}\exp(\lambda \langle v, X \rangle) \leq \exp(\frac{\lambda^2}{2}v^\top \Sigma v)$ for all $\lambda \geq 0$, $v \in S^{d-1}$ where Σ is the covariance matrix of X .

Then, with probability $1 - \delta$.

$$\|X\| \leq \sqrt{\text{Tr}\Sigma} + \sqrt{2\lambda_{\max}(\Sigma) \log(\frac{1}{\delta})}$$

Proof. **Preliminaries**

- Reason for $\sqrt{\text{Tr}\Sigma}$. Note that

$$\begin{aligned}\mathbb{E}\|X\|_2 &\leq \sqrt{\mathbb{E}\|X\|_2^2} \\ &= \sqrt{\mathbb{E}X^\top X} \\ &= \sqrt{\mathbb{E}\text{Tr}(X^\top X)} \\ &= \sqrt{\text{Tr}(\mathbb{E}XX^\top)} \\ &= \sqrt{\text{Tr}(\Sigma)}\end{aligned}$$

Note that $X^\top X = \text{Tr}(X^\top X) = \text{Tr}(XX^\top) = \text{Tr}(\Sigma)$.

- KL-divergence: Suppose $\Theta \subseteq \mathbb{R}^d$ and ρ and π are densities on Θ . Then

$$\text{KL}(\rho\|\pi) = \mathbb{E}_{\theta \sim \rho} \log \frac{\rho(\theta)}{\pi(\theta)} = \int_{\Theta} \rho(\theta) \log \frac{\rho(\theta)}{\pi(\theta)} d\theta$$

We can show that $\text{KL}(\rho\|\pi) \geq 0$.

$$\begin{aligned} -\text{KL}(\rho\|\pi) &= \mathbb{E}_{\theta \sim \rho} \log(\pi(\theta)/\rho(\theta)) \\ (\text{Jensen's}) &\leq \log \mathbb{E}_{\theta \sim \rho} (\pi(\theta)/\rho(\theta)) \\ &\leq \log 1 = 0 \end{aligned}$$

Additionally, $\pi = \rho$ if and only if $\text{KL}(\rho\|\pi) = 0$.

- Variational identity: Change of measure / density./ If ρ, π are two densities on Θ and for $h(\theta)$ we have $\mathbb{E}_{\theta \sim \pi} h(\theta) < \infty$ then

$$\log \mathbb{E}_{\theta \sim \pi} \exp(h(\theta)) = \sup_{\rho: \text{KL}(\rho\|\pi) < \infty} (\mathbb{E}_{\theta \sim \pi} h(\theta) - \text{KL}(\rho\|\pi))$$

Proof. of claim: Let $\pi'(\theta) = \frac{\pi(\theta) \exp(h(\theta))}{\mathbb{E}_{\theta \sim \pi} \exp(h(\theta))}$. Then π' is a valid density (denominator was to ensure integration to 1). Now fix ρ .

$$\begin{aligned} 0 \leq \text{KL}(\rho\|\pi') &= \mathbb{E}_{\theta \sim \rho} \log \left(\frac{\rho \mathbb{E}_{\theta \sim \pi} \exp(h)}{\pi \exp(h)} \right) \\ &= \text{KL}(\rho\|\pi) + \log(\mathbb{E}_{\theta \sim \rho} \exp(h)) - \mathbb{E}_{\theta \sim \rho} h(\theta) \quad (*) \end{aligned}$$

Thus

$$\log \mathbb{E}_{\theta \sim \pi} \exp(h(\theta)) \geq \sup_{\rho: \text{KL}(\rho\|\pi) < \infty} (\mathbb{E}_{\theta \sim \pi} h(\theta) - \text{KL}(\rho\|\pi))$$

When $\rho = \pi$ the $\geq$ becomes $=$ so we have

$$\log \mathbb{E}_{\theta \sim \pi} \exp(h(\theta)) = \sup_{\rho: \text{KL}(\rho\|\pi) < \infty} (\mathbb{E}_{\theta \sim \pi} h(\theta) - \text{KL}(\rho\|\pi))$$

Intuition (change of measure / exponential tilting). The quantity

$$\log \mathbb{E}_{\theta \sim \pi} \exp(h(\theta))$$

can be viewed as a “soft maximum” of h under the reference density π : large values of $h(\theta)$ contribute exponentially more to the expectation. The variational identity says this soft maximum can be written as an optimization over alternative densities ρ :

$$\log \mathbb{E}_{\theta \sim \pi} \exp(h) = \sup_{\rho} \left(\mathbb{E}_{\theta \sim \rho} h(\theta) - \text{KL}(\rho \| \pi) \right).$$

Here $\mathbb{E}_{\rho} h$ rewards placing mass where h is large, while $\text{KL}(\rho \| \pi)$ penalizes moving too far from the baseline π . So the supremum balances “fit to h ” versus “complexity relative to π .”

What is π' ? The density

$$\pi'(\theta) = \frac{\pi(\theta) \exp(h(\theta))}{\mathbb{E}_{\theta \sim \pi} \exp(h(\theta))}$$

is the *exponentially tilted* (or *Gibbs*) version of π : it reweights π by the factor $\exp(h(\theta))$, then renormalizes. Equivalently, π' is the *optimizer* of the variational problem: among all ρ , it maximizes $\mathbb{E}_{\rho} h - \text{KL}(\rho \| \pi)$. In this sense, π' is the “best” change of measure that concentrates on large- h regions while paying the smallest KL cost from π .

- Tail bounds via variational inequality. Assume X is a random vector and $\Theta \subseteq \mathbb{R}^d$; and there is a function

$$\mathbb{E} \exp(f(X, \theta)) < \infty$$

for all $\theta \in \Theta$, then if π is a fixed density of Θ (think of as prior) with probability $1 - \delta$, for any ρ with $\text{KL}(\rho \| \pi) < \infty$ we have

$$\mathbb{E}_{\theta \sim \rho} \exp(f(X, \theta)) \leq \mathbb{E}_{\theta \sim \rho} \log(\mathbb{E}_X \exp(f(X, \theta))) + \text{KL}(\rho \| \pi) + \log(1/\delta)$$

Proof.

Recall the change of measure formula we have above involving h , ρ and π . We will use that in the following proof.

We will define (think of X as a parameter)

$$\begin{aligned} h(\theta, X) &= f(x, \theta) - \log \mathbb{E} \exp(f(X, \theta)) \\ &= \end{aligned}$$

Note that

$$\begin{aligned}\mathbb{E}_X \exp(h(\theta, X)) &= \mathbb{E}_X \exp(f(x, \theta) - \log \mathbb{E} \exp(f(X, \theta))) \\ &= \frac{\mathbb{E}_X \exp(f(\theta, X))}{\mathbb{E}_X \exp(f((X, \theta))} = 1\end{aligned}$$

Then

$$\begin{aligned}& \exp(\sup_{\rho} (\mathbb{E}_{\theta \sim \rho} f(X, \theta) - \mathbb{E}_{\theta \sim \rho} \mathbb{E}_X (f(X, \theta) - \text{KL}(\rho \parallel \theta)))) \\ (\text{change of measure}) \quad &= \mathbb{E}_X \mathbb{E}_{\theta \sim \pi} \exp(h(\theta)) \\ &= \mathbb{E}_{\theta \sim \pi} \mathbb{E}_X \exp(h(\theta)) \\ &= \mathbb{E}_{\theta \sim \pi} 1\end{aligned}$$

If $\mathbb{E} \exp(Y) = 1$ then by Chernoff's method we have with probability $1 - \delta$,

$$Y \leq \log(1/\delta)$$

I think it would be good to go through this and clean it up a bit to absorb it more.

Towards the proof of theorem 1:

$$\|X\| = \sup_{v \in S^{d-1}} \langle X, v \rangle$$

Let $\beta > 0$ be a parameter and let $\pi(\theta) = \mathcal{N}(0, I_d/\beta)$ and let $\rho_v = \mathcal{N}(v, I_d/\beta)$ indexed by $v \in S^{d-1}$. Let $f(X, \theta) = \langle X, \theta \rangle$ where $\theta \sim \rho_v$. Then

$$\mathbb{E}_{\theta \sim \rho_v} f(X, \theta) = \mathbb{E}_{\theta \sim \rho_v} \langle X, \theta \rangle = \langle X, v \rangle$$

Then

$$\sup_{\rho_v} \mathbb{E}_{\theta \sim \rho_v} \langle X, \theta \rangle = \sup_{\rho_v} \langle X, v \rangle = \|X\|$$

Next day continuation: : Define $f : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ as

$$f(x, \theta) = \alpha \langle x, \theta \rangle$$

where α is some fixed parameter. Let $\pi = \mathcal{N}(0, \frac{1}{\beta} I_d)$, $\beta > 0$ is fixed. Then $\rho_v = \mathcal{N}(v, \frac{1}{\beta} I_d)$ where $v \in S^{d-1}$ is fixed. Then by the lemma above

$$\mathbb{E}_{\theta \sim \rho_v} \alpha \langle X, \theta \rangle = \alpha \langle X, \mathbb{E}_{\theta \sim \rho_v} \theta \rangle = \alpha \langle X, v \rangle$$

Taking the supremum we have

$$\sup_{v \in S^{d-1}} \mathbb{E}_{\theta \sim \rho_v} \alpha \langle X, \theta \rangle = \alpha \sup_{v \in S^{d-1}} \langle X, v \rangle = \alpha \|X\|$$

$$(**) \quad \text{KL}(\rho \parallel \pi) = \frac{\beta \|v\|^2}{2} = \beta/2$$

The next part is the exponential moment:

$$\begin{aligned} \mathbb{E}_X \exp(\alpha \langle X, \theta \rangle) &= \mathbb{E}_X \exp(\alpha \|\theta\| \langle X, \theta / \|\theta\| \rangle) \\ &\leq \exp(\alpha^2/2 \|\theta\|^2 (\theta^\top / \|\theta\|) \Sigma (\theta / \|\theta\|)) \\ &= \exp(\alpha^2/2 \theta^\top \Sigma \theta) \quad (***) \end{aligned}$$

That was by the hypothesis on X . Then

$$\begin{aligned} (***) &\leq \mathbb{E}_{\theta \sim \rho_v} \alpha^2/2 \theta^\top \Sigma \theta \\ &= \alpha^2/2 \mathbb{E}_{\theta \sim \rho_v} (\theta - v + v)^\top \Sigma (\theta - v + v) \\ &= \alpha^2/2 \mathbb{E}_{\theta \sim \rho_v} v^\top \Sigma v + \alpha^2/2 \mathbb{E}_{\theta \sim \rho_v} (\theta - v)^\top \Sigma (\theta - v) \\ &= \alpha^2/2 \mathbb{E}_{\theta \sim \rho_v} v^\top \Sigma v + \alpha^2/2 \mathbb{E}_{\theta \sim \rho_v} \text{Tr}((\theta - v)^\top \Sigma (\theta - v)) \\ &= \alpha^2/2 v^\top \Sigma v + \alpha^2/2 \text{Tr}(\Sigma \mathbb{E}_{\theta \sim \rho_v} (\theta - v)(\theta - v)^\top) \\ &\leq \alpha^2/2 \lambda_{\max}(\Sigma) + \alpha^2/2 \text{Tr}(\Sigma)/\beta \end{aligned}$$

Plugging everything in we have

$$\alpha \|X\| \leq \alpha^2/2 (\lambda_{\max}(\Sigma) + \frac{\text{Tr}(\Sigma)}{\beta}) + \beta/2 + \log(1/\delta)$$

3.2 Gaussian Concentration Inequality

Theorem 3.9. If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -Lipschitz (i.e. for all $x, y \in \mathbb{R}^d \implies |f(x) - f(y)| \leq L \|x - y\|_2$) and $X \sim \mathcal{N}(0, I_d)$ then

$$\mathbb{P}(f(X) - \mathbb{E}f(x) \geq t) \leq \exp(-\frac{t^2}{2L^2})$$

and

$$\mathbb{P}(|f(X) - \mathbb{E}f(x)| \geq t) \leq 2 \exp(-\frac{t^2}{2L^2})$$

Proof. (With sub-optimal constants.) First assume f is differentiable; the general Lipschitz case follows by standard smoothing/approximation.

Claim. Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be convex and let $X, Y \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_d)$. Then

$$\mathbb{E} \phi(f(X) - \mathbb{E}f(X)) \leq \mathbb{E} \phi\left(\frac{\pi}{2} \langle \nabla f(X), Y \rangle\right).$$

Proof of Claim. Fix $\theta \in [0, \pi/2]$ and define

$$w(\theta) = X \sin \theta + Y \cos \theta, \quad w'(\theta) = X \cos \theta - Y \sin \theta.$$

Then $(w(\theta), w'(\theta))$ is jointly Gaussian (as a linear map of (X, Y)). Moreover,

$$\text{Cov}(w(\theta)) = \sin^2 \theta I_d + \cos^2 \theta I_d = I_d, \quad \text{Cov}(w'(\theta)) = \cos^2 \theta I_d + \sin^2 \theta I_d = I_d,$$

and the cross-covariance is

$$\text{Cov}(w(\theta), w'(\theta)) = \sin \theta \cos \theta I_d - \cos \theta \sin \theta I_d = 0.$$

Hence $w(\theta) \sim \mathcal{N}(0, I_d)$, $w'(\theta) \sim \mathcal{N}(0, I_d)$, and since they are jointly Gaussian and uncorrelated, they are independent. In particular,

$$(w(\theta), w'(\theta)) \stackrel{d}{=} (X, Y).$$

By identical distribution, $\mathbb{E}f(X) = \mathbb{E}f(Y)$. Condition on X and apply Jensen in Y :

$$\begin{aligned} \mathbb{E} \phi(f(X) - \mathbb{E}f(X)) &= \mathbb{E} \phi(f(X) - \mathbb{E}f(Y)) \\ &= \mathbb{E} \left[\phi(\mathbb{E}_Y[f(X) - f(Y) \mid X]) \right] \\ &\leq \mathbb{E} \left[\mathbb{E}_Y[\phi(f(X) - f(Y)) \mid X] \right] \\ &= \mathbb{E} \phi(f(X) - f(Y)). \end{aligned}$$

Now use the fundamental theorem of calculus along the path $w(\theta)$:

$$f(X) - f(Y) = f(w(\pi/2)) - f(w(0)) = \int_0^{\pi/2} \frac{d}{d\theta} f(w(\theta)) d\theta = \int_0^{\pi/2} \langle \nabla f(w(\theta)), w'(\theta) \rangle d\theta.$$

Let $\Theta \sim \text{Unif}(0, \pi/2)$ independent of (X, Y) . Then

$$\int_0^{\pi/2} \langle \nabla f(w(\theta)), w'(\theta) \rangle d\theta = \frac{\pi}{2} \mathbb{E}_\Theta [\langle \nabla f(w(\Theta)), w'(\Theta) \rangle \mid X, Y].$$

Apply Jensen in Θ (conditioning on X, Y):

$$\begin{aligned} \phi \left(\int_0^{\pi/2} \langle \nabla f(w(\theta)), w'(\theta) \rangle d\theta \right) &= \phi \left(\frac{\pi}{2} \mathbb{E}_\Theta [\langle \nabla f(w(\Theta)), w'(\Theta) \rangle \mid X, Y] \right) \\ &\leq \mathbb{E}_\Theta \left[\phi \left(\frac{\pi}{2} \langle \nabla f(w(\Theta)), w'(\Theta) \rangle \right) \mid X, Y \right]. \end{aligned}$$

Taking expectations yields

$$\mathbb{E} \phi(f(X) - f(Y)) \leq \mathbb{E}_\Theta \mathbb{E}_{X,Y} \phi \left(\frac{\pi}{2} \langle \nabla f(w(\Theta)), w'(\Theta) \rangle \right).$$

Finally, for each fixed θ , we have $(w(\theta), w'(\theta)) \stackrel{d}{=} (X, Y)$, so

$$\mathbb{E}_{X,Y} \phi\left(\frac{\pi}{2} \langle \nabla f(w(\theta)), w'(\theta) \rangle\right) = \mathbb{E}_{X,Y} \phi\left(\frac{\pi}{2} \langle \nabla f(X), Y \rangle\right),$$

and therefore the Θ -average is the same. Combining everything gives

$$\mathbb{E} \phi(f(X) - \mathbb{E}f(X)) \leq \mathbb{E} \phi\left(\frac{\pi}{2} \langle \nabla f(X), Y \rangle\right),$$

as claimed.

Now choose $\phi(u) = e^{\lambda u}$ with $\lambda > 0$. The claim gives

$$\begin{aligned} \mathbb{E} \exp(\lambda(f(X) - \mathbb{E}f(X))) &\leq \mathbb{E} \exp\left(\lambda \frac{\pi}{2} \langle \nabla f(X), Y \rangle\right) \\ &= \mathbb{E}_X \left[\mathbb{E}_Y \exp\left(\lambda \frac{\pi}{2} \langle \nabla f(X), Y \rangle\right) \right]. \end{aligned}$$

Condition on X . Since $Y \sim \mathcal{N}(0, I_d)$, for any fixed vector $a \in \mathbb{R}^d$, $\langle a, Y \rangle \sim \mathcal{N}(0, \|a\|_2^2)$, hence

$$\mathbb{E}_Y e^{\langle u, Y \rangle} = \exp\left(\frac{\|u\|_2^2}{2}\right).$$

Apply this with $u = \lambda \frac{\pi}{2} \nabla f(X)$ to get

$$\begin{aligned} \mathbb{E}_Y \exp\left(\lambda \frac{\pi}{2} \langle \nabla f(X), Y \rangle\right) &= \exp\left(\frac{1}{2} \left\| \lambda \frac{\pi}{2} \nabla f(X) \right\|_2^2\right) \\ &= \exp\left(\frac{\lambda^2 \pi^2}{8} \|\nabla f(X)\|_2^2\right). \end{aligned}$$

Therefore

$$\mathbb{E} \exp(\lambda(f(X) - \mathbb{E}f(X))) \leq \mathbb{E}_X \exp\left(\frac{\lambda^2 \pi^2}{8} \|\nabla f(X)\|_2^2\right).$$

This is the missing last step: bound $\|\nabla f(X)\|_2$ by L . If f is L -Lipschitz and differentiable, then $\|\nabla f(x)\|_2 \leq L$ for all x (e.g. by the directional derivative bound: $|\langle \nabla f(x), v \rangle| = \lim_{t \rightarrow 0} \frac{|f(x+tv) - f(x)|}{t} \leq L\|v\|_2$ for all v , hence $\|\nabla f(x)\|_2 \leq L$). Thus $\|\nabla f(X)\|_2^2 \leq L^2$ almost surely, and

$$\mathbb{E} \exp(\lambda(f(X) - \mathbb{E}f(X))) \leq \exp\left(\frac{\lambda^2 \pi^2}{8} L^2\right).$$

Now apply Chernoff's bound: for any $\lambda > 0$,

$$\mathbb{P}(f(X) - \mathbb{E}f(X) \geq t) \leq \exp\left(-\lambda t + \frac{\lambda^2 \pi^2}{8} L^2\right).$$

Minimize the RHS over λ ; the optimizer is $\lambda^* = \frac{4t}{\pi^2 L^2}$, giving

$$\mathbb{P}(f(X) - \mathbb{E}f(X) \geq t) \leq \exp\left(-\frac{2}{\pi^2} \frac{t^2}{L^2}\right).$$

The same bound holds for $\mathbb{P}(f(X) - \mathbb{E}f(X) \leq -t)$ by applying the argument to $-f$, and the two-sided inequality follows by a union bound:

$$\mathbb{P}(|f(X) - \mathbb{E}f(X)| \geq t) \leq 2 \exp\left(-\frac{2}{\pi^2} \frac{t^2}{L^2}\right).$$

This proves Gaussian concentration with a sub-optimal constant. ■

Example 3.10. Suppose $X \sim \mathcal{N}(0, \Sigma)$ where Σ is invertible. $f(X) = \|X\|_2$. If $Z \sim \mathcal{N}(0, I_d)$ then $X = \Sigma^{1/2}Z$. Then

$$f(X) = \|\Sigma^{1/2}Z\|_2$$

We try to find the Lipschitz constant

$$\begin{aligned} |f(x) - f(y)| &= | \|\Sigma^{1/2}x\|_2 - \|\Sigma^{1/2}y\|_2 | \\ &\leq \|\Sigma^{1/2}(x - y)\| \\ &\leq \lambda_{\max}(\Sigma^{1/2}) \|x - y\|_2 \\ &= \sqrt{\lambda_{\max}(\Sigma)} \|x - y\|_2 \end{aligned}$$

Thus

$$\mathbb{P}(|\|X\| - \mathbb{E}\|X\|| \geq t) \leq 2 \exp\left(-\frac{t^2}{2\lambda_{\max}(\Sigma)}\right)$$

This implies with probability $1 - \delta$

$$|\|X\| - \mathbb{E}\|X\|| \leq \sqrt{2\lambda_{\max}(\Sigma) \log(2/\delta)}$$

Thus

$$\|X\| \leq \sqrt{\text{Tr}(\Sigma)} + \sqrt{2\lambda_{\max}(\Sigma) \log(2/\delta)}$$

When we had the sub-gaussian concentration, we had a one sided bound. This case gives us a two sided bound since we are actually working with Gaussians. ■

4 Matrices

Definition 4.1. If $A \in \mathbb{R}^{m \times n}$, its singular value decomposition (SVD) can be written as

$$A = \sum_{i=1}^r \sigma_i u_i v_i^\top, \quad r = \text{rank}(A),$$

where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ are the singular values, $u_i \in \mathbb{R}^m$ are the left singular vectors, and $v_i \in \mathbb{R}^n$ are the right singular vectors. The vectors $\{u_i\}_{i=1}^r$ are orthonormal eigenvectors of $AA^\top$, and $\{v_i\}_{i=1}^r$ are orthonormal eigenvectors of $A^\top A$, with

$$AA^\top u_i = \sigma_i^2 u_i, \quad A^\top A v_i = \sigma_i^2 v_i.$$

Equivalently, $\sigma_i = \sqrt{\lambda_i}$ where λ_i is an eigenvalue of $A^\top A$ (or $AA^\top$).

If A is square and invertible with $A \in \mathbb{R}^{m \times m}$ then

$$A^{-1} = \sum_{i=1}^m \frac{1}{\sigma_i} v_i u_i^\top$$

Definition 4.2 (Operator norm). Let $A \in \mathbb{R}^{n \times n}$. The (spectral/operator) norm is

$$\|A\|_{\text{op}} := \sup_{v \in S^{n-1}} \|Av\|_2.$$

Equivalently,

$$\|A\|_{\text{op}} = \sup_{u, v \in S^{n-1}} u^\top A v.$$

If A is symmetric, then

$$\|A\|_{\text{op}} = \sup_{v \in S^{n-1}} |v^\top A v|.$$

Finally, $\|A\|_{\text{op}}$ equals the largest singular value of A :

$$\|A\|_{\text{op}} = \sigma_1, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0.$$

Definition 4.3 (ε -cover and covering number). Let $(\mathcal{X}, \rho)$ be a metric space and let $S \subseteq \mathcal{X}$. A set $T \subseteq \mathcal{X}$ is called an ε -cover (or ε -net) of S with respect to ρ if for every $x \in S$ there exists $t \in T$ such that

$$\rho(x, t) \leq \varepsilon.$$

Equivalently, $S \subseteq \bigcup_{t \in T} B_\rho(t, \varepsilon)$, where $B_\rho(t, \varepsilon) := \{x \in \mathcal{X} : \rho(x, t) \leq \varepsilon\}$.

The ε -covering number of S (with respect to ρ) is

$$\mathcal{N}(S, \rho, \varepsilon) := \min\{|T| : T \text{ is an } \varepsilon\text{-cover of } S \text{ w.r.t. } \rho\},$$

with the convention $\mathcal{N}(S, \rho, \varepsilon) = \infty$ if no finite ε -cover exists.

Definition 4.4 (Packing number). The packing number $\mathcal{P}(\mathcal{S}, \rho, \epsilon)$ is the largest number of points in S such that for any pair $\theta_i, \theta_j \in \mathcal{P}$, $\rho(\theta_i, \theta_j) > \epsilon$. Here $\mathcal{P}$ is a packing.

Lemma 4.5. For any set S and $\epsilon > 0$, we have

$$\mathcal{P}(S, \rho, 2\epsilon) \leq \mathcal{N}(S, \rho, \epsilon) \leq \mathcal{P}(S, \rho, \epsilon)$$

Proof. For the second inequality, any maximal packing (can't add another point) is a covering.

Take a max ϵ -packing and assume for the sake of contradiction it is not an ϵ -net. Then there is some $\theta \in \mathcal{S}$ over ϵ away from every element of the packing. Thus we can add another point to the “max” packing which is a contradiction. Thus the max packing must be an ϵ -net.

Thus the minimal covering number $\mathcal{N}(S, \rho, \epsilon)$ is bounded above by the packing number $\mathcal{P}(S, \rho, \epsilon)$.

For the first inequality, start with an ϵ -net $\mathcal{N}$.

Then take a 2ϵ packing $\mathcal{P} \subseteq S$. No two distinct packing points can lie in the same ϵ . If $p_1, p_2 \in \mathcal{P}$ did lie in the same ϵ -ball centered at v then

$$\rho(p_1, p_2) \leq \rho(p_1, v) + \rho(v, p_2) \leq 2\epsilon$$

contradicting the fact that points in the packing must be at least 2ϵ apart. Since the ϵ -net balls cover S , every packing point is in an ϵ ball centered at a point in the net. But we just showed they are each in a distinct ball. Thus $\mathcal{P}(S, \rho, 2\epsilon) \leq \mathcal{N}(S, \rho, \epsilon)$. ■

Example 4.6 (Covering number of Euclidean Ball). We notate

$$\mathcal{N}(B_2^d, \epsilon) = \mathcal{N}(B_2^d, \|\cdot\|_2, \epsilon)$$

For any $\epsilon \in (0, 1)$

$$\epsilon^{-d} \leq \mathcal{N}(B_2^d, \epsilon) \leq (1 + \frac{2}{\epsilon})^d$$

Proof. Volume comparison argument.

- First inequality, find the number of ϵ balls you can fit inside the ball of radius ϵ . (Number of ϵ -balls) $\cdot \text{Vol}(B_2^d(0, \epsilon)) \geq \text{Vol}(B_2^d(0, 1))$.
- Second inequality, take a maximal ϵ packing P . Then by the triangle inequality, the $\epsilon/2$ -balls centered at a $p \in P$ will be disjoint. Moreover, each one is contained in the $(1 + \epsilon/2)$ unit ball. Thus $|P| \cdot \text{Vol}(B_2^d(0, \epsilon/2)) \leq \text{Vol}(B_2^d(0, 1 + \epsilon/2))$. Then $|P| \leq (1 + 2/\epsilon)^d$. Since a ϵ -maximal packing is an ϵ -net, $\mathcal{N}(B_2^d, \epsilon) \leq (1 + 2/\epsilon)^d$.

■

Theorem 4.7. Suppose $X \in \mathbb{R}^{m \times n}$ is a random matrix with independent zero mean elements X_{ij} where the max gaussian norm of all elements is bounded by K . Then with probability $1 - \delta$

$$\|X\|_{\text{op}} \leq C \cdot K \cdot (\sqrt{n} + \sqrt{m} + \sqrt{\log(1/\delta)})$$

Proof. Recall

$$\|X\|_{\text{op}} = \sup_{u \in S^{m-1}, v \in S^{n-1}} u^\top X v.$$

Step 1: epsilon-net reduction. Fix $\varepsilon \in (0, 1/2)$. Let $\mathcal{M}$ be an ε -net of S^{m-1} and $\mathcal{N}$ an ε -net of S^{n-1} . A standard net argument gives

$$\|X\|_{\text{op}} \leq \frac{1}{1 - 2\varepsilon} \max_{u \in \mathcal{M}, v \in \mathcal{N}} u^\top X v.$$

(Proof sketch: for any u, v choose $u_0 \in \mathcal{M}, v_0 \in \mathcal{N}$ with $\|u - u_0\|_2 \leq \varepsilon, \|v - v_0\|_2 \leq \varepsilon$, and expand $u^\top X v - u_0^\top X v_0$; then use $\|Xv\|_2 \leq \|X\|_{\text{op}}$.)

Moreover, there exist ε -nets with cardinalities

$$|\mathcal{M}| \leq \left(1 + \frac{2}{\varepsilon}\right)^m, \quad |\mathcal{N}| \leq \left(1 + \frac{2}{\varepsilon}\right)^n.$$

Step 2: a fixed bilinear form is sub-Gaussian. Fix $u \in S^{m-1}$ and $v \in S^{n-1}$ and define

$$Z := u^\top X v = \sum_{i=1}^m \sum_{j=1}^n u_i v_j X_{ij}.$$

This is a sum of independent mean-zero sub-Gaussian random variables. A standard closure property of sub-Gaussian norms implies there is an absolute constant $C_0 > 0$ such that

$$\|Z\|_{\psi_2} \leq C_0 \left(\sum_{i,j} u_i^2 v_j^2 \|X_{ij}\|_{\psi_2}^2 \right)^{1/2} \leq C_0 K \left(\sum_i u_i^2 \right)^{1/2} \left(\sum_j v_j^2 \right)^{1/2} = C_0 K.$$

Hence there is an absolute constant $c_0 > 0$ such that for all $t \geq 0$,

$$\mathbb{P}(|u^\top X v| \geq t) \leq 2 \exp\left(-c_0 \frac{t^2}{K^2}\right).$$

Step 3: union bound over the nets. By the union bound,

$$\begin{aligned} \mathbb{P}\left(\max_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top X v| \geq t\right) &\leq |\mathcal{M}| |\mathcal{N}| \cdot 2 \exp\left(-c_0 \frac{t^2}{K^2}\right) \\ &\leq 2 \exp\left((m+n) \log\left(1 + \frac{2}{\varepsilon}\right) - c_0 \frac{t^2}{K^2}\right). \end{aligned}$$

Choose $\varepsilon = 1/4$, so $\log(1 + 2/\varepsilon) = \log 9$. Now choose

$$t = \frac{K}{\sqrt{c_0}} \sqrt{(m+n) \log 9 + \log(2/\delta)}.$$

Then the previous display is at most δ .

Step 4: conclude for the operator norm. On this event (probability at least $1 - \delta$),

$$\|X\|_{\text{op}} \leq \frac{1}{1 - 2\varepsilon} \max_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top X v| \leq 2t = \frac{2K}{\sqrt{c_0}} \sqrt{(m+n) \log 9 + \log(2/\delta)}.$$

Finally, use $\sqrt{m+n} \leq \sqrt{m} + \sqrt{n}$ and absorb the constants (including $\log 9$ and $\log 2$) into a single absolute constant C to obtain

$$\|X\|_{\text{op}} \leq CK \left(\sqrt{m} + \sqrt{n} + \sqrt{\log(1/\delta)} \right).$$

■

Remark 4.8 (reminder: epsilon-net argument and proper vs. improper covers). •

Epsilon-net control of the Euclidean norm. For $x \in \mathbb{R}^d$,

$$\|x\|_2 = \sup_{v \in S^{d-1}} \langle x, v \rangle.$$

Let $N \subset S^{d-1}$ be an ε -net in Euclidean distance, i.e. for every $v \in S^{d-1}$ there exists $v_N \in N$ with $\|v - v_N\|_2 \leq \varepsilon$. Then for any $x \in \mathbb{R}^d$ and any $v \in S^{d-1}$,

$$\langle x, v \rangle = \langle x, v_N \rangle + \langle x, v - v_N \rangle \leq \langle x, v_N \rangle + \|x\|_2 \|v - v_N\|_2 \leq \langle x, v_N \rangle + \varepsilon \|x\|_2.$$

Taking the supremum over $v \in S^{d-1}$ gives

$$\|x\|_2 \leq \sup_{w \in N} \langle x, w \rangle + \varepsilon \|x\|_2,$$

hence (for $\varepsilon < 1$)

$$\|x\|_2 \leq \frac{1}{1 - \varepsilon} \sup_{w \in N} \langle x, w \rangle.$$

- **Proper vs. improper covers.** Let (Y, ρ) be a metric space and let $N_{\text{imp}}(Y, \rho, \varepsilon)$ denote the minimum number of radius- ε balls whose union contains Y (centers may be anywhere), and let $N_{\text{prop}}(Y, \rho, \varepsilon)$ denote the minimum number of radius- ε balls whose centers are required to lie in Y .

Clearly,

$$N_{\text{imp}}(Y, \rho, \varepsilon) \leq N_{\text{prop}}(Y, \rho, \varepsilon).$$

Conversely, any improper $\varepsilon/2$ -cover can be converted into a proper ε -cover: if $\{B(x_k, \varepsilon/2)\}_{k=1}^N$ covers Y (with possibly $x_k \notin Y$), then for each k with $Y \cap B(x_k, \varepsilon/2) \neq \emptyset$ choose a point $y_k \in Y \cap B(x_k, \varepsilon/2)$. For any $z \in B(x_k, \varepsilon/2)$,

$$\rho(z, y_k) \leq \rho(z, x_k) + \rho(x_k, y_k) \leq \varepsilon/2 + \varepsilon/2 = \varepsilon,$$

so $B(x_k, \varepsilon/2) \subseteq B(y_k, \varepsilon)$. Therefore,

$$N_{\text{prop}}(Y, \rho, \varepsilon) \leq N_{\text{imp}}(Y, \rho, \varepsilon/2).$$

Example 4.9. The following is an example

1. (IID Normal / Wigner Type) If $X_{ij} \stackrel{\text{iid}}{\sim} N(0, 1)$ and $X \in \mathbb{R}^{n \times n}$ then

$$\|X\| \leq C(\sqrt{n} + \sqrt{\log(1/\delta)})$$

with probability $1 - \delta$.

2. IID symmetric. We define the elements on and above the diagonal. X_{ij} are independent when $i \leq j$ and $X = X^\top$. On the off diagonal we have $X_{ij} \stackrel{\text{iid}}{\sim} N(0, 1)$ and $X_{ii} \stackrel{\text{iid}}{\sim} N(0, 2)$. Define $Y_{ij} = X_{ij}$ for $i < j$. $Y_{ii} = X_{ii}/2$ and $Y_{ij} = 0$ for $j < i$. Then $X = Y + Y^\top$. Then

$$\|X\|_{\text{op}} = \|Y + Y^\top\|_{\text{op}} \leq \|Y\|_{\text{op}} + \|Y^\top\|_{\text{op}} = 2\|Y\|_{\text{op}}$$

and then we can apply the last result to $\|Y\|_{\text{op}}$.

Proposition 4.10 (Concentration for the covariance matrix). If $X_1, \dots, X_n$ are independent vectors in $\mathbb{R}^d$; $\mathbb{E}X = 0$ then define the covariance matrix $\Sigma = \mathbb{E}X_i X_i^\top$.

We can estimate the covariance matrix via

$$\hat{\Sigma}_n = \frac{1}{n} \sum X_i X_i^\top$$

We are interested in bounding

$$\left\| \hat{\Sigma}_n - \Sigma \right\|_{\text{op}}$$

Note that the elements of this matrix are not independent so the previous theorem will not work :)

Definition 4.11. The effective rank of Σ is

$$r(\Sigma) = \frac{\text{Tr}(\Sigma)}{\|\Sigma\|_{\text{op}}}$$

If $\Sigma = I_d$ then $r(I_d) = d$. In general $r(\Sigma) \leq d$. Think of a PCA example where most of the variance of the matrix describes is in a small subspace of $\mathbb{R}^d$.

Theorem 4.12. If $X_1, \dots, X_n$ are independent zero mean, with

$$\mathbb{E}[\exp(\lambda \langle v, X \rangle)] \leq \exp(\frac{\lambda^2}{2} v^\top \Sigma v)$$

then

$$\left\| \hat{\Sigma}_n - \Sigma \right\|_{\text{op}} \leq C \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{r(\Sigma)}{n}} + \sqrt{\frac{\|\Sigma\|_{\text{op}} \log(1/\delta)}{n}} \right)$$

whenever $n \geq C(r(\Sigma) + \log(1/\delta))$.

If we drop the $n \geq C(r(\Sigma) + \log(1/\delta))$ requirement then

$$\left\| \hat{\Sigma}_n - \Sigma \right\|_{\text{op}} \leq C \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{r(\Sigma)}{n}} + \sqrt{\frac{\|\Sigma\|_{\text{op}} \log(1/\delta)}{n}} + \frac{r(\Sigma)}{n} + \frac{\log(1/\delta)}{n} \right)$$

which has the sub-gaussian and sub-exponential tails.

4.1 Matrix Concentration Inequalities

Let $A \in \mathbb{R}^{d \times d}$ be symmetric with spectral decomposition

$$A = \sum_{i=1}^d \lambda_i u_i u_i^\top,$$

where $\lambda_i \in \mathbb{R}$ and $\{u_i\}$ is an orthonormal basis. For a function $f : \mathbb{R} \rightarrow \mathbb{R}$ define the matrix functional calculus

$$f(A) := \sum_{i=1}^d f(\lambda_i) u_i u_i^\top.$$

1. **Transfer principle (functional calculus + Loewner order).** If $A \succeq 0$ then $\lambda_i \geq 0$ for all i . Many scalar inequalities for f can be transferred to matrix inequalities.

A basic example: if f is *operator monotone* on an interval $I \subset \mathbb{R}$, and A, B are symmetric with spectra contained in I , then

$$A \preceq B \implies f(A) \preceq f(B).$$

(Here $A \preceq B$ means $B - A \succeq 0$.)

In class when he wrote this down again he wrote

For functions $f \leq g$ for all $|x| \leq K$ and X symmetric then $f(X) \preceq g(X)$ for $\|X\|_{\text{op}} \leq K$.

2. **Log is order-preserving on positive definite matrices.** If $X \succeq Y \succ 0$, then $\log(X) \succeq \log(Y)$ (the matrix logarithm is defined via functional calculus on $(0, \infty)$).

3. **Exponentials do not generally factor.** In general,

$$\exp(A + B) \neq \exp(A) \exp(B),$$

unless A and B commute (i.e. $AB = BA$).

4. **Golden–Thompson inequality.** For symmetric A, B ,

$$\mathrm{tr}(\exp(A + B)) \leq \mathrm{tr}(\exp(A) \exp(B)).$$

5. **Lieb’s concavity theorem (a key matrix Jensen tool).** Fix a symmetric matrix $H \in \mathbb{R}^{d \times d}$ and define, for $A \succ 0$,

$$\psi(A) := \mathrm{tr}(\exp(H + \log A)).$$

Then ψ is concave on the positive definite A-S: for $A, B \succ 0$ and $\alpha \in [0, 1]$,

$$\psi(\alpha A + (1 - \alpha)B) \geq \alpha \psi(A) + (1 - \alpha) \psi(B).$$

In the scalar case ($d = 1$), if $H = h \in \mathbb{R}$ and $A = a > 0$, then

$$\psi(a) = \exp(h + \log a) = e^h a,$$

so concavity is trivial.

A standard application (matrix Jensen): if Z is a random symmetric matrix, then with $A := \exp(Z) \succ 0$,

$$\mathbb{E} \mathrm{tr}(\exp(H + Z)) = \mathbb{E} \mathrm{tr}(\exp(H + \log A)) \leq \mathrm{tr}(\exp(H + \log(\mathbb{E}A))) = \mathrm{tr}(\exp(H + \log(\mathbb{E}e^Z))).$$

This is one of the main steps that lets us mimic scalar MGF/Chernoff arguments in the matrix setting.

Remark 4.13 (Some matrix properties). • If $0 \prec X \preceq Y$ then $\log X \preceq \log Y$. The same does not have to work for other transformations. For example $X \preceq Y$ does not imply $\exp(X) \preceq \exp(Y)$.

- However, we do have $\mathrm{Tr}(\exp(X)) \leq \mathrm{Tr}(\exp(Y))$
- Another similar result is the Golden–Thompson inequality which is stated above

Proposition 4.14 (Matrix Bernstein inequality). Suppose $X_1, \dots, X_n$ are independent zero mean symmetric matrices with $\|X_i\|_{\text{op}} \leq K$, then for any $t \geq 0$

$$\mathbb{P} \left(\left\| \sum_i X_i \right\|_{\text{op}} \geq t \right) \leq 2d \exp \left(\frac{-t^2/2}{\|\sum_i \mathbb{E} X_i^2\|_{\text{op}} + Kt/3} \right)$$

Proof. Application of Lieb's. If X is a random matrix then

$$\mathbb{E} \text{Tr}(\exp(H + X)) \leq \text{Tr}(\exp(H + \log \mathbb{E} \exp(X)))$$

by Jensen's + Lieb's

Let $S = \sum_{i=1}^n X_i$. Then

$$\|S\|_{\text{op}} = \max\{\lambda_{\max}(S), \lambda_{\max}(-S)\}$$

Then

$$\begin{aligned} \mathbb{P}(\lambda \cdot \lambda_{\max}(S) \geq \lambda t) &\leq \frac{\mathbb{E} \exp(\lambda \cdot \lambda_{\max}(S))}{\exp(\lambda t)} \\ &= \frac{\mathbb{E} \lambda_{\max}(\exp(\lambda S))}{\exp(\lambda t)} \\ &\leq \frac{\mathbb{E} \text{Tr}(\exp(\lambda S))}{\exp(\lambda t)} \end{aligned}$$

Note that $\exp(\lambda S)$ is PSD. Put $S_j = \sum_{i=1}^j X_i$. Also

$$\begin{aligned} \mathbb{E} \text{Tr}(\exp(\lambda S)) &= \mathbb{E} \text{Tr}(\exp(\lambda(S_{n-1} + X_n))) \\ &\quad \mathbb{E} (\mathbb{E} [\text{Tr}(\exp(\lambda(S_{n-1} + X_n))] \mid X_1, \dots, X_n]) \\ \text{(Lieb's step)} \quad &\leq \mathbb{E} [\text{Tr}(\exp(\lambda S_{n-1} + \log \mathbb{E} \exp(\lambda X_n)))] \\ (*) \quad &\leq \mathbb{E} \text{Tr}(\exp(\lambda S_{n-1} + g(\lambda) \mathbb{E} X_n^2)) \\ &\leq \text{Tr}(\exp(g(\lambda) \sum_{i=1}^n \mathbb{E} X_i^2)) \\ &\leq d \exp(\lambda_{\max}(g(\lambda) \sum_{i=1}^n \mathbb{E} X_i^2)) \\ &= d \exp(g(\lambda) \sum_{i=1}^n \mathbb{E} X_i^2) \end{aligned}$$

Line (*) is because:

Claim:

$$\mathbb{E} \exp(\lambda X_n) \preceq \exp(g(\lambda) \mathbb{E}[X_n^2])$$

where $g(\lambda) = \frac{\lambda^2/2}{1-\lambda K/3}$ due to the $d = 1$ case.

Thus

$$\log(\mathbb{E} \exp(\lambda X_n)) \preceq g(\lambda) \mathbb{E} X_n^2$$

Optimize over λ to get the final result. By the union bound for $\lambda_{\max}(-S)$ we prove the claim. $\blacksquare$

Corollary 4.15 (of matrix bernstein).

$$\mathbb{E} \left\| \sum_{i=1}^n X_i \right\|_{\text{op}} \leq \inf_{\lambda > 0} \frac{1}{\lambda} \log \mathbb{E} \exp(\lambda \left\| \sum_{i=1}^n X_i \right\|_{\text{op}}) \leq 2d \exp(g(\lambda) \left\| \sum_{i=1}^n X_i \right\|_{\text{op}})$$

Another bound on the LHS is

$$\sqrt{2 \log(2d) \left\| \sum_{i=1}^n X_i \right\|_{\text{op}}} + \frac{\log(2d)K}{3}$$

It is a matrix analogue of the Klitchnines inequality.

Proposition 4.16 (Matrix Bernstein inequality). Suppose $X_1, \dots, X_n$ are independent, mean-zero, symmetric $d \times d$ random matrices with $\|X_i\|_{\text{op}} \leq K$ almost surely. Define the variance proxy

$$V := \left\| \sum_{i=1}^n \mathbb{E}[X_i^2] \right\|_{\text{op}}.$$

Then for all $t \geq 0$,

$$\mathbb{P} \left(\left\| \sum_{i=1}^n X_i \right\|_{\text{op}} \geq t \right) \leq 2d \exp \left(-\frac{t^2/2}{V + Kt/3} \right).$$

Proof. Let $S := \sum_{i=1}^n X_i$. Since S is symmetric,

$$\|S\|_{\text{op}} = \max\{\lambda_{\max}(S), \lambda_{\max}(-S)\}.$$

Fix $\lambda > 0$. By Markov's inequality,

$$\begin{aligned} \mathbb{P}(\lambda_{\max}(S) \geq t) &= \mathbb{P}(e^{\lambda \lambda_{\max}(S)} \geq e^{\lambda t}) \\ &\leq e^{-\lambda t} \mathbb{E} e^{\lambda \lambda_{\max}(S)} \\ &= e^{-\lambda t} \mathbb{E} \lambda_{\max}(e^{\lambda S}) \\ &\leq e^{-\lambda t} \mathbb{E} \text{Tr}(e^{\lambda S}), \end{aligned}$$

using $\lambda_{\max}(A) \leq \text{Tr}(A)$ for $A \succeq 0$.

Next, set partial sums $S_k := \sum_{i=1}^k X_i$ (so $S_n = S$). Iterating conditional expectation and applying Lieb's concavity inequality in the standard way yields

$$\mathbb{E}\text{Tr}(e^{\lambda S}) \leq \text{Tr}\left(\exp\left(\sum_{i=1}^n \log \mathbb{E}e^{\lambda X_i}\right)\right).$$

We now use the mgf bound: for $0 \leq \lambda < 3/K$,

$$\log \mathbb{E}e^{\lambda X_i} \preceq g(\lambda) \mathbb{E}[X_i^2], \quad g(\lambda) := \frac{\lambda^2/2}{1 - \lambda K/3}.$$

(Equivalently, $\mathbb{E}e^{\lambda X_i} \preceq \exp(g(\lambda)\mathbb{E}[X_i^2])$.)

Therefore,

$$\begin{aligned} \mathbb{E}\text{Tr}(e^{\lambda S}) &\leq \text{Tr}\left(\exp\left(g(\lambda) \sum_{i=1}^n \mathbb{E}[X_i^2]\right)\right) \\ &\leq d \exp\left(g(\lambda) \left\| \sum_{i=1}^n \mathbb{E}[X_i^2] \right\|_{\text{op}}\right) = d e^{g(\lambda)V}. \end{aligned}$$

Combining with the earlier tail step gives, for $0 \leq \lambda < 3/K$,

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq d \exp(-\lambda t + g(\lambda)V).$$

The same bound holds for $\mathbb{P}(\lambda_{\max}(-S) \geq t)$ (apply the argument to $-X_i$), hence by a union bound,

$$\mathbb{P}(\|S\|_{\text{op}} \geq t) \leq 2d \exp(-\lambda t + g(\lambda)V).$$

Optimizing over $\lambda \in (0, 3/K)$ yields

$$\mathbb{P}(\|S\|_{\text{op}} \geq t) \leq 2d \exp\left(-\frac{t^2/2}{V + Kt/3}\right).$$

■

Corollary 4.17 (Expectation bound (from matrix Bernstein)). With $S = \sum_{i=1}^n X_i$ and $V = \left\| \sum_{i=1}^n \mathbb{E}[X_i^2] \right\|_{\text{op}}$ as above,

$$\mathbb{E}\|S\|_{\text{op}} \leq \sqrt{2V \log(2d)} + \frac{K}{3} \log(2d).$$

Remark 4.18 (Rectangular Matrices). If $A \in \mathbb{R}^{d_1 \times d_2}$ then put $\tilde{A} = \begin{bmatrix} 0 & A \\ A^\top & 0 \end{bmatrix} \in \mathbb{R}^{(d_1+d_2) \times (d_1+d_2)}$ where

$$\|A\|_{\text{op}} = \lambda_{\max}(\tilde{A})$$

5 Gaussian Process

Definition 5.1 (Gaussian Process). A collection $(X_t)_{t \in T}$ is a Gaussian Process if for any finite set $T_0 \subseteq T$, $T_0 = \{t_1, \dots, t_n\}$ we have $(X_{t_1}, \dots, X_{t_n})$ has multivariate Gaussian distribution.

Canonical Gaussian process in $\mathbb{R}^d$:

Let $g \sim \mathcal{N}(0, I_d)$. Let $T \subseteq \mathbb{R}^d$. Then put $X_t = \langle g, t \rangle$. Then

$$\mathbb{E}X_t = \langle \mathbb{E}g, t \rangle = 0$$

and

$$(\langle g, t_1 \rangle, \dots, \langle g, t_n \rangle)$$

can be expressed as Ag for some matrix A . A linear transformation of a multivariate gaussian is also a multivariate gaussian. Thus $(X_t)_{t \in T}$ is a Gaussian process.

Want to study $\sup_{t \in T} X_t$ or $\mathbb{E} \sup_{t \in T} X_t$. One example we have already come across is $X \sim N(0, I_d)$ then

Or $X \sim N(0, \Sigma)$ then $X = \Sigma^{1/2}Z$ where $Z \sim N(0, I_d)$. Then

$$\mathbb{E} \|X\| = \mathbb{E} \sup_{t \in S^{d-1}} \langle t, X \rangle = \mathbb{E} \sup_{t \in S^{d-1}} \langle t, \Sigma^{1/2}Z \rangle = \sup_{t \in \Sigma^{1/2}S^{d-1}} \langle t, Z \rangle$$

Lemma 5.2 (Slepian's Lemma). If (X_t) and (Y_t) are two zero mean Gaussian processes ($\mathbb{E}X_t = \mathbb{E}Y_t = 0$ for all $t \in T$) such that

1. $\mathbb{E}X_t^2 = \mathbb{E}Y_t^2$ for all $t \in T$ and *Comment on this assumption: Define $d(t, s) = \sqrt{\mathbb{E}(X_t - X_s)^2}$. Think of this as an induced distance on the elements of the process. In the canonical gaussian process, for $t, s \in \mathbb{R}^d$,*

$$d(t, s) = \sqrt{\mathbb{E}(\langle g, t \rangle - \langle g, s \rangle)^2} = \sqrt{\mathbb{E}(\langle g, t - s \rangle)^2} = \sqrt{\|t - s\|_2^2}$$

because

$$\langle g, t - s \rangle \sim N(0, \|t - s\|_2^2)$$

2. $\mathbb{E}[X_t - X_s]^2 \leq \mathbb{E}[Y_t - Y_s]^2$ *Comment on this assumption:*

Then for all $\tau > 0$

$$\mathbb{P}(\sup_{t \in T} X_t \geq \tau) \leq \mathbb{P}(\sup_{t \in T} Y_t \geq \tau)$$

and

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t$$

Proof. A simplification to make our proof a bit lighter.

- The index set T is finite.
- Then $(X_t)_{t \in T}$ can be viewed as a random vector with $\text{Cov}(X) = \Sigma$, $X \sim \mathcal{N}(0, \Sigma^X)$. (same for Y). $Y \sim \mathcal{N}(0, \Sigma^Y)$

,

Let $|T| = n$. Consider $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $f(x) = \mathbf{1}\{\max_{i \in [n]} \{x_i\} \leq \tau\}$.

Without loss of generality we may assume X and Y are independent (the quantities involved in hypothesis are all real numbers).

Let $Z(u) = \sqrt{u}X + \sqrt{1-u}Y$ for $u \in [0, 1]$. Then $Z(0) = Y$ and $Z(1) = X$. Assume that we proved that

- $\mathbb{E}f(Z(u))$ is non-decreasing
- $\mathbb{E}[f(Z(1))] \geq \mathbb{E}[f(Z(0))]$

Then $\mathbb{E}[f(X)] \geq \mathbb{E}[f(Y)] \iff \mathbb{P}(\max_i X_i \leq \tau) \geq \mathbb{P}(\max_i Y_i \leq \tau)$ if and only if

$$\mathbb{P}(\max_i X_i \geq \tau) \leq \mathbb{P}(\max_i Y_i \geq \tau)$$

We state a result from probability theory: **Stein's Identity**.

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function and $X \sim \mathcal{N}(0, 1)$ then

$$\mathbb{E}f'(X) = \mathbb{E}Xf(X)$$

Suppose f has finite support. That is there exists $c_1 > 0$ such $f(x) = 0$ when $|x| \geq c_1$. Let $g(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$. Then

$$\begin{aligned} \mathbb{E}f'(X) &= \int_{\mathbb{R}} f'(x)g(x) dx \\ &= f(x)g(x) \Big|_{-\infty}^{\infty} - \int_{\mathbb{R}} g'(x)f(x) dx \\ &= 0 - \int_{\mathbb{R}} g'(x)f(x) dx \\ &= \int_{\mathbb{R}} xg(x)f(x) dx = \mathbb{E}[xf(x)] \end{aligned}$$

Lemma 5.3. If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is twice differentiable then

$$\frac{d}{du} \mathbb{E}f(Z(u)) = \frac{1}{Z} \sum_{i,j} (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E} \frac{\partial^2}{\partial x_j \partial x_i} f(z(u))$$

Follows from Multivariate Stein's identity for $X \sim \mathcal{N}(0, \Sigma)$, $X \in \mathbb{R}^d$; A generalization of Stein's identity is

$$\Sigma \mathbb{E} \nabla f(X) = \mathbb{E} X f(X); \quad f : \mathbb{R}^d \rightarrow \mathbb{R}$$

Sketch of Proof. Define $Z(u) = \sqrt{u} X + \sqrt{1-u} Y \in \mathbb{R}^d$, so the i -th component is

$$Z_i(u) = \sqrt{u} X_i + \sqrt{1-u} Y_i.$$

Then $f(Z(u))$ is just f evaluated at the vector $Z(u)$. By the **chain rule**:

$$\frac{d}{du} f(Z(u)) = \sum_{i=1}^d \left. \frac{\partial f}{\partial z_i} \right|_{z=Z(u)} \cdot \frac{dZ_i}{du}$$

where the first factor is the i -th partial derivative of f evaluated at $Z(u)$, and the second factor is just differentiating the i -th component with respect to u :

$$\frac{dZ_i}{du} = \frac{X_i}{2\sqrt{u}} - \frac{Y_i}{2\sqrt{1-u}}.$$

Putting it together:

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \mathbb{E} \sum_{i=1}^d \left(\frac{X_i}{2\sqrt{u}} - \frac{Y_i}{2\sqrt{1-u}} \right) \left. \frac{\partial f}{\partial z_i} \right|_{z=Z(u)}$$

We handle the two terms separately. For the X term, condition on Y so that $X \sim \mathcal{N}(0, \Sigma^X)$ is the only source of randomness. Define $h(X) = \left. \frac{\partial f}{\partial z_i} \right|_{z=Z(u)}$. By the multivariate Stein's identity $\Sigma \mathbb{E} \nabla f(X) = \mathbb{E}[X f(X)]$, applied componentwise:

$$\mathbb{E}[X_i h(X)] = \sum_{j=1}^d \Sigma_{ij}^X \mathbb{E} \left[\frac{\partial h}{\partial x_j}(X) \right].$$

Since $Z(u) = \sqrt{u} X + \sqrt{1-u} Y$, the chain rule gives $\frac{\partial h}{\partial x_j} = \sqrt{u} \frac{\partial^2 f}{\partial z_j \partial z_i} \Big|_{z=Z(u)}$, so the X term contributes:

$$\frac{1}{2\sqrt{u}} \sum_{i=1}^d \mathbb{E} \left[X_i \left. \frac{\partial f}{\partial z_i} \right|_{z=Z(u)} \right] = \frac{1}{2} \sum_{i,j} \Sigma_{ij}^X \mathbb{E} \left. \frac{\partial^2 f}{\partial z_j \partial z_i} \right|_{z=Z(u)}.$$

The identical argument applied to $Y \sim \mathcal{N}(0, \Sigma^Y)$ gives the Y term as $\frac{1}{2} \sum_{i,j} \Sigma_{ij}^Y \mathbb{E} \left. \frac{\partial^2 f}{\partial z_j \partial z_i} \right|_{z=Z(u)}$. Combining:

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \frac{1}{2} \sum_{i,j} (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E} \left. \frac{\partial^2 f}{\partial z_j \partial z_i} \right|_{z=Z(u)}.$$

Now we will approximate our $h(x_i) = \mathbf{1}\{x_i \leq \tau\}$ with $\tilde{g}(x_i)$ which is smooth with L_∞ accuracy ϵ . Let $g(x) = \prod_{i=1}^n \tilde{g}(x_i)$. Then

$$\frac{\partial g(x)}{\partial x_i \partial x_j} = \tilde{g}'(x_i) \tilde{g}'(x_j) \prod_{k \neq i, j} \tilde{g}(x_k) \geq 0$$

Go over all of this again and confirm how it completes the proof of Slepian's lemma. Right now it feels like jumbled garb. ■

Lemma 5.4 (Sudakov–Fernique comparison lemma). Let $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ be centered Gaussian processes. Assume that for all $s, t \in T$,

$$\mathbb{E}(X_t - X_s)^2 \leq \mathbb{E}(Y_t - Y_s)^2.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t.$$

Remark 5.5 (Applications of Slepian's and Sudakov-Fernique Inequalities). Some examples of applying these lemmas:

1. Consider $X \in \mathbb{R}^{m \times n}$. such that $X_{ij} \sim N(0, 1)$ and independent. Then

$$\mathbb{E} \|X\|_{\text{op}} \leq C(\sqrt{n} + \sqrt{m})$$

by results from a previous lecture (ϵ -nets + concentration). The Sudakov-Fenique lemma gives

$$\mathbb{E} \|X\|_{\text{op}} \leq \sqrt{n} + \sqrt{m}$$

In particular when $m = n$

$$\mathbb{E} \|X\|_{\text{op}} \leq n\sqrt{n}$$

We have

$$\|X\|_{\text{op}} = \sup_{u \in S^{m-1}, v \in S^{n-1}} u^\top X v$$

and will consider $u^\top X v$ as a Gaussian process indexed by $(u, v) \in S^{m-1} \times S^{n-1}$ + zero mean.

Let $u, w \in S^{m-1}$ and $v, z \in S^{n-1}$. Then

$$\begin{aligned}
\mathbb{E}(u^\top X v - w^\top X z)^2 &= \mathbb{E}\left(\sum_{i,j} u_i v_j X_{ij} - \sum_{i,j} w_i z_j X_{ij}\right)^2 \\
&= \mathbb{E}\left(\sum_{i,j} [(u_i v_j - w_i z_j) X_{ij}]\right)^2 \\
&= \mathbb{E} \sum_{i,j} (u_i v_j - w_i z_j)^2 X_{ij}^2 + 0 \\
&= \sum_{i,j} (u_i v_j - w_i z_j)^2 \\
&\leq \|u - w\|_2^2 + \|v - z\|_2^2
\end{aligned}$$

Now denote $Y_{(u,v)} = \langle g_1, u \rangle + \langle g_2, v \rangle$ where $g_1 \sim N(0, I_m)$ and $g_2 \sim N(0, I_n)$ where g_1 and g_2 are independent.

Now check the hypothesis of the Sudakov-Fernique theorem

$$\mathbb{E}[Y_{(u,v)} - Y_{(w,z)}]^2 = \|u - w\|^2 + \|v - z\|^2$$

Then we can apply the theorem:

$$\begin{aligned}
\mathbb{E} \|X\|_{\text{op}} &\leq \mathbb{E} \sup_{(u,v)} Y_{(u,v)} \\
&\leq \mathbb{E} \sup_{u \in S^{m-1}} \langle g_1, u \rangle + \mathbb{E} \sup_{v \in S^{n-1}} \langle g_2, v \rangle \\
&= \mathbb{E}[\|g_1\|_2] + \mathbb{E}[\|g_2\|_2] \leq \sqrt{\mathbb{E} \|g_1\|_2^2} + \sqrt{\mathbb{E} \|g_2\|_2^2} \\
&= \sqrt{m} + \sqrt{n}
\end{aligned}$$

2. Recall Gaussian concentration: if $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -Lipschitz (w.r.t. $\|\cdot\|_2$) and $g \sim \mathcal{N}(0, I_d)$, then for all $t \geq 0$,

$$\mathbb{P}(f(g) - \mathbb{E}f(g) \geq t) \leq \exp\left(-\frac{t^2}{2L^2}\right).$$

Let X have i.i.d. $\mathcal{N}(0, 1)$ entries and identify $\text{vec}(X) \in \mathbb{R}^{n^2}$, so $\text{vec}(X) \sim \mathcal{N}(0, I_{n^2})$. Define $f : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$ by $f(\text{vec}(X)) = \|X\|_{\text{op}}$. For matrices $X, Y \in \mathbb{R}^{n \times n}$,

$$|\|X\|_{\text{op}} - \|Y\|_{\text{op}}| \leq \|X - Y\|_{\text{op}} \leq \|X - Y\|_F = \|\text{vec}(X) - \text{vec}(Y)\|_2,$$

so f is 1-Lipschitz. Therefore, for all $t \geq 0$,

$$\mathbb{P}(\|X\|_{\text{op}} - \mathbb{E}\|X\|_{\text{op}} \geq t) \leq \exp\left(-\frac{t^2}{2}\right).$$

Equivalently, with probability at least $1 - \delta$,

$$\|X\|_{\text{op}} \leq \mathbb{E}\|X\|_{\text{op}} + \sqrt{2 \log(1/\delta)}.$$

Using the standard bound $\mathbb{E}\|X\|_{\text{op}} \leq 2\sqrt{n}$, we obtain

$$\mathbb{P}\left(\|X\|_{\text{op}} \leq 2\sqrt{n} + \sqrt{2 \log(1/\delta)}\right) \geq 1 - \delta.$$

5.1 Sudakov's minoration and examples

Theorem 5.6 (Sudakov's minoration). If X_t is a zero mean gaussian process and $d(t, s) = \sqrt{\mathbb{E}[X_t - X_s]^2}$ there exists a constant $c > 0$ such that for any $\epsilon > 0$

$$\mathbb{E} \sup_{t \in T} X_t \geq c\epsilon \sqrt{\log \mathcal{N}(T, d, \epsilon)}$$

Proof. Consider the max packing of T at scale $\epsilon > 0$ call it P_ϵ . Then trivially,

$$\mathbb{E} \sup_{t \in T} X_t \geq \mathbb{E} \sup_{t \in P_\epsilon} X_t$$

Since we have a max-packing, for all $t, s \in P_\epsilon$

$$\mathbb{E}[X_t - X_s]^2 \geq \epsilon^2$$

If for each $t \in P_\epsilon$, set g_t to be a gaussian random variable in $\mathbb{R}^d$ then set $Y_t = g_t \frac{\epsilon}{\sqrt{2}}$. Then

$$g_1, \dots, g_{|P_\epsilon|}$$

Then

$$\mathbb{E}(Y_t - Y_s)^2 = \frac{\epsilon^2}{2} \mathbb{E}[g_t - g_s]^2 = \epsilon^2$$

Then

$$\begin{aligned} \mathbb{E} \sup_{t \in T} X_t &\geq \mathbb{E} \sup_{t \in P_\epsilon} X_t \\ &= \mathbb{E} \max_{t \in P_\epsilon} X_t \\ (\text{By Sudakov-Fernique}) &\geq \mathbb{E} \max_{t \in P_\epsilon} g_t \epsilon / \sqrt{2} \\ &\geq c\epsilon \sqrt{\log |P_\epsilon|} \\ &\geq c\epsilon \sqrt{\log \mathcal{N}(T, d, \epsilon)} \end{aligned}$$

■

Bounding the covering number using Sudakov's Minoration: Let $T = \Sigma^{1/2}S^{d-1}$ where $\Sigma \succ 0$. Let $X_t = \langle g, t \rangle$. Now,

1. Upper Bound:

$$\begin{aligned}\mathbb{E} \sup_{t \in \Sigma^{1/2}S^{d-1}} X_t &= \mathbb{E} \sup_{t \in \Sigma^{1/2}S^{d-1}} \langle g, t \rangle \\ &= \mathbb{E} \sup_{t \in S^{d-1}} \langle g, \Sigma^{1/2}t \rangle \\ &= \mathbb{E}[\|Z\|_2] \leq \text{Tr}(\Sigma)\end{aligned}$$

Then

$$\sqrt{\text{Tr}(\Sigma)} \geq c\epsilon \sqrt{\log(\mathcal{N}(\Sigma^{1/2}S^{d-1}, \|\cdot\|_2, \epsilon))}$$

Thus

$$\mathcal{N}(\Sigma^{1/2}S^{d-1}, \|\cdot\|_2, \epsilon) \leq \exp\left(\frac{c^\top \text{Tr}(\Sigma)}{\epsilon^2}\right)$$

2. Covering the L_1 ball with L_2 balls.

$$B_1^d = \{x \in \mathbb{R}^d : \|x\|_1 \leq 1\}$$

$$B_2^d = \{x \in \mathbb{R}^d : \|x\|_2 \leq 1\}$$

Let $T = B_1^d$. Then $\mathbb{E} \sup_{t \in T} \langle g, t \rangle = \mathbb{E}[\|g\|_\infty] \leq \sqrt{2 \log(d)}$. Using the same strategy as in example 1 we have

$$\log \mathcal{N}(B_1^d, \|\cdot\|_2, \epsilon) \leq c \frac{2 \log(d)}{\epsilon^2}$$

This bound is bad when ϵ is small. In that case better to put the L_1 in a big L_2 ball and cover the L_2 ball. To get the right expression in the small $\epsilon > 0$ regime, better to use the volumetric argument strategy. (packing, change radius of balls to $\epsilon/2$, expand L_1 ball by $1 + \epsilon/2$ factor and compare volumes)

Comparing this bound to a different one we saw (by volumetric argument)

$$\log \mathcal{N}(B_2^d, \|\cdot\|_2, \epsilon) \leq d \log(1 + \frac{2}{\epsilon})$$

To upper bound the covering number, we consider an ϵ -max packing. Then reducing the radius to $\epsilon/2$ balls so that all the $\epsilon/2$ balls centered at points in the max packing do not intersect. Let x be an element of the max packing.

Then

$$x + \epsilon/2 B_2^d \subseteq B_1^d + \epsilon/2 B_2^d$$

Using norm relations, we also have

$$B_1^d \subseteq B_2^d \subseteq \sqrt{d} B_1^d$$

Note these sums of sets are called *Minkowski Sums*.

Thus

$$x + \epsilon/2 B_2^d \subseteq B_1^d \left(1 + \frac{\sqrt{d}\epsilon}{2}\right)$$

Then

$$\mathcal{M}(B_1^d, \|\cdot\|_2, \epsilon) \leq \frac{\text{Vol}(B_1^d \left(1 + \frac{\sqrt{d}\epsilon}{2}\right))}{\text{Vol}(B_2^d \epsilon/2)} = \frac{(1 + \frac{\sqrt{d}\epsilon}{2})^d \text{Vol}(B_1^d)}{(\epsilon/2)^d \text{Vol}(B_2^d)} \leq \frac{(1 + \frac{\sqrt{d}\epsilon}{2})^d}{(\epsilon/2)^d} \left(\frac{c}{\sqrt{d}}\right)^d$$

Thus

$$\log \mathcal{N}(B_2^d, \|\cdot\|_2, \epsilon) \leq d \log(c(1 + \frac{1}{\epsilon\sqrt{d}})) \wedge \frac{c \log(2d)}{\epsilon^2}$$

Check how these bounds compare in the $\epsilon \sim \frac{1}{\sqrt{d}}$ regime.

Proposition 5.7 (Dual Norm of Sudakov Minoration). Let $T \subseteq \mathbb{R}^d$ be a centrally symmetric convex set, closed. Let the polar set be $T^\circ = \{x \in \mathbb{R}^d : \sup_{y \in T} \langle x, y \rangle \leq 1\}$.

As an example

$$(B_2^d)^\circ = B_2^d$$

Interestingly,

$$(B_p^d)^\circ = B_q^d, \quad \frac{1}{p} + \frac{1}{q} = 1$$

There exists $c > 0$ such that for every $\epsilon > 0$

$$c\epsilon \sqrt{\log(\mathcal{N}(B_2^d, \epsilon T^\circ))} \leq \mathbb{E} \sup_{t \in T} \langle g, t \rangle$$

Proof.

■

Example 5.8. A consequence of the dual Sudakov minoration is

$$\log(\mathcal{N}(B_2^d, \epsilon B_\infty^d)) \leq \frac{c \log(2d)}{\epsilon^2}$$

Remark 5.9 (conjecture). . If T and K are centrally symmetric convex sets then there exists an absolute constants $c, C > 0$ so that

$$c \log(\mathcal{N}(T, K)) \leq \log(\mathcal{N}(K^\circ, T^\circ)) \leq C \log(\mathcal{N}(T, K))$$

and

5.2 Gaussian Width

Definition 5.10 (Gaussian Width). Let $T \subseteq \mathbb{R}^d$. The Gaussian width of T is

$$w(T) = \mathbb{E} \sup_{t \in T} \langle g, t \rangle$$

where $g \sim N(0, I_d)$. It is the expected supremum of the canonical Gaussian process indexed by T .

Proposition 5.11.

1. $w(T)$ is finite if and only if T is bounded (i.e. $\|t\|_2 \leq b$ for all $t \in T$). For the forward direction:

$$w(T) = \mathbb{E} \sup_{t \in T} \langle g, t \rangle \leq \mathbb{E} \sup_{t \in T} \|g\|_2 \|t\|_2 \leq b\sqrt{d}.$$

For the reverse direction, suppose T is unbounded, so there exist $t_n \in T$ with $\|t_n\|_2 \rightarrow \infty$. Let E_n be the event that $\langle g, t_n \rangle \geq \frac{1}{10} \|g\|_2 \|t_n\|_2$. Since $\langle g, t_n \rangle / (\|g\|_2 \|t_n\|_2) = \cos \theta$ where θ is the angle between g and t_n , and g is rotationally symmetric, $\mathbb{P}(E_n) = \mathbb{P}(\cos \theta \geq \frac{1}{10}) > 0$ uniformly in n . Therefore

$$w(T) = \mathbb{E} \sup_{t \in T} \langle g, t \rangle \geq \mathbb{E} [\langle g, t_n \rangle \mathbf{1}_{E_n}] \geq \frac{1}{10} \|t_n\|_2 \mathbb{E} [\|g\|_2 \mathbf{1}_{E_n}] \rightarrow \infty,$$

where the last factor $\mathbb{E}[\|g\|_2 \mathbf{1}_{E_n}]$ is a positive constant independent of n .

2. $w(T + K) = w(T) + w(K)$. Indeed,

$$\begin{aligned} w(T + K) &= \mathbb{E} \sup_{t \in T, k \in K} \langle t + k, g \rangle \\ &= \mathbb{E} \sup_{t \in T, k \in K} (\langle t, g \rangle + \langle k, g \rangle) \\ &= \mathbb{E} \sup_{t \in T} \langle t, g \rangle + \mathbb{E} \sup_{k \in K} \langle k, g \rangle = w(T) + w(K). \end{aligned}$$

3. $w(\alpha T) = |\alpha| w(T)$ for all $\alpha \in \mathbb{R}$. For $\alpha \geq 0$ this is clear. For $\alpha < 0$, use the symmetry $g \stackrel{d}{=} -g$:

$$w(\alpha T) = \mathbb{E} \sup_{x \in T} \langle \alpha x, g \rangle = \mathbb{E} \sup_{x \in T} \langle \alpha x, -g \rangle = \mathbb{E} \sup_{x \in T} \langle -\alpha x, g \rangle = |\alpha| w(T).$$

4. $w(T) = \frac{1}{2} w(T - T)$. This follows from properties 2 and 3. One also has

$$w(\text{conv}(T)) = w(T),$$

since the supremum of a linear function over a convex hull is attained at an extreme point.

5. If T is centrally symmetric then $T - T = \text{conv}(T)$. If $a \in \mathbb{R}^d$ then $w(T + a) = w(T)$ since $\mathbb{E} \langle g, a \rangle = 0$.
6. If T is finite and $\text{diam} T = \max_{t,s \in T} \|t - s\|_2$. Then

$$w(T) \leq \sqrt{\log |T|} \text{diam} T$$

For any $x \in T - T$, $\|x\|_2 \leq \text{diam} T$.

$$\begin{aligned} w(T) &= \frac{1}{2} w(T - T) \\ &= \frac{1}{2} \sup_{x \in T - T} \langle x, g \rangle \leq \frac{1}{2} \mathbb{E} \max\{g_1, \dots, g_{|T|^2}\} \leq \frac{1}{2} \sqrt{\log |T|^2} \text{diam} T \end{aligned}$$

where $g_i \sim N(0, \|x_i\|_2^2)$ since

$$\langle x, g \rangle \sim N(0, \|x\|_2^2)$$

its variance is less than or equal to $\text{diam} T$.

Example 5.12.

- $w(B_2^d) = \mathbb{E} \sup_{x \in B_2^d} \langle x, g \rangle = \mathbb{E} \|g\|_2 \leq \sqrt{d}$
- $w(B_1^d) = \mathbb{E} \sup_{x \in B_1^d} \langle x, g \rangle = \mathbb{E} \|g\|_\infty \leq \sqrt{2 \log(2d)}$
- $w(B_\infty^d) = \mathbb{E} \sup_{x \in B_\infty^d} \langle x, g \rangle = \mathbb{E} \|g\|_1 = \sum_{i=1}^n \mathbb{E} |g_i| = d \sqrt{\frac{2}{\pi}}$.

Proposition 5.13. For any $T \subseteq \mathbb{R}^d$, there exist absolute constants $c_1, c_2 > 0$ such that

$$c_1 \text{diam}(T) \leq w(T) \leq c_2 \sqrt{d} \cdot \text{diam}(T).$$

Lower bound. There exist $x, y \in T$ with $\|x - y\|_2 = \text{diam}(T)$. Then

$$\begin{aligned} w(T) &= \frac{1}{2} w(T - T) = \frac{1}{2} \mathbb{E} \sup_{t,s \in T} \langle g, t - s \rangle \\ &\geq \frac{1}{2} \mathbb{E} \max(0, \langle g, x - y \rangle) = \frac{1}{2} \mathbb{E} |Z|, \end{aligned}$$

where $Z = \langle g, x - y \rangle \sim \mathcal{N}(0, \text{diam}(T)^2)$. Since $\mathbb{E} |Z| = \sqrt{2/\pi} \text{diam}(T)$, we get

$$w(T) \geq \sqrt{\frac{1}{2\pi}} \text{diam}(T).$$

Upper bound. Since $T - T \subseteq 2 \operatorname{diam}(T) \cdot B_2^d$, we can bound $w(T - T)$ by the Gaussian width of the enclosing ball:

$$w(T - T) = \mathbb{E} \sup_{v \in T - T} \langle g, v \rangle \leq \mathbb{E} \sup_{\|v\|_2 \leq 2 \operatorname{diam}(T)} \langle g, v \rangle = 2 \operatorname{diam}(T) \cdot \mathbb{E} \|g\|_2.$$

By Jensen's inequality, $\mathbb{E} \|g\|_2 \leq \sqrt{\mathbb{E} \|g\|_2^2} = \sqrt{d}$, so

$$w(T) = \frac{1}{2} w(T - T) \leq \sqrt{d} \cdot \operatorname{diam}(T).$$

5.3 Gaussian Process: Supremums

Proposition 5.14. Consider a zero mean process $(X_t)_{t \in T}$ such that for all $t, s \in T$,

$$\|X - t - X_s\|_{\psi_2} \leq d(t, s)$$

Example: $X_t = \langle t, g \rangle$. Then

$$\|X - t - X_s\|_{\psi_2} = \|\langle t - s, g \rangle\|_{\psi_2} \leq C \|t - s\|_2$$

by Cauchy Stewarz. Just define d to be a scaled version of the Euclidean distance.

The Generic chaining construction:

Consider an admissible sequence of sets.

$$(T_k)_{k=0}^\infty; T_k \subseteq T : T_0 = 1$$

where $|T_0| = 1$ and $|T_k| \leq 2^{2^k}$.

$$2^{2^{k+1}} / 2^{2^k} = 2^{2^{2^k} - 2^k} = 2^{2^k}$$

On set $k + 1$ the size of the set can increase by a factor of 2^{2^k} from the size of set k . In other words, you square the size of the previous set.

For each $t \in T$ define

$$d(t, T_k) = \inf_{x \in T_k} d(t, x)$$

Put

$$\gamma_2(T) = \inf_{(T_k) \text{ admissible}} \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k)$$

The result is that

$$\mathbb{E} \sup_{t \in T} \sup_{t \in T} (X_t) \leq C \gamma_2(T)$$

Chaining (Generic) Construction

Suppose we have a centered process $(X_t)_{t \in T}$ and assume $\|X_t - X_s\|_{\psi_2} \leq Ld(t, s)$ for all $t, s \in T$ where d is a metric on T and L is a constant.

An example, consider the canonical Gaussian process $X_t = \langle g, t \rangle$ where $g \sim \mathcal{N}(0, I_d)$, $t \in T \subseteq \mathbb{R}^n$. Then $\|X_t - X_s\|_{\psi_2} = c \|t - s\|_2$ where c is some absolute constant.

Proposition 5.15 (properties of γ_2).

1. If we replace $d(t, s)$ by $d'(t, s)$ where $d(t, s) \leq d'(t, s)$ then

$$\gamma_2(T, d) \leq \gamma_2(T, d')$$

2. Similarly, if we replace $d(t, s)$ by $Ld(t, s)$ for $L > 0$ then

$$\gamma_2(T, Ld) \leq L\gamma_2(T, d)$$

Theorem 5.16. Under the assumptions above

$$\mathbb{E} \sup_{t \in T} X_t \leq C\gamma_2(T)$$

Proof. For simple T is finite, there is an $(T_k)_{k=0}^\infty$ which is an admissible sequence such that at some very large k_0 : for all $k \geq k_0$, $T_k = T$. Put $\{t_0\} = T_0$. Then for all $t \in T$, let $\pi_k(t)$ denote a closest element to t in T_k .

$$X_t - X_{t_0} = \overbrace{\sum_{k=0}^{\infty} X_{\pi_{k+1}(t)} - X_{\pi_k(t)}}^{\star}$$

For $k = 0$, $X_{\pi_k(t)} = X_{t_0}$. Then we telescope until $t \in T_k$ and then the rest of the terms are 0.

To bound $\star$, we consider the event: for all $k \geq 0$, for all $a_k \in T_{k+1}$ and $b_k \in T_k$,

$$|X_{a_k} - X_{b_k}| \leq Cu \cdot 2^{k/2} d(a_k, b_k)$$

Our goal is to show that this is a high probability event. Call this the good event A .

Fix $a \in T_{k+1}$ and $b \in T_k$. Then

$$\|X_a - X_b\|_{\psi_2} \leq d(a, b)$$

by assumption.

Thus by a consequence of the bound of the ψ_2 norm,

$$\mathbb{P}(|X_a - X_b| \geq cu2^{k/2}d(a, b)) \leq 2 \exp\left(\frac{-c'u^22^k d^2(a, b)}{d^2(a, b)}\right) = \exp(-c'u^22^k)$$

where c' is another absolute constant.

Then assuming $u \geq 1$

$$\begin{aligned} \mathbb{P}(A^c) &\leq \sum_{k=0}^{\infty} \overbrace{|T_{k+1}| \cdot |T_k|}^{2^{3 \cdot 2^k}} \cdot 2 \exp(-c'u^22^k) \\ &= \sum_{k=0}^{\infty} 2^{3 \cdot 2^k + 1} \cdot \exp(-c'u^22^k) \\ &= \sum_{k=0}^{\infty} \cdot \exp(\log(2)(3 \cdot 2^k + 1) - c'u^22^k) \\ &\leq \sum_{k=0}^{\infty} \cdot \exp(6 \cdot 2^k \log(2) - c'u^22^k) \\ \text{provided } c' \text{ is large enough } &\leq \sum_{k=0}^{\infty} \cdot \exp(-c''u^22^k) \\ &\leq 2 \cdot \exp(-u^2) \end{aligned}$$

First line is by a union bound.

Then on the event A ,

$$\begin{aligned} X_t - X_{t_0} &\leq \sum_{k=0}^{\infty} Cu2^{k/2}d(\pi_{k+1}(t), \pi_k(t)) \\ &\leq \sum_{k=0}^{\infty} Cu2^{k/2}(d(\pi_{k+1}(t), t) + d(\pi_k(t), t)) \\ &\leq C' \sum_{k=0}^{\infty} u2^{k/2}d(\pi_k(t), t) \end{aligned}$$

Define $M = \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(\pi_k(t), t)$, which is a deterministic quantity depending on the admissible sequence (T_k) . We showed that on the event A (which depends on

u),

$$\sup_{t \in T} (X_t - X_{t_0}) \leq C' u \cdot M$$

and that for $u \geq 1$,

$$\mathbb{P}(A^c) \leq 2 \exp(-u^2).$$

Equivalently, for all $u \geq 1$,

$$\mathbb{P} \left(\sup_{t \in T} (X_t - X_{t_0}) > C' u \cdot M \right) \leq 2 \exp(-u^2).$$

Now apply the tail integration formula. Set $Z = \sup_{t \in T} (X_t - X_{t_0})$. Then

$$\begin{aligned} \mathbb{E}[Z] &= \int_0^\infty \mathbb{P}(Z > v) dv \\ &= C' M \int_0^\infty \mathbb{P}(Z > C' u \cdot M) du \quad (\text{substituting } v = C' u M) \\ &\leq C' M \left(\int_0^1 1 du + \int_1^\infty 2 \exp(-u^2) du \right) \\ &\leq C' M (1 + \sqrt{\pi}) \\ &\leq C'' M. \end{aligned}$$

The split at $u = 1$ is because our tail bound only applies for $u \geq 1$; for $u < 1$ we use the trivial bound $\mathbb{P}(\cdot) \leq 1$. The Gaussian integral $\int_1^\infty 2e^{-u^2} du \leq \int_0^\infty 2e^{-u^2} du = \sqrt{\pi}$.

Since $M = \sup_{t \in T} \sum_{k=0}^\infty 2^{k/2} d(\pi_k(t), t) \leq \sup_{t \in T} \sum_{k=0}^\infty 2^{k/2} d(t, T_k)$, we have

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E}[X_{t_0}] + C'' \sup_{t \in T} \sum_{k=0}^\infty 2^{k/2} d(t, T_k).$$

Since X_{t_0} is a centered Gaussian, $\mathbb{E}[X_{t_0}] = 0$. Taking the inf over all admissible sequences (T_k) gives

$$\mathbb{E} \sup_{t \in T} X_t \leq C'' \gamma_2(T).$$

■

Theorem 5.17 (Majorizing Measure theorem, Talagrand). If $(X_t)_{t \in T}$ is a zero mean Gaussian process with $d(t, s) = \|X_t - X_s\|_{L_2}$ then

$$c\gamma_2(T) \leq \mathbb{E} \sup_{t \in T} X_t \leq C\gamma_2(T)$$

where c and C are absolute constants.

Remark 5.18 (Applications and Questions). Check it out!

1. **Talagrand's Comparison Theorem:** Assume $(X_t)_{t \in T}$ is a zero mean sub-gaussian process and $(Y_t)_{t \in T}$ is a zero mean gaussian processes such that

$$\|X_t - X_s\|_{\psi_2} \leq C \sqrt{\mathbb{E}[Y_t - Y_s]^2} := Cd(t, s)$$

for all $s, t \in T$. Then $\mathbb{E} \sup_{t \in T} X_t \leq C' \mathbb{E} \sup_{t \in T} Y_t$. This result has not been proved without using the majorizing measure theorem.

Proof. From the majorizing measure result,

$$\mathbb{E} \sup_{t \in T} X_t \leq C'' \gamma_2(T) \leq C''' \mathbb{E} \sup_{t \in T} Y_t$$

Where did the sub-gaussian result come in? ■

2. If $T \subseteq \mathbb{R}^d$ and $d(t, s) = \|t - s\|_2$ then

$$c\gamma_2(T) \leq \gamma_2(\text{conv}(T)) \leq C\gamma_2(T)$$

Proof. Use the majoring measure theorem and that the taking a convex hull of T has the same gaussian width as T . ■

Proposition 5.19 (Dudley Entropy Integrand). We want to think about

$$\gamma_2(T) = \inf_{(T_k)} \sup_{t \in T} \sum_{k=1}^{\infty} 2^{k/2} d(t, T_k)$$

Consider the relaxation

$$D(T) = \inf_{(T_k)} \sum_{k=1}^{\infty} 2^{k/2} \sup_{t \in T} d(t, T_k)$$

Clearly, $\gamma_2(T) \leq D(T)$.

Remark 5.20 (Tensorization). Moving the supremum inside the sum decouples the levels: in γ_2 , a single $t \in T$ must be simultaneously worst-case across all scales k , whereas in $D(T)$ each level k independently picks its own worst-case point $t_k \in T$. This is a *tensorization* — the sup over the diagonal $\{(t, t, \dots) : t \in T\} \subset T^{\mathbb{N}}$ is relaxed to a sup over the full product $T^{\mathbb{N}}$, and $D(T)$ factors cleanly across levels. The payoff is that each term $\sup_{t \in T} d(t, T_k)$ (essentially a covering number at scale k) can be optimized independently, which is why $D(T)$ recovers the Dudley entropy integral. The cost is that $D(T)$ may strictly overestimate $\gamma_2(T)$, since the true worst-case t for the full sum need not be worst-case at any individual level.

Define the entropy numbers

$$\epsilon_k = \inf\{\epsilon > 0 : \mathcal{N}(T, d, \epsilon) \leq 2^{2^k}\}$$

Then it is not too hard to see an equivalent expression for $D(T)$ is given by

$$D(T) = \sum_{k=0}^{\infty} 2^{k/2} \epsilon_k$$

The Dudley interval is defined as

$$D(T) \leq C \int_0^{\text{diam}(T)} \sqrt{\log(\mathcal{N}(T, d, \epsilon))} d\epsilon$$

The motivation for the $\sqrt{\log}$ is that

$$\sqrt{\log(2^{2^k})} \approx 2^{k/2}$$

Example 5.21. Consider a $g \sim N(0, I_d)$ and we are interested in $\mathbb{E} \|g\|_2$. By Cauchy-Schwarz we have

$$\mathbb{E} \|g\|_2 \leq \sqrt{\mathbb{E} \|g\|_2^2} = \sqrt{d}$$

since $\|g\|_2^2$ is a sum of d squared independent univariate standard gaussians.

Now we try to use Dudley. Consider the norm as supremum of a gaussian process indexed by the unit sphere. We use

$$D(T) :$$

$$d(t, s) = \sqrt{\mathbb{E}(\langle t, g \rangle - \langle s, g \rangle)^2} = \|t - s\|_2$$

$$\begin{aligned}
\mathbb{E} \|g\|_2 &= \mathbb{E} \sup_{t \in S^{d-1}} \langle t, g \rangle \\
&\leq C \int_0^2 \sqrt{\log(\mathcal{N}(T, d, \epsilon))} d\epsilon \\
&\leq C \int_0^2 \sqrt{\log(1 + \frac{2}{\epsilon})^d} d\epsilon \\
&\leq C \int_0^2 \sqrt{\log(1 + \frac{2}{\epsilon})} d\epsilon \\
&= C\sqrt{d} \int_0^2 \sqrt{\log(1 + \frac{2}{\epsilon})} d\epsilon \\
&\leq C'\sqrt{d}
\end{aligned}$$

We can say $\log(1 + \frac{2}{\epsilon}) \leq \frac{1}{\sqrt{\epsilon}}$ for all ϵ small enough which is how we went from C to C' (showing the integral in the second to last line is some finite constant).

Proposition 5.22 (Gap between Dudley and γ_2 for an ellipsoid). Consider T as an ellipsoid. Can write

$$T = \Sigma B_2^d$$

for a PSD Σ . Then

$$\mathbb{E} \sup_{t \in T} \langle t, g \rangle \leq \sqrt{\text{Tr}(\Sigma)}$$

which is a dimension free bound (assuming a decay of eigenvalues).

Claim if $T \subseteq \mathbb{R}^d$ then

$$\gamma_2(T) \leq D(T) \leq C\gamma_2(T) \cdot \log(d)$$

Proof.

$$\gamma_2(T) = \inf_{(T_k)} \sup_{x \in T} \sum_{k=0}^{\infty} 2^{k/2} d(x, T_k) \geq \inf_{(T_k)} \sup_{x \in T} 2^{m/2} d(x, T_m) = 2^{m/2} \epsilon_m$$

for any $m \geq 0$.

Next, by putting T in an L_2 ball we have

$$\mathcal{N}(T, \|\cdot\|_2, \epsilon) \leq \left(1 + \frac{\text{diam} T}{\epsilon}\right)^2$$

Our point budget for ϵ_k can satisfy (in order to cover T)

$$2^{2^k} = \left(1 + \frac{\text{diam} T}{\epsilon_k}\right)^2$$

Such an ϵ'_k is an upper bound on ϵ_k , we can always have $\epsilon_k \leq \text{diam} T$.

We can rearrange to get

$$\epsilon'_k = 2\text{diam}(T)2^{(-2^k \cdot \frac{1}{d})}$$

$$\begin{aligned} D(T) &= \sum_{k=0}^{\infty} 2^{k/2} \epsilon_k \\ &= \sum_{k=0}^{\lceil c \log(d) \rceil} 2^{k/2} \epsilon_k + \sum_{k=\lceil c \log(d) \rceil+1}^{\infty} 2^{k/2} \epsilon_k \end{aligned}$$

$$\begin{aligned} \text{by the lower bound on } \gamma \text{ we just proved } &\leq \lceil c \log(d) \rceil \gamma_2(T) + \sum_{k=\lceil c \log(d) \rceil+1}^{\infty} 2^{k/2} \epsilon_k \\ &\leq \lceil c \log(d) \rceil \gamma_2(T) + \sum_{k=\lceil c \log(d) \rceil+1}^{\infty} 2^{k/2} 2\text{diam}(T) 2^{(-2^k \cdot \frac{1}{d})} \\ &\leq C' (\log(d) \gamma_2(T) + \text{diam}(T)) \\ &\leq C''(\gamma_2(T)) \end{aligned}$$

Where we choose $c > 0$ so that $2^k/d \geq 2^{k/2}$ for all $k \geq \lceil c \log(d) \rceil + 1$.

We also used in the last bit

$$\text{diam}(T) \leq C_1 w(T) \leq C_2 \gamma_2(T)$$

■

Remark 5.23. To show we cannot improve the $\log d$ rate for this bound consider the B_1^d ball. We can show

$$\gamma_2(T) \asymp \sqrt{\log d}$$

and

$$D(T) \asymp (\log d')^{3/2} \asymp \log(d) \gamma_2(T)$$

6 Empirical Process Theory

Motivating Example

Example 6.1. Suppose X is a random variable and $F(x)$ is the cdf of X . We sometimes consider the empirical cdf when given an iid sample $X_1, \dots, X_n$:

$$F_n(x) = \sum_{i=1}^n 1\{X_i \leq x\}$$

We know $F_n(x) \approx F(x)$ but how good is this approximation for a given point?

Using Hoeffding we have

$$\mathbb{P}(|F_n(x) - F(x)| \geq t) \leq 2 \exp(-2nt^2)$$

Put another way, with probability $1 - \delta$ we have

$$|F_n(x) - F(x)| \leq \sqrt{\frac{\log(2/\delta)}{2n}}$$

Kolmogorov-Smirnov Statistic

$$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)|$$

How do we analyze these quantities non-asymptotically?

Definition 6.2 (Empirical Process). Consider the class $\mathcal{F} = \{f : \psi \rightarrow \mathbb{R}\}$ and $X_1, \dots, X_n$ are iid rvs in ψ . Then

$$\left(\mathbb{E}f(X) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right)_{f \in \mathcal{F}}$$

is an empirical process.

The cdf example we saw above can be seen as an empirical process where the class $\mathcal{F}$ is given by

$$\mathcal{F} = \{y \mapsto 1\{y \leq x\} : x \in \mathbb{R}\}$$

Proposition 6.3 (Symmetrization).

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left(\mathbb{E} f(X) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right) \leq 2 \cdot \mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i f(X_i) \right)$$

where ϵ_i are iid Rademacher signs (random variables). Sometimes the RHS is called the Rademacher complexity.

Also we have

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left(\mathbb{E} \frac{1}{n} \sum_{i=1}^n f(X_i) - f(X) \right) \leq 2 \cdot \mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i f(X_i) \right)$$

and

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \mathbb{E} f(X) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right| \leq 2 \cdot \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \epsilon_i f(X_i) \right|$$

Proof. Let $X'_1, \dots, X'_n$ be another iid sample. Then

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}} \left(\mathbb{E} f(X) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right) &= \mathbb{E} \sup_{f \in \mathcal{F}} \left(\mathbb{E} \left(\frac{1}{n} \sum_{i=1}^n f(X'_i) \right) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right) \\ &\leq \mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n (f(X'_i) - f(X_i)) \right) \\ &= \mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i (f(X'_i) - f(X_i)) \right) \\ &\leq 2 \mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i f(X_i) \right) \end{aligned}$$

■

Example 6.4 (Kolmogorov-Smirnov).

$$\begin{aligned}
\mathbb{E} \sup_{x \in \mathbb{R}} |F_n(x) - F(x)| &\leq 2 \mathbb{E} \sup_{x \in \mathbb{R}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i 1\{X_i \leq x\} \right) \\
&= \mathbb{E} \mathbb{E} \left(\sup_{x \in \mathbb{R}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i 1\{X_i \leq x\} \right) \mid X_1, \dots, X_n \right) \\
&= \mathbb{E} \mathbb{E} \left(\max_{\text{over } n+1 \text{ values}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i 1\{X_i \leq x\} \right) \mid X_1, \dots, X_n \right)
\end{aligned}$$

The conclusion of this example was the expected supremum was upper bounded by

$$2 \sqrt{\frac{2 \log(2(n+1))}{n}}$$

Proposition 6.5. Assume $X_1, \dots, X_n$ are independent random variables and $\mathcal{F}$ is a class of functions bounded absolutely by $\ell \in \mathbb{R}_+$. Then with probability $1 - \delta$

$$\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n (f(X_i) - Ef(X)) \right| \leq \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n (f(X_i) - Ef(X)) \right| + \ell \sqrt{\frac{2 \log(1/\delta)}{n}}$$

Proof. Let $g(X_1, \dots, X_n) = \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n (f(X_i) - Ef(X)) \right|$. Now we want to check that the bounded differences condition holds.

Consider $x_1, \dots, x_n, x'_i$. Then

$$\left| \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n f(x_i) - Ef(x) - \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i \neq i'} f(x_i) + f(x'_i) - Ef(x) \right| \leq \sup_{f \in \mathcal{F}} \left| \frac{1}{n} (f(x_i) - f(x'_i)) \right| \leq \frac{2\ell}{n}$$

since $\sup_f A(f) - \sup_f B(f) \leq \sup_f (A(f) - B(f))$.

Then by the bounded differences result, with probability at least $1 - \delta$

$$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \leq 2 \sqrt{\frac{2 \log(2(n+1))}{n}} + 2 \sqrt{\frac{2 \log(1/\delta)}{n}}$$

As a corollary, using the uniform Glivenko-Cantelli property we have

$$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \rightarrow 0$$

almost surely. ■

6.1 Shatter Function and VC Dimension

Remark 6.6. Reminder:

$$\mathbb{E} \sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \leq 2 \mathbb{E} \sup_{x \in \mathbb{R}} \left| \frac{1}{n} \sum_{i=1}^n \epsilon_i \mathbf{1}\{X_i \leq x\} \right|$$

Given $X_1, \dots, X_n$, the supremum turns into a max over $n+1$ elements since the bound is constant for all x satisfying $X_{(i)} \leq x < X_{(i+1)}$ for any i .

Let's generalize this observation: let $\mathcal{X}$ be some space and $\mathcal{A}$ be the collection of events $A \in \mathcal{A}$, $A \subseteq \mathcal{X}$. For example

$$\mathbb{R} = \mathcal{X}, \mathcal{A} = \{(-\infty, t] : t \in \mathbb{R}\}$$

Define the Shatter / Growth function

$$S_{\mathcal{A}}(n) = \max_{x_1, \dots, x_n \in \mathcal{X}} \left| \overbrace{\{(\mathbf{1}(x_1 \in A), \dots, \mathbf{1}(x_n \in A)) : A \in \mathcal{A}\}}^{\in \{0,1\}^n} \right|$$

As an example, fix $x_1 < \dots, < x_n$. Then as we consider the different values of $A \in \mathcal{A}$ we have

$$S_{\mathcal{A}}(n) = n + 1$$

since the realizations of the binary vectors can only be

$$(0, \dots, 0), (1, 0, \dots, 0), \dots, (1, \dots, 1)$$

In cases where the x_i are not increasing or overlapping we get fewer realises binary vectors (or the same number) but not more.

Theorem 6.7. Let $X_1, \dots, X_n$ be iid random objects in $\mathcal{X}$. Consider a collection of events $\mathcal{A}$ with the growth function $S_{\mathcal{A}}(n)$, denoting

$$P(A) := \mathbb{P}(X_i \in A)$$

and

$$P_n(A) := \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \in A)$$

Then

$$\sup_{A \in \mathcal{A}} |P_n(A) - P(A)| \leq 2 \sqrt{\frac{2 \log(2S_{\mathcal{A}}(n))}{n}} + 2 \sqrt{\frac{2 \log(1/\delta)}{n}}$$

Proposition 6.8 (Properties of Growth function). Props:

1. $S_{\mathcal{A}}(n) \leq 2^n$
2. $S_{\mathcal{A}}(n) \leq |\mathcal{A}|$

Definition 6.9 (Vapnik-Chervonenkis dimension). The VC-Dimension of $\mathcal{A}$ is the largest integer d such that the growth function

$$S_{\mathcal{A}}(d) = 2^d$$

Example 6.10. Consider

$$\mathbb{R} = \mathcal{X}, \mathcal{A} = \{(-\infty, t] : t \in \mathbb{R}\}$$

Note that $d = 0 \iff |\mathcal{A}| = 1$.

For our particular example, we start by asking if $d \geq 1$. Can we find some point such that both (1) and (0) are realized. Yes. Fix any $x \in \mathbb{R}$ and then choose $y < x$ to make $\mathbf{1}(x \in (-\infty, y)) = 0$ and another $w \geq x$ to get the 1. Thus

$$S_{\mathcal{A}}(1) = 2^1$$

so $d \geq 1$. For consider two points, we can achieve

$$(0, 0), (1, 0), (1, 1)$$

but not (0, 1). Thus the VC-dimension is 1.

In other example, we take $\mathcal{A} = \{[a, b], a, b \in \mathbb{R}\}$. By similar reasoning we can show the VC-dimension is 2 since (1, 0, 1) cannot be realized.

Lemma 6.11 (Sauer-Shelah-Vapnik-Chervonenkis). If $\mathcal{A}$ has VC-dimension d then

$$S_{\mathcal{A}}(n) \leq \sum_{k=0}^d \binom{n}{k}$$

If additionally, $n \geq d$ then

$$\sum_{k=0}^d \binom{n}{k} \leq \left(\frac{en}{d}\right)^d$$

If the VC-dimension is infinite, then $S_{\mathcal{A}}(n) = 2^n$.

Thus the growth function is either polynomial in n or exponential (no other options).

Classes of sets with finite VC-dimension are called VC-classes of sets.

Corollary 6.12 (of the last theorem). If $\mathcal{A}$ has VC dimension d then if $n \geq d$,

$$\sup_{A \in \mathcal{A}} |P_n(A) - P(A)| \leq 2\sqrt{\frac{4d \log(\frac{en}{d})}{n}} + 2\sqrt{\frac{2 \log(1/\delta)}{n}}$$

Corollary 6.13. Uniform laws of large numbers ($\mathcal{A}$ is Glivenko-Cantelli)

$$\sup_{A \in \mathcal{A}} |P_n(A) - P(A)| \rightarrow 0$$

almost surely if equivalent to VC dimension of $\mathcal{A}$ being finite.

Proof. We have only so far showed that when d is finite, the uniform law of large numbers holds. ■

Proposition 6.14. If $X = \mathbb{R}^p$ and $\mathcal{A}$ is the class of all closed half-spaces, then the VC dimension of $\mathcal{A}$ is $p + 1$.

Proof. We show the VC dimension equals $p + 1$ by establishing matching upper and lower bounds.

Lower bound (VC dim $\geq p + 1$). We exhibit $p + 1$ points that are shattered by $\mathcal{A}$. Consider the vertices of a standard p -simplex: take $x_0 = \mathbf{0}$ and $x_i = e_i$ for $i = 1, \dots, p$, where e_i denotes the i -th standard basis vector. These $p + 1$ points are *affinely independent*, meaning no point lies in the affine hull of the others.

We claim these points are shattered. Fix any subset $S \subseteq \{x_0, x_1, \dots, x_p\}$. We must produce a closed half-space containing exactly the points in S . Since $x_0, \dots, x_p$ are affinely independent, there is a unique affine function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ (i.e., $f(x) = w^\top x + b$)

for some $w \in \mathbb{R}^p$, $b \in \mathbb{R}$) taking prescribed values at each x_i . In particular, choose f so that

$$f(x_i) = \begin{cases} +1 & x_i \in S, \\ -1 & x_i \notin S. \end{cases}$$

Then the closed half-space $H = \{x \in \mathbb{R}^p : f(x) \geq 0\}$ satisfies $H \cap \{x_0, \dots, x_p\} = S$. Since S was arbitrary, the $p + 1$ points are shattered.

Upper bound (VC dim $\leq p + 1$). We show that no set of $p + 2$ points can be shattered by $\mathcal{A}$. The key tool is the following classical result.

Theorem 6.15 (Radon's Theorem). Any set of $p + 2$ points in $\mathbb{R}^p$ can be partitioned into two disjoint non-empty subsets A and B such that $\text{conv}(A) \cap \text{conv}(B) \neq \emptyset$.

Now let $\{x_1, \dots, x_{p+2}\} \subset \mathbb{R}^p$ be arbitrary. By Radon's theorem, there exists a partition $A \sqcup B = \{x_1, \dots, x_{p+2}\}$ with $\text{conv}(A) \cap \text{conv}(B) \neq \emptyset$; pick a point $z \in \text{conv}(A) \cap \text{conv}(B)$.

We claim the labeling that places $+1$ on A and 0 on B cannot be realized by any half-space. Suppose for contradiction that there exists a closed half-space $H = \{x : w^\top x \leq b\}$ with $A \subseteq H$ and $B \cap H = \emptyset$ (i.e., $B \subseteq H^c$). Since H is convex and $A \subseteq H$, we have $\text{conv}(A) \subseteq H$, so $z \in H$. On the other hand, the open half-space $H^c = \{x : w^\top x > b\}$ is also convex, and $B \subseteq H^c$ implies $\text{conv}(B) \subseteq H^c$, so $z \in H^c$. This contradicts $z \in H$.

Therefore, no $p + 2$ points can be shattered, and we conclude $\text{VCdim}(\mathcal{A}) \leq p + 1$.

Combining both bounds gives $\text{VCdim}(\mathcal{A}) = p + 1$. ■

Alternate Construction Let $X = \mathbb{R}^p$; for $\mathcal{A}$ consisting of homogenous half-spaces (dividing plane passing through 0) then VC dimension of $\mathcal{A}$ is p .

Tightness VC phenomena is tight for these half space constructions.

Example 6.16. Axis aligned rectangles

$$\mathcal{A} = \{(a, b) \times (c, d) : a, b, c, d \in \mathbb{R}\}$$

The VC dimension is 4.

Heuristically The VC dimension is the minimum number of parameters that describe a function in that class. So for the rectangle example the number of parameters is 4.

Remark 6.17. The shatter function is

$$s_{\mathcal{A}}(n) = \max |\{(\mathbf{1}\{x_1 \in A\}, \dots, \mathbf{1}\{x_n \in A\}) : A \in \mathcal{A}\}|$$

and the VC dimension of $\mathcal{A}$ is the largest d such that $s_{\mathcal{A}}(d) = 2^d$.

In words, $\mathcal{A}$ "shatters" some set of size d but no set of size $d + 1$.

Proof. Sketch of the proof of the Sauer-Shelah-VC result Let $x_1, \dots, x_n \in X$. Without loss of generality assume $n \geq d$ (what happens when $n < d$?).

We chose of n points so that the max in the definition of $s_{\mathcal{A}}(n)$ is achieved.

Form a binary matrix V whose rows are the binary vectors

$$(\mathbf{1}\{x_1 \in A\}, \dots, \mathbf{1}\{x_n \in A\})_{A \in \mathcal{A}}$$

So the number of rows in V equals $s_{\mathcal{A}}(n)$. We want to bound this.

Define a procedure called shifting on V . Repeat until nothing changes:

1. pick a column j , scan each row.
2. If the entry is zero do nothing
3. if the entry is 1, replace it with 0 unless doing so would create a duplicate row.

Call the result V^* - the matrix is immune to shifting (no more 1's can be zeroed out without creating duplicate rows).

Key Observations:

1. The number of rows is preserved:

$$\#\text{rows}(V) = \#\text{rows}(V^*)$$

Since we only change entries when we create duplicates, each operation preserves the row count.

2. Any set of columns shattered by V^* is also shattered by V . Why? Shifting can only remove rows (by making 1s into 0s). If every 2^k pattern on k columns appears in V^* , one can show each pattern must have a parent in V covering it too (the argument traces back which row the shifted row came from).

Note that when we say a set of k columns is shattered by V we mean that every possible binary string of length k appears as some row in the sub-matrix formed

by those columns. Then the VC-dimension of V is the size of the largest shattered column set.

It suffices to show that one shift step cannot create a newly shattered set. After a shift step the matrix V loses a row r and gains a row r' which was not already present in the matrix. Thus the number of unique rows is the same. Suppose the column we altered is c and the shattered set we want to consider is S . Consider the cases of $c \in S$ and $c \notin S$.

3. Therefore

$$\text{VC-dim}(V^*) \leq (V) = \text{VC-dim}(U \text{ restricted to } \{x_1, \dots, x_n\}) \leq \text{VC-dim}(U) = d$$

The Crucial Claim V^* has no row with more than d ones. Proof by contradiction. Suppose some row $v \in V^*$ has at least $d + 1$ ones, say in columns $j_1, \dots, j_{d+1}$.

- V^* is immune to shifting so each of those 1s cannot be zeroed out without creating a duplicate row.
- This means that for each j_i there exists another row in V^* which agrees with v on all other columns but has a zero in column j_i .
- Iterating this argument across all 2^{d+1} subsets of $\{j_1, \dots, j_{d+1}\}$, one can show all 2^{d+1} binary patterns on those columns are present in V^*
- That means $\text{VC-dim}(V^*) \geq d + 1$
- This is a contradiction

So every row of V^* has at most d ones. the number of such vectors is exactly

$$\sum_{k=0}^d \binom{n}{k}$$

Thus

$$S_{\mathcal{A}}(n) = \#\text{rows}(V) = \#\text{rows}(V^*) \leq \sum_{k=0}^d \binom{n}{k}$$

The second inequality is a standard binomial sum bound. ■

6.2 Covering Numbers for VC-Classes

Definition 6.18. Given a class $\mathcal{F}$ of functions mapping $X \rightarrow \{0, 1\}$, we define

$$\text{VC-dim}(\mathcal{F}) = \max \{d : \exists x_1, \dots, x_d \text{ s.t. } |\{(f(x_1), \dots, f(x_d)) : f \in \mathcal{F}\}| = 2^d\}.$$

Recall that for a set class $\mathcal{A}$, we take $f_A(x) = \mathbf{1}\{x \in A\}$ for each $A \in \mathcal{A}$, so the VC-dimension of $\mathcal{A}$ as a function class agrees with the combinatorial shattering definition. The quantity $|\{(f(x_1), \dots, f(x_n)) : f \in \mathcal{F}\}|$ is called the *growth function* $S_{\mathcal{F}}(n)$, and measures how many distinct labelings of n points $\mathcal{F}$ can produce. A priori $S_{\mathcal{F}}(n) \leq 2^n$; Sauer–Shelah says this blows up exponentially only if $\text{VC-dim}(\mathcal{F}) = \infty$.

Theorem 6.19 (Sauer–Shelah). If $\mathcal{F}$ has $\text{VC-dim}(\mathcal{F}) = d$, then for all $x_1, \dots, x_n \in X$,

$$S_{\mathcal{F}}(n) = |\{(f(x_1), \dots, f(x_n)) : f \in \mathcal{F}\}| \leq \sum_{k=0}^d \binom{n}{k} \leq \left(\frac{en}{d}\right)^d.$$

In particular, the growth function transitions from exponential to polynomial at $n = d$: for $n \leq d$ we have $S_{\mathcal{F}}(n) = 2^n$, while for $n > d$ the bound is polynomial in n . This *phase transition* is the key combinatorial fact underlying much of statistical learning theory.

Example 6.20. The Sauer–Shelah bound is sharp: it is achieved by the class $\mathcal{F}$ consisting of all functions equal to 1 on at most d points of X . Indeed, this class has VC-dimension exactly d , and on any n points it realizes exactly $\sum_{k=0}^d \binom{n}{k}$ distinct labelings (one for each subset of size $\leq d$).

Definition 6.21. Given $x_1, \dots, x_n \in X$ and $f, g \in \mathcal{F}$, define the empirical metric

$$e_n(f, g) = \sqrt{\frac{1}{n} \sum_{i=1}^n \mathbf{1}\{f(x_i) \neq g(x_i)\}},$$

where we used that f, g are binary so $(f(x_i) - g(x_i))^2 = \mathbf{1}\{f(x_i) \neq g(x_i)\}$. This is the square root of the empirical Hamming distance, and should be thought of as a *non-random* (fixed data) pseudometric on $\mathcal{F}$: two functions are close if they disagree on few of the observed points. Note e_n depends on the sample $x_1, \dots, x_n$ but not on any labels, so this is purely a complexity measure of $\mathcal{F}$.

Lemma 6.22 (Covering numbers for VC-classes). If $\mathcal{F}$ is a class of binary functions with $\text{VC-dim}(\mathcal{F}) = d$, then

$$\mathcal{N}(\mathcal{F}, e_n, \epsilon) \leq \left(\frac{c}{\epsilon}\right)^{4d}$$

for all $\epsilon \in (0, 1)$ and all choices of $x_1, \dots, x_n$, where c is an absolute constant.

Proof. We bound the covering number via the packing number, using $\mathcal{N}(\mathcal{F}, e_n, \epsilon) \leq \mathcal{M}(\mathcal{F}, e_n, \epsilon)$. Let V_ϵ be a maximal ϵ -packing, so $|V_\epsilon| = \mathcal{M}(\mathcal{F}, e_n, \epsilon)$. Represent each $f \in V_\epsilon$ by its evaluation vector $(f(x_1), \dots, f(x_n)) \in \{0, 1\}^n$.

Step 1: From packing to Hamming distance. For any $u \neq v \in V_\epsilon$, the packing condition $e_n(u, v) \geq \epsilon$ gives

$$|\{i \in [n] : u_i \neq v_i\}| \geq n\epsilon^2,$$

so each pair disagrees on at least a ϵ^2 -fraction of coordinates.

Step 2: Probabilistic witness set. Let $Y_1, \dots, Y_k \stackrel{iid}{\sim} \text{Unif}([n])$ and set $K = \{Y_1, \dots, Y_k\}$. We say K *witnesses* the pair (u, v) if $\exists i \in K$ with $u_i \neq v_i$.

For any fixed pair $u \neq v$, since u and v disagree on at least $n\epsilon^2$ coordinates,

$$\mathbb{P}(Y_1 \text{ does not witness } (u, v)) = 1 - \frac{|\{i : u_i \neq v_i\}|}{n} \leq 1 - \epsilon^2.$$

Since $Y_1, \dots, Y_k$ are independent,

$$\mathbb{P}(K \text{ does not witness } (u, v)) \leq (1 - \epsilon^2)^k \leq e^{-\epsilon^2 k}.$$

Applying a union bound over all $\binom{|V_\epsilon|}{2} \leq |V_\epsilon|^2/2$ pairs,

$$\mathbb{P}(\exists u \neq v \in V_\epsilon : K \text{ does not witness } (u, v)) \leq \frac{|V_\epsilon|^2}{2} e^{-\epsilon^2 k}.$$

Choosing $k = \left\lceil \frac{2 \log |V_\epsilon|}{\epsilon^2} \right\rceil$ makes the right-hand side at most $\frac{1}{2} < 1$. Hence there exists a deterministic realization of $K \subseteq [n]$ with $|K| \leq k$ that witnesses *every* pair in V_ϵ .

Corollary: If we project V_ϵ onto K no two of the projected vectors will be the same. Thus $|V_\epsilon|_K = |V_\epsilon|$.

When $k \geq d$, the Sauer-Shelah result implies

$$x = |V_\epsilon| \leq \left(\frac{ek}{d}\right)^d \leq \left(\frac{e2 \log(|V_\epsilon|)/\epsilon^2}{d}\right)^d \leq \left(\frac{ek}{d}\right)^d \leq \left(\frac{e2|V_\epsilon|^{1/(2d)}/(2d\epsilon^2)}{d}\right)^d$$

Thus

$$x \leq \left(\frac{e}{d^2\epsilon^2}\right)^{2d} \leq \left(\frac{c}{\epsilon}\right)^{4d}$$

since so $(e/d^2) \leq c$ for some constant $c > 0$.

If $k < d$ then

$$2 \log |V_\epsilon| / \epsilon^2 < d$$

so

$$|V|_\epsilon \leq e^{\epsilon^2 d/2} \leq e^{d/2} \leq (c/\epsilon)^{4d}$$

for c large enough (holds for all $\epsilon \in (0, 1)$).

■

Remark 6.23. The bound $\mathcal{N}(\mathcal{F}, e_n, \epsilon) \leq (c/\epsilon)^{4d}$ is remarkable for several reasons.

- The bound is *independent of n* : no matter how many sample points we use to define e_n , the covering number depends only on d and ϵ . This is surprising because e_n is a metric on an infinite function class.
- The dependence on ϵ mirrors the volumetric bound for covering a Euclidean ball in $\mathbb{R}^d$, where $\mathcal{N}(B^d, \|\cdot\|, \epsilon) \asymp (1/\epsilon)^d$. The extra factor of 4 in the exponent is the price of working with a combinatorial rather than a geometric notion of dimension.
- As a consequence, one can plug this into Dudley's entropy integral

$$\mathbb{E} \sup_{f \in \mathcal{F}} |G_n(f)| \lesssim \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, e_n, \epsilon)} d\epsilon \lesssim \sqrt{d}$$

to obtain $O(\sqrt{d/n})$ uniform convergence rates, recovering the classical VC generalization bound from a metric entropy perspective.

Corollary 6.24 (Uniform law of large numbers for VC classes). Let $\mathcal{F}$ be a class of binary functions with $\text{VC-dim}(\mathcal{F}) = d$, and let $X_1, \dots, X_n \stackrel{iid}{\sim} P$. Then

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right) \leq c' \sqrt{\frac{d}{n}}.$$

Proof. Step 1: Symmetrization. By the symmetrization inequality,

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_i f(X_i) - \mathbb{E} f(X) \right) \leq \mathbb{E}_X \mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_i \epsilon_i f(X_i),$$

where $\epsilon_1, \dots, \epsilon_n \stackrel{iid}{\sim} \text{Unif}(\{-1, +1\})$ are independent of the X_i . It now suffices to bound the inner Rademacher expectation for fixed $X_1, \dots, X_n$.

Step 2: Sub-Gaussian increments. Fix $X_1, \dots, X_n$ and consider the stochastic process $Z_f = \frac{1}{n} \sum_i \epsilon_i f(X_i)$ indexed by $f \in \mathcal{F}$, with randomness only in the ϵ_i . For any $f, g \in \mathcal{F}$,

$$Z_f - Z_g = \frac{1}{n} \sum_i \epsilon_i (f(X_i) - g(X_i)).$$

Claim: $\|\epsilon_i\|_{\psi_2} = \frac{1}{\sqrt{\log 2}}$. We have $\mathbb{E} \exp(\epsilon_i^2/t^2) = \exp(1/t^2) = 2$ when $t = \frac{1}{\sqrt{\log 2}}$ which proves the claim.

Since $\|\epsilon_i\|_{\psi_2} = \frac{1}{\sqrt{\log 2}}$ and the ϵ_i are independent, the sub-Gaussian triangle inequality gives

$$\|Z_f - Z_g\|_{\psi_2} \leq \frac{1}{n\sqrt{\log 2}} \sqrt{\sum_i (f(X_i) - g(X_i))^2} = \frac{1}{\sqrt{\log 2}} \cdot \frac{e_n(f, g)}{\sqrt{n}},$$

where $e_n(f, g) = \sqrt{\frac{1}{n} \sum_i (f(X_i) - g(X_i))^2}$ is the empirical L^2 metric. Thus (Z_f) is a sub-Gaussian process with respect to the scaled metric $d(f, g) := \frac{e_n(f, g)}{\sqrt{n}}$, up to an absolute constant.

Step 2.5: Generic Chaining. Since (Z_f) is a sub-Gaussian process with respect to d , the generic chaining upper bound gives $\mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} Z_f \lesssim \gamma_2(\mathcal{F}, d)$. We then bound γ_2 above by Dudley's entropy integral:

Step 3: Dudley's entropy integral. Since f, g are binary, the diameter of $\mathcal{F}$ under d satisfies $\text{diam}_d(\mathcal{F}) \leq \frac{1}{\sqrt{n}}$. Applying Dudley's bound and substituting $\epsilon = u\sqrt{n}$,

$$\begin{aligned} \mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} Z_f &\lesssim \int_0^{1/\sqrt{n}} \sqrt{\log \mathcal{N}(\mathcal{F}, d, u)} du \\ &= \int_0^{1/\sqrt{n}} \sqrt{\log \mathcal{N}(\mathcal{F}, e_n, u\sqrt{n})} du \\ &= \frac{1}{\sqrt{n}} \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, e_n, \epsilon)} d\epsilon. \end{aligned}$$

Step 4: Plug in the VC covering number bound. By the VC covering number lemma, $\mathcal{N}(\mathcal{F}, e_n, \epsilon) \leq (c/\epsilon)^{4d}$ uniformly in the sample points. Therefore

$$\int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, e_n, \epsilon)} d\epsilon \leq \int_0^1 \sqrt{4d \log(c/\epsilon)} d\epsilon = 2\sqrt{d} \int_0^1 \sqrt{\log(c/\epsilon)} d\epsilon \lesssim \sqrt{d},$$

where the last integral is a finite absolute constant (it is of the form $\int_0^1 \sqrt{\log(1/\epsilon)} d\epsilon < \infty$). Combining with Step 3,

$$\mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} Z_f \lesssim \sqrt{\frac{d}{n}}.$$

Taking the outer expectation over $X_1, \dots, X_n$ (the bound does not depend on the specific sample) completes the proof. $\blacksquare$

Remark 6.25 (Roadmap: from VC dimension to uniform convergence). The proof chains the following reductions:

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_i f(X_i) - \mathbb{E} f(X) \right) &\xrightarrow{\text{symmetrization}} \mathbb{E}_X \mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_i \epsilon_i f(X_i) \\ &\xrightarrow{\text{sub-Gaussian increments}} \frac{1}{\sqrt{n}} \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, e_n, \epsilon)} d\epsilon \\ &\xrightarrow{\text{Sauer-Shelah}} c \sqrt{d/n}. \end{aligned}$$

The second arrow converts a probabilistic problem into a geometric one. The VC structure enters only at the last step; Steps 1–2 hold for any sub-Gaussian process indexed by a metric space. Put into words:

1. **Symmetrization.** Replaces the empirical process $\frac{1}{n} \sum_i f(X_i) - \mathbb{E} f(X)$ with a Rademacher process $\frac{1}{n} \sum_i \epsilon_i f(X_i)$, freezing the sample points so that the only remaining randomness is in the signs ϵ_i . This reduces the problem to controlling a random process over a fixed index set.
2. **Sub-Gaussian increments.** The frozen-sample process (Z_f) has increments governed by the empirical L^2 metric e_n , making it a sub-Gaussian process. This is the bridge from probability to geometry: the problem now lives entirely in the metric space $(\mathcal{F}, e_n)$.
3. **Dudley's entropy integral.** The expected supremum of a sub-Gaussian process is controlled by the metric entropy of the index set at all scales:

$$\mathbb{E} \sup_{f \in \mathcal{F}} Z_f \lesssim \frac{1}{\sqrt{n}} \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, e_n, \epsilon)} d\epsilon.$$

The problem is now purely geometric: how complex is $\mathcal{F}$ under e_n ?

4. **VC dimension via Sauer-Shelah.** The VC covering number bound $\mathcal{N}(\mathcal{F}, e_n, \epsilon) \leq (c/\epsilon)^{4d}$ makes the entropy integral finite and of order $\sqrt{d}$, yielding the final rate $\sqrt{d/n}$.

Corollary 6.26 (1; The high probability version of the uniform law of large numbers for VC Classes). With probability at least $1 - \delta$

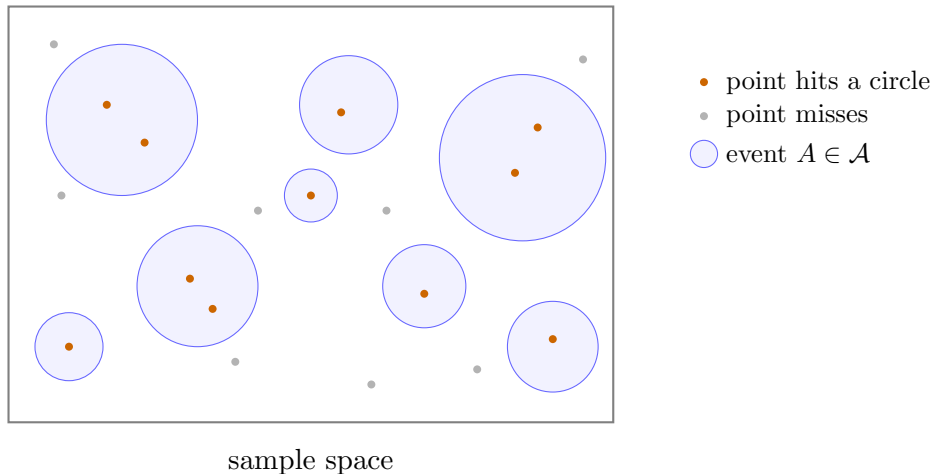
$$\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n (f(X_i) - \mathbb{E}f(X)) \right| \leq C \left(\sqrt{d/n} + \frac{\sqrt{\log(1/\delta)}}{n} \right)$$

Hint: apply the bounded differences inequality. Change one point in the sample X_i to X'_i and see how the increment changes. The $\sqrt{d/n}$ is the expectation and the $\log(1/\delta)/n$ is the sub-gaussian tail.

Example 6.27 (Uniform convergence for classes of sets). Let $\mathcal{A}$ be a class of events (e.g. all discs in $\mathbb{R}^2$) with $\text{VC-dim}(\mathcal{A}) = d$, and define the empirical measure $P_n(A) = \frac{1}{n} \sum_i \mathbf{1}_{X_i \in A}$. Applying the corollary with $f = \mathbf{1}_A$ gives: with probability at least $1 - \delta$,

$$\sup_{A \in \mathcal{A}} |P_n(A) - P(A)| \leq C \left(\sqrt{\frac{d}{n}} + \frac{\sqrt{\log(1/\delta)}}{n} \right).$$

Diagram: rectangle full of many disjoint circles of varying sizes. Add a bunch of random points in the rectangle. All circles have radius at least ϵ . If we want to construct a finite number of points so that every circle is hit by a point, can show the VC dimension of the circle set is finite. Put a uniform measure on the rectangle and sample m points at random. Simultaneously for all circles the number of points that hit a circle minus the true probability of hitting a circle (with radius r this is πr^2). If $m \geq \pi \epsilon^2 / 2$ then the empirical measure will not be zero with high probability. I.e. we will hit at least one circle.



Corollary 6.28 (DKW, Dvoretzky-Kiefer-Wolfowitz inequality). If $F(x)$ is a CDF, $F_n(X)$ -empirical cdf with probability $1 - \delta$

$$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \leq C \left(\sqrt{\frac{1}{n}} + \frac{\sqrt{\log(1/\delta)}}{n} \right) \leq C' \sqrt{\frac{\log(2\delta)}{n}}$$

Proof. Take $\mathcal{F} = \{u \mapsto \mathbf{1}\{y \leq x\} : x \in \mathbb{R}\}$. Then

$$\mathbb{E} \mathbf{1}\{X \leq x\} = F(x)$$

and

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{x_i \leq x\}$$

Recall that the VC-dimension of $\mathcal{F}$ is 1.

Now use the previous corollary with $d = 1$. ■

Remark 6.29. Sharp constant for DKW result. Massart (1990) For $\epsilon > 0$

$$\mathbb{P}(\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \geq \epsilon) \leq 2 \exp(-2n\epsilon^2)$$

Thus with prob at least $1 - \delta$

$$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \leq \sqrt{\frac{\log(2/\delta)}{2n}}$$

Corollary 6.30 (statistical learning theory). If $\mathcal{F}$ is a set of classifiers with VC-dim = d and we observe an iid sample $(X_i, Y_i)_{i=1}^n$; $Y_i \in \{0, 1\}$. We define the risk to be

$$R(f) = \mathbb{P}(f(X) \neq Y)$$

Recall this is what we get for risk under the 0/1 loss: $L(f(x), y) = \mathbf{1}\{f(x) \neq y\}$. The empirical risk is

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{f(X_i) \neq Y_i\}$$

We want to construct $\hat{f}$ based on the sample such that

$$R(\hat{f}) - R(f^*) = R(\hat{f}) - \inf_f R(f)$$

Remember that $f^\star$ is called the bayes predictor. We decide to have

$$\hat{f} = \arg \min_{f \in \mathcal{F}} R_n(f)$$

an empirical risk minimizer. Then

$$\begin{aligned} R(\hat{f}) - R(f^\star) &= R(\hat{f}) - R_n(\hat{f}) + R_n(\hat{f}) + R_n(f^\star) - R_n(f^\star) - R(f^\star) \\ &\leq R(\hat{f}) - R_n(\hat{f}) + R_n(f^\star) - R(f^\star) \\ &\leq 2 \sup_{f \in \mathcal{F}} |R(f) - R_n(f)| \\ &= 2 \sup_{f \in \mathcal{F}} \left| P(f(X) \neq Y) - \frac{1}{n} \sum_i \mathbf{1}\{f(X_i) \neq Y_i\} \right| \\ &\leq C(\sqrt{d/n} + \sqrt{\frac{\log(1/\delta)}{n}}) \end{aligned}$$

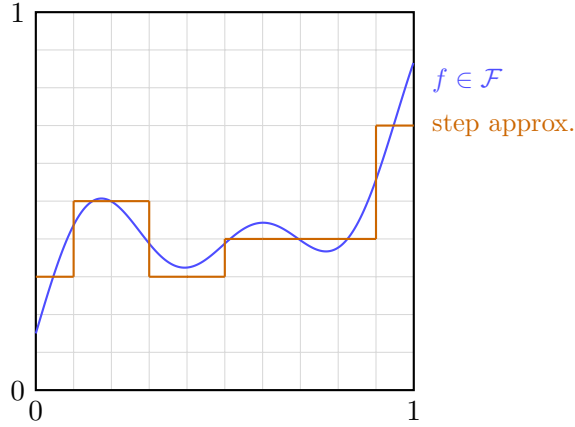
We relied on the claim that the VC dimension of a class $(x, y) \mapsto \mathbf{1}\{f(x) \neq y\}$ is eq to the VC dimension of $\mathcal{F}$. I'm a bit confused on this last step where the VC dimension gets added in.

Nikita ended by thinking about how this result would apply when the space of classifiers we are considering is the half-spaces (like support vector machines). In this case you know the VC dimension exactly.

Example 6.31. The class of 1-Lipschitz functions $[0, 1] \rightarrow [0, 1]$.

Call it $\mathcal{F}$. Seems like getting a grip on the covering number may be very difficult but actually we can do this.

Picture: unit square. over lay an even 10 x 10 grid. only mark 0 and 1 on the y axis and x axis. Lipschitz functions are not allowed to grow with slope larger than 1 anywhere. In each box on the grid, a function will stay in the boxes near where it is as it moves to the right. We can approximate these functions with step functions and the number of steps can be computed. Picture has a step function approximating the test function.



Let's work towards the covering number for this class of functions: $\mathcal{N}(\mathcal{F}, L_2(P_n), \epsilon)$ where

$$\|f - g\|_{L_2(P_n)} = \sqrt{\frac{1}{n} \sum_{i=1}^n (f(x_i) - g(x_i))^2} \leq \sup_{x \in [0,1]} |f(x) - g(x)| = \|f - g\|_{L_\infty}$$

The covering we choose does not have to be Lipschitz. We can go through our non-Lipschitz functions to approximate Lipschitz functions with the same number of Lipschitz functions. This is an application of the triangle inequality.

Make the grid size in the picture above have notches evenly spaced by $\epsilon/2$ increments. Then the number of boxes in the image is bounded by $\left(\frac{2}{\epsilon}\right)^{\frac{2}{\epsilon}}$. This is also a bound for the covering number. We will use this bound when the Dudley integral bound is used:

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E}(X) \right| &\leq \frac{2}{\sqrt{n}} \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n \epsilon_i (f(X_i) - \mathbb{E}(X)) \right| \\ &\leq \frac{C}{\sqrt{n}} \mathbb{E} \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(P_n)}, \epsilon)} d\epsilon \\ &\leq \frac{C}{\sqrt{n}} \mathbb{E} \int_0^1 \sqrt{\frac{2}{\epsilon} \log\left(\frac{2}{\epsilon}\right)} d\epsilon \\ &\leq \frac{C'}{\sqrt{n}} \end{aligned}$$

Improvement: If we construct a step function from left to right, we have at most $2/\epsilon$ choices for the starting height at $x = 0$. For each of the remaining $(2/\epsilon) - 1$ steps, the 1-Lipschitz constraint dictates that the function can only move up by one grid box, down by one grid box, or stay in the same box. This leaves at most 3 choices per step.

Thus, the number of such step paths (and our bound for the covering number) is roughly bounded by $\frac{2}{\epsilon} 3^{2/\epsilon}$. This improves the exponent's base from $2/\epsilon$ to a constant 3. We will use this tighter $O(1/\epsilon)$ entropy bound in the Dudley integral:

$$\begin{aligned}
\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right| &\leq \frac{2}{\sqrt{n}} \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n \epsilon_i (f(X_i) - \mathbb{E} f(X)) \right| \\
&\leq \frac{C}{\sqrt{n}} \mathbb{E} \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(P_n)}, \epsilon)} d\epsilon \\
&\leq \frac{C}{\sqrt{n}} \mathbb{E} \int_0^1 \sqrt{\log(2/\epsilon) + \frac{2}{\epsilon} \log 3} d\epsilon \\
&\leq \frac{C'}{\sqrt{n}}
\end{aligned}$$

6.3 Covering Numbers and unit convergence

In the VC case: $\mathcal{N}(\mathcal{F}, L_2(P_n), \epsilon) \leq (\frac{\epsilon}{\epsilon})^{4d}$.

In the Lipschitz case: $\log \mathcal{N}(\mathcal{F}, L_n(P_n), \epsilon) \leq (2/\epsilon) \log(2/\epsilon)$.

Definition 6.32 (Parametric Class of functions). A class of functions $\mathcal{F}$ is a parametric class of functions if

$$\sup_P \log \mathcal{N}(\mathcal{F}, L_2(P), \epsilon) \leq C \cdot p \cdot \log(1/\epsilon)$$

for some $p > 0$ which plays the role of dimension.

A class of functions $\mathcal{F}$ is a non-parametric class of functions if

$$\sup_P \log \mathcal{N}(\mathcal{F}, L_2(P), \epsilon) \leq C \cdot p \cdot (1/\epsilon)^p$$

for some $p > 0$ which plays the role of dimension.

We are taking the supremum over all measures P .

Proposition 6.33. For parametric classes:

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right| \leq \frac{C'}{\sqrt{n}} \int_0^1 p \log(1/\epsilon) d\epsilon \leq C'' \sqrt{\frac{p}{n}}$$

Non-parametric scenario. First consider $0 < p < 2$. Then

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right| \leq \frac{C'}{\sqrt{n}} \int_0^1 (1/\epsilon)^{p/2} d\epsilon = \frac{C_p}{\sqrt{n}}$$

Proposition 6.34. If $\mathcal{F}$ consists of $[0, 1]$ valued functions then

$$\mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \left(\sum_{i=1}^n f(X_i) - \mathbb{E}(X) \right) \right| \leq C \left(\eta + \frac{1}{\sqrt{n}} \int_{\eta/4}^1 \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), \epsilon)} d\epsilon \right)$$

Go back to the $p = 2$ case of the last proposition. Then using the result we just mentioned we get

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right| &\leq C \left(\eta + \frac{1}{\sqrt{n}} \int_{\eta/4}^1 \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), \epsilon)} d\epsilon \right) \\ &\leq C (\eta + \log(4/\eta)/\sqrt{n}) \\ &\leq C' \frac{\log n}{\sqrt{n}} \end{aligned}$$

Now for the $p > 2$ case:

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right| &\leq C \left(\eta + \frac{1}{\sqrt{n}} \int_{\eta/4}^1 \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), \epsilon)} d\epsilon \right) \\ &\leq C \left(\eta + \frac{1}{\sqrt{n}} \int_{\eta/4}^1 \left(\frac{1}{\epsilon} \right)^{p/2} d\epsilon \right) \\ &\leq C_p \left(\eta + \frac{1}{\sqrt{n}} (\eta/4)^{1-p/2} - \frac{1}{\sqrt{n}} \right) \end{aligned}$$

Optimizing over $\eta \in [0, 1]$ yields the bound (for $\eta = n^{-1/p}$)

$$\leq C'_p (n^{-1/p} + n^{-1/p}) = 2C'_p \frac{1}{n^{1/p}}$$

Thus as p growth the rate of convergence gets worse.

We call $p < 2$ the Donsker regime and $p \geq 2$ the non-Donsker regime.

7 Concentration Inequalities for Gaussian and Rademacher Processes

Consider a gaussian process indexed by $T \subseteq \mathbb{R}^d$ and $\phi_1, \dots, \phi_d$ are 1-Lipschitz functions:

$$\mathbb{E} \sup_{t \in T} \sum_{i=1}^n g_i \phi_i(t_i) \leq \mathbb{E} \sup_{t \in T} \sum_{i=1}^n g_i t_i$$

Proof. Sudakov-Fernique: Let $X_t = \sum_{i=1}^n g_i \phi_i(t_i)$; $Y_t = \sum_{i=1}^n g_i t_i$

Then

$$\mathbb{E}[X_t - X_s]^2 = \mathbb{E} \left(\sum_i g_i (\phi_i(t_i) - \phi_i(s_i))^2 \right) = \sum_i (\phi_i(t_i) - \phi_i(s_i))^2 \leq \sum_i (t_i - s_i)^2$$

Then the result follows by Sudakov-Fernique. This was discussed in these notes near Slepian's lemma. ■

Ledoux-Talagrand construction If $\phi_1, \dots, \phi_d$ are 1-Lipschitz mapping 0 to 0. then

$$\mathbb{E} \sup_{t \in T} \left| \sum_i \epsilon_i \phi_i(t_i) \right| \leq 2 \mathbb{E} \sup_{t \in T} \left| \sum_i \epsilon_i t_i \right|$$

Another prop where there are no absolute values: If $\phi_1, \dots, \phi_d$ are 1-Lipschitz functions then

$$\mathbb{E} \sup_{t \in T} \sum_i \epsilon_i \phi_i(t_i) \leq \mathbb{E} \sup_{t \in T} \sum_i \epsilon_i t_i$$

The proof of the claim which has the absolute value bars is harder than the one without (by a lot).

Proof. Write

$$\begin{aligned}
\mathbb{E} \sup_{t \in T} \sum_i \epsilon_i \phi_i(t_i) &= \frac{1}{2} \mathbb{E} \sup_{t \in T} \left(\sum_{i=1}^{d-1} \epsilon_i \phi_i(t_i) + \phi_d(t_d) \right) + \frac{1}{2} \mathbb{E} \left(\sup_{t \in T} \sum_{i=1}^{d-1} \epsilon_i \phi_i(t_i) - \phi_d(t_d) \right) \\
&= \frac{1}{2} \mathbb{E} \sup_{t, s \in T} \left(\sum_{i=1}^{d-1} \epsilon_i \phi_i(t_i) + \epsilon_i \phi_i(s_i) + \phi_d(t_d) - \phi_d(s_d) \right) \\
&\leq \frac{1}{2} \mathbb{E} \sup_{t, s \in T} \left(\sum_{i=1}^{d-1} \epsilon_i \phi_i(t_i) + \epsilon_i \phi_i(s_i) + |t_d - s_d| \right) \\
(\text{by symmetry}) &\leq \frac{1}{2} \mathbb{E} \sup_{t, s \in T} \left(\sum_{i=1}^{d-1} \epsilon_i \phi_i(t_i) + \epsilon_i \phi_i(s_i) + t_d - s_d \right) \\
&= \frac{1}{2} \mathbb{E} \sup_{t \in T} \left(\sum_{i=1}^{d-1} \epsilon_i \phi_i(t_i) + t_d \right) + \frac{1}{2} \mathbb{E} \sup_{t \in T} \left(\sum_{i=1}^{d-1} \epsilon_i \phi_i(t_i) - t_d \right) \\
&= \mathbb{E} \sup_{t \in T} \left(\sum_{i=1}^{d-1} \epsilon_i \phi_i(t_i) + \epsilon_d t_d \right)
\end{aligned}$$

Then iterate that procedure to attain the desired result.

The reason we could seemingly swap $\mathbb{E}$ with supremums here is because this particular expectation can be written as a finite sum of the ϵ_i s evaluated at their 2^n different configurations. ■

7.1 Application: Excess Risk in Statistical Learning

Setup

1. $\ell(\cdot, \cdot)$ is a loss function
2. $f(X)$ is a prediction
3. Y is a response
4. Loss of a prediction is measured with $\ell(f(X), Y)$
5. Risk is $\mathbb{E}_{X,Y} \ell(f(X), Y)$
6. Empirical risk of a sample $(X_1, Y_1), \dots, (X_n, Y_n)$ is $R_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(X_i), Y_i)$

Examples of loss

1. Square loss $\ell(f(X), Y) = (f(X) - Y)^2$. If $Y \in [0, 1]$ and $f \in [0, 1]$ then this function is 2-Lipschitz since

$$|\ell(u, Y) - \ell(w, Y)| = |u - w| \cdot |u - Y + w - Y| \leq 2|u - w|$$

2. Binary Loss $\ell(f(X), Y) = \mathbf{1}(f(X) \neq Y)$
3. Absolute loss $\ell(f(X), Y) = |f(X) - Y|$
4. Logistic loss $\ell(f(X), Y) = \log(1 + \exp(-Yf(X)))$

Given a collection of functions $\mathcal{F}$ define the empirical risk minimizer

$$\hat{f} = \arg \min_{f \in \mathcal{F}} R_n(f)$$

and this is a random object since the sample is random. The excess risk of $\hat{f}$ is

$$R(\hat{f}) - \inf_{f \in \mathcal{F}} R(f)$$

Let's think about the expected excess risk

$$\begin{aligned} \mathbb{E} \left(R(\hat{f}) - \inf_{f \in \mathcal{F}} R(f) \right) &= \mathbb{E}[R(\hat{f}) - R_n(\hat{f}) + R_n(\hat{f}) + R_n(f^*) - R_n(f^*) - R(f^*)] \\ &\leq \mathbb{E}[R(\hat{f}) - R_n(\hat{f}) + R_n(f^*) - R(f^*)] \\ &\leq 2\mathbb{E} \sup_{f \in \mathcal{F}} |R(f) - R_n(f)| \\ (\text{symmetrization step}) &\leq 4\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \epsilon_i \ell(f(X_i), Y_i) \right| \end{aligned}$$

Another route is

$$\begin{aligned} \mathbb{E}[R(\hat{f}) - R_n(\hat{f}) + R_n(f^*) - R(f^*)] &\leq \mathbb{E} \sup_{f \in \mathcal{F}} (R(f) - R_n(f)) \\ &\leq 2\mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i \ell(f(X_i), Y_i) \right) \end{aligned}$$

We now assume that $u \mapsto \ell(u, Y)$ is L -Lipschitz for all Y -s.

Then

$$2\mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i \ell(f(X_i), Y_i) \right) \leq 2L\mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i f(X_i) \right)$$

The term $\mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i f(X_i) \right)$ is the Rademacher complexity of $\mathcal{F}$ and it measures how well $\mathcal{F}$ can line up with the 2^n possible directions of the realizations of ϵ . More expressive class = larger rademacher complexity.

7.2 Rademacher Complexity

We just defined the Rademacher complexity above. We showed that it can be a way to bound the excess risk of the empirical risk minimizer.

Example 7.1 (bounding rademacher complexity in some cases).

1. If $\mathcal{F}$ is a class of binary functions with VC-dim = d then

$$\mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \epsilon_i f(x_i) \leq C \sqrt{d/n}$$

2. $\mathcal{F} = \{x \mapsto \langle x, w \rangle : w \in B_2^d\}$ then

$$\begin{aligned} \mathbb{E} \sup_{w \in B_2^d} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i \langle x_i, w \rangle \right) &= \frac{1}{\sqrt{n}} \mathbb{E} \sup_{w \in B_2^d} \left(\left\langle \sum_{i=1}^n \epsilon_i x_i / \sqrt{n}, w \right\rangle \right) \\ &= \frac{1}{\sqrt{n}} \mathbb{E} \left\| \sum_{i=1}^n \epsilon_i x_i / \sqrt{n} \right\|_2 \\ &\leq \frac{1}{\sqrt{n}} \sqrt{\mathbb{E} \left\| \sum_{i=1}^n \epsilon_i x_i / \sqrt{n} \right\|_2^2} \\ &\leq \frac{1}{\sqrt{n}} \sqrt{\sum_{i=1}^n \|x_i / \sqrt{n}\|_2^2} \\ &\leq \frac{1}{\sqrt{n}} \sqrt{\max_i \|x_i\|_2^2} \\ &= \frac{\max \|x_i\|_2}{\sqrt{n}} \end{aligned}$$

What if we do not want to use ϵ_i s? Maybe the X_i 's themselves have a nice distribution like gaussian.

Entropy with bracketing: given two functions f, g such that $f \leq g$ on a domain $\mathcal{X}$, we say the bracket $[f, g]$ consists of all such functions h such that

$$f(x) \leq h(x) \leq g(x)$$

for all $x \in \mathcal{X}$.

An ϵ -bracket with respect to $L_2(P)$ if

$$[f, g] : \|f - g\|_{L_2(P)} \leq \epsilon$$

Define

$$\mathcal{N}_{[]}(\mathcal{F}, L_2(P), \epsilon)$$

is the minimum number of ϵ -brackets with respect to $L_2(P)$ which cover f . The bracket $\mathcal{F}$ does not have to be in the model class $\mathcal{F}$.

Theorem 7.2. If $\mathcal{F}$ is a class of functions such that for all $f \in \mathcal{F}$, $\|f\|_{L_\infty(P)} \leq m$,

$$\mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f(X_i) - \mathbb{E}f(X)) \leq \frac{C}{\sqrt{n}} \int_0^m \sqrt{\log \mathcal{N}_{[]}(\mathcal{F}, L_2(P), \epsilon)} d\epsilon$$

can also write a version with absolute values. It is cool to compare this result with what you would get by using symmetrization + Dudley:

$$\mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f(X_i) - \mathbb{E}f(X)) \leq \frac{C}{\sqrt{n}} \mathbb{E}_{X_1, \dots, X_n} \int_0^{\sup_{x \in \mathcal{X}, f \in \mathcal{F}} |f(x)|} \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), \epsilon)} d\epsilon$$

The first version depends on the true measure P which can be a good thing in some cases giving a better bound than the second case where you are using the empirical measure.

7.3 Mean Estimation

Let $X_1, \dots, X_n$ be i.i.d. random variables in $\mathbb{R}$ with $\mathbb{E}[X_1] = \mu$. How can we estimate μ ?

If X_i is σ -subgaussian, meaning $\mathbb{E}[e^{\lambda(X_i - \mu)}] \leq e^{\lambda^2 \sigma^2 / 2}$ for all $\lambda \in \mathbb{R}$, then

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n (X_i - \mu)\right| \geq t\right) \leq \exp\left(-\frac{nt^2}{2\sigma^2}\right),$$

so with probability at least $1 - \delta$,

$$\left|\frac{1}{n} \sum_{i=1}^n (X_i - \mu)\right| \leq \sqrt{\frac{2\sigma^2 \log(1/\delta)}{n}},$$

a $\log(1/\delta)$ rate. In comparison, Chebyshev's inequality gives

$$\left|\frac{1}{n} \sum_{i=1}^n (X_i - \mu)\right| \leq \sqrt{\frac{\text{Var}(X_1)}{n\delta}},$$

which is polynomial in $1/\delta$.

Definition 7.3 (Median of Means Estimator). Given n data points, partition them uniformly at random into k blocks $B_1, \dots, B_k$ of size m , where $n = km$. (Independence between blocks is required for the proposition below.) Define the block means

$$\bar{X}_j = \frac{1}{m} \sum_{i \in B_j} X_i,$$

and let the *median-of-means* (MoM) estimator be

$$\hat{\mu} := \text{Median}(\bar{X}_1, \dots, \bar{X}_k).$$

Proposition 7.4. Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X_1) = \sigma^2$. Set $k = \lceil 8 \log(1/\delta) \rceil$ and $m = n/k$. Then the MoM estimator satisfies, with probability at least $1 - \delta$,

$$|\hat{\mu} - \mu| \leq C \sqrt{\frac{\sigma^2 \log(1/\delta)}{n}}.$$

Proof. Assume $n \geq 8 \log(1/\delta)$, so that $m \geq 1$. For each block B_j , Chebyshev's inequality gives

$$\mathbb{P} \left(\left| \frac{1}{m} \sum_{i \in B_j} (X_i - \mu) \right| \geq t \right) \leq \frac{\text{Var}(X_1)}{mt^2} = \frac{\sigma^2}{mt^2}.$$

Setting $t = 2\sqrt{\sigma^2/m}$ yields an upper bound of $\frac{1}{4}$, so with probability at least $\frac{3}{4}$,

$$|\bar{X}_j - \mu| \leq 2\sqrt{\frac{\sigma^2}{m}}.$$

”Good event for a block”

The probability that at least $K/2$ blocks, the good event did not happen, let Y be the number of bad-block events.. we have $\mathbb{E}Y = \frac{1K}{4}$ and by Hoeffding's inequality

$$\mathbb{P}(Y \geq K/2)$$

$$\begin{aligned} \mathbb{P}(Y \geq K/2) &= \mathbb{P}(Y - \frac{1}{4}K \geq K/2 - \frac{1}{4}K) \\ &\leq \exp\left(\frac{-2(K/4)^2}{K}\right) \\ &= \exp(-K/8) \\ &\leq \delta \end{aligned}$$

The number of good blocks is more than $K/2$ with probability at least $1 - \delta$. Thus

$$|\hat{\mu} - \mu| \leq 2\sqrt{\frac{\sigma^2}{m}} = 2\sqrt{\frac{\sigma^2}{m}} \leq C\sqrt{\frac{\sigma^2 \log(1/\delta)}{n}}$$

■

AI Summary: The MoM estimator achieves the same $\sqrt{\sigma^2 \log(1/\delta)/n}$ rate as the sample mean under subgaussian assumptions, yet requires only a finite variance assumption — a substantially weaker condition. This makes it attractive for heavy-tailed data where the sample mean can behave poorly. The primary drawback is practical: the random partitioning into blocks introduces additional randomness, the constant C is not sharp, and the estimator is rarely used in place of the sample mean in low-noise settings where finite variance is not a binding constraint. Extensions of the MoM idea to higher-dimensional settings (e.g., mean estimation in $\mathbb{R}^d$) are more technically involved and remain an active area of research.

7.4 Estimating the moments of high dimension distributions

If X is a random vector in $\mathbb{R}^d$ and $v \in S^{d-1}$. Assume $\mathbb{E}X = 0$. Let $p \geq 2$ be an integer. He wrote down $\mathcal{N}(\mu, \Sigma)$ on the side. We talked about how analyzing a specific direction $\langle X, v \rangle$ is not high dimensional problem. Getting a uniform bound in all directions is the hard part. Then

$$\mathbb{E} |\langle X, v \rangle|^p =: \|\langle X, v \rangle\|_{L_p}^p$$

Can we construct estimator for this quantity (moment) which is uniform over $v \in S^{d-1}$?

We assume that $\|\langle X, v \rangle\|_{L_{2p}}^{2p} < \infty$ (apparently this implies $\sup_{v \in S^{d-1}} \|\langle X, v \rangle\|_{L_{2p}}^{2p} < \infty$). (like a finite variance assumption).

Hypercontractivity assumption: for all $v \in S^{d-1}$ we have

$$\mathbb{E} |\langle X, v \rangle|^{2p} \leq L(\mathbb{E} |\langle X, v \rangle|^p)^2$$

We also wrote down

$$\|X\|_{L_p} \leq C_1 \sqrt{p} \|X\|_{\psi_2} \leq C_2 \sqrt{p} \text{Var}(X)$$

and this always holds for a Gaussian distribution.

Assume that p is even.

Proposition 7.5. Assume $X_1, \dots, X_n$ are iid in $\mathbb{R}^d$; $\mathbb{E}X_1 = 0$. Any hypercontractive: for all $v \in S^{d-1}$ we have

$$\mathbb{E} |\langle X, v \rangle|^{2p} \leq L(\mathbb{E} |\langle X, v \rangle|^p)^2$$

and p is even, then k is the number of blocks with $n = km$ and we want $k = C(d + \log(1/\delta))$. Put

$$\hat{\mu}_v = \text{median}\left(\frac{1}{m} \sum_{i \in D_j} |\langle x_i, v \rangle|^p\right)_{j=1}^k$$

Then with probability $1 - \delta$ simultaneously for all $v \in S^{d-1}$

$$|\hat{\mu}_v - \mathbb{E}|\langle X, v \rangle|^p| \leq C'_{L,p} \mathbb{E}|\langle X, v \rangle|^p \sqrt{\frac{d + \log(1/\delta)}{n}}$$

Cor: Assuming that $n \geq C''(d + \log(1/\delta))$ we have

$$\frac{1}{2} \mathbb{E}|\langle X, v \rangle|^p \leq \hat{\mu}_v \leq \frac{3}{2} \mathbb{E}|\langle X, v \rangle|^p$$

Proof. Fix $v \in S^{d-1}$ and a block B_j . Consider the probability the block is bad:

$$\begin{aligned} \mathbb{P}\left(\frac{1}{m} \sum_{i \in B_j} (|\langle X, v \rangle|^p - \mathbb{E}|\langle X, v \rangle|^p) \geq t\right) &\leq \frac{\mathbb{E}|\langle X, v \rangle|^{2p}}{mt^2} \\ &\leq \frac{L(\mathbb{E}|\langle X, v \rangle|^p)^2}{mt^2} \end{aligned}$$

Want to have $\frac{L(\mathbb{E}|\langle X, v \rangle|^p)^2}{mt^2} = \frac{1}{8}$ so $t = \frac{2\sqrt{2}\sqrt{L}\mathbb{E}|\langle X, v \rangle|^p}{\sqrt{m}}$.

Consider the class of binary random variables:

$$f_{v, B_j} = \mathbf{1}[\text{Good Event}(v, B_j)] = \mathbf{1}\left\{\frac{1}{m} \sum_{i \in B_j} (|\langle X, v \rangle|^p - \mathbb{E}|\langle X, v \rangle|^p) \leq \frac{2\sqrt{2}\sqrt{L}\mathbb{E}|\langle X, v \rangle|^p}{\sqrt{m}}\right\}$$

Then $\mathbb{E}[f_v, B_j] \geq \frac{7}{8}$. The MoM- μ is "OK" when for all v (simultaneously) is at least $k/2$

$$\sum_{j=1}^K f_{v, B_j} > k/2$$

Bad event $\sum_{j=1}^K f_{v, B_j} \leq k/2$

$$\sum_{j=1}^K (f_{v, B_j} - \mathbb{E}f_{v, B_j}) \leq k/2 - \frac{7}{8}k = -3/8k$$

Warren's Lemma The VC-dim of the binary class induced by polynomials of degree at most p of d variables is at most $Cd \log(ep)$.

If $y_1, \dots$ are the coef ; $v_1, \dots$ - are the variables

$$\mathbf{1}[v_1^p y_1 + v_2^p y_2 + \dots + v_1^{p-1} v_2 y_j + \dots + \dots + y_n \geq 0]$$

(think of splitting a space above and below the polynomial instead of a plane like we did in SVM). Then since p is even

$$|\langle X, v \rangle|^p - \mathbb{E} |\langle X, v \rangle|^p$$

can be expanded to see that it is a polynomial of degree at most p of variables $v_1, \dots, v_d$. By Warren's lemma and the above observation uniformly over $v \in S^{d-1}$ we have

$$\left| \sum_{j=1}^k (f_{v, B_j} - \mathbb{E} f_{v, B_j}) \right| \leq C \left(\sqrt{\frac{Cd \log(ep) + \log(1/\delta)}{k}} \right) k = C' \sqrt{kCd \log(ep) + \log(1/\delta)}$$

■

8 Non Parametric Least Squares and Localization

Consider $X_1, \dots, X_n \in \mathcal{X}$ to be fixed (non-random design). Then

$$Y_i = f^*(X_i) + \sigma w_i$$

with $w_1, \dots, w_n \stackrel{\text{iid}}{\sim} N(0, 1)$. Assume $f^* \in \mathcal{F}$ where $\mathcal{F}$ is the function class / model class. In non-parametric regression $\mathcal{F}$ is very large.

For any $g : \mathcal{X} \rightarrow \mathbb{R}$, we define

$$\|g\|_n^2 = \frac{1}{n} \sum_{i=1}^n g(x_i)^2$$

For $f : \mathcal{X} \rightarrow \mathbb{R}$ we then have

$$\|Y - f\|_n^2 = \frac{1}{n} \sum_{i=1}^N (Y_i - f(X_i))^2$$

Assume that

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \|Y - f\|_n^2.$$

Symmetrization step:

$$\frac{1}{2} \left\| \hat{f} - f \right\|_n^2 \leq \frac{\sigma}{n} \sum_{i=1}^n w_i (\hat{f}(X_i) - f^*(X_i))$$

Proof:

$$\left\| Y - \hat{f} \right\|_n^2 \leq \left\| Y - f^* \right\|_n^2$$

and also use

$$Y - \hat{f} = (f^* - \hat{f}) + \sigma w$$

Definition 8.1. $\mathcal{F}$ is star shaped around f^* if

$$f^* + \alpha(f - f^*) \in \mathcal{F} \quad \forall f \in \mathcal{F}, \alpha \in [0, 1]$$

If $\mathcal{F}$ is star shaped around f^* then $\mathcal{F}^* = \mathcal{F} - \{f^*\}$ is star shaped around 0.

Definition 8.2 (localized gaussian complexity).

$$G_n(\delta, \mathcal{F}^*) = \mathbb{E}_w \left(\sup_{g \in \mathcal{F}^*, \|g\|_n \leq \delta} \frac{1}{n} \sum_{i=1}^n w_i g(x_i) \right)$$

Lemma 8.3. If $\mathcal{F}^*$ is star shaped then

$$\delta \mapsto \frac{G_n(\delta; \mathcal{F}^*)}{\delta}$$

is non increasing (or non-decreasing one of the two) The set

$$\{\delta > 0 : \frac{G_n(\delta; \mathcal{F}^*)}{\delta} \leq \frac{\delta}{2\sigma}\}$$

is non-empty.

Proof. $0 < \delta \leq t$. Fix $g \in \mathcal{F}^*$

he wrote this down too fast and i didn't get it loll. ask abdullah ■

Definition 8.4 (valid radius). We call δ_n a valid radius if

$$\frac{G_n(\delta_n; \mathcal{F}^*)}{\delta_n} \leq \frac{\delta_n}{2\sigma}$$

If δ_n is valid then all $\delta \geq \delta_n$ are valid since G_n is non-increasing in delta.

8.1 next section

Theorem 8.5. Assume $\mathcal{F}^*$ is star shaped and δ_n is a valid radius then

$$\mathbb{P}(\|\hat{f} - f^*\|_n^2 \geq 16t\delta_n) \leq \exp\left(-\frac{nt\delta_n}{2\sigma^2}\right)$$

Proof. Define

$$\hat{\Delta} = \hat{f} - f^* \in \mathcal{F}^*$$

Then

$$\frac{1}{2} \|\Delta\|_2^2 \leq \sigma/n \sum_{i=1}^n w_i \hat{\Delta}(x_i) \stackrel{?}{\asymp} \|\hat{\Delta}\|_2$$

Fix $u \geq \delta_n$. Then define

$$A(u) = \{\exists g \in \mathcal{F}^* \text{ such that } \|g\|_n \geq u \text{ and } \left| \frac{\sigma}{n} \sum_i w_i g(x_i) \right| \geq 2u \|g\|_n\}$$

Then

$$\mathbb{P}(A(u)) \leq \exp\left(-\frac{nu^2}{2\sigma^2}\right)$$

Define

$$Z_n(u) = \sup_{g \in \mathcal{F}^*, \|g\|_n \leq u} \left| \frac{\sigma}{n} \sum_i w_i g(x_i) \right|$$

When $A(u)$ occurs, we find $g \in \mathcal{F}^*$ with $\|g\|_n \geq u$ and $\left| \frac{\sigma}{n} \sum_i w_i g(x_i) \right| \geq 2u \|g\|_n$. Define

$$\tilde{g} = \frac{u}{\|g\|_n} g$$

so $\tilde{g} \in \mathcal{F}^*$ and $\|\tilde{g}\|_n = u$. So

$$Z_n(u) \geq \left| \frac{\sigma}{n} \sum_i w_i \tilde{g}(x_i) \right| = \frac{u}{\|g\|_n} \left| \frac{\sigma}{n} \sum_i w_i g(x_i) \right| \geq 2u^2$$

Thus

$$A(u) \subseteq \{Z_n(u) \geq 2u^2\}$$

Write Z_n as a function of its u and the $w = (w_1, \dots, w_n)$. We have

$$\begin{aligned} |Z_n(u; w) - Z_n(u; w')| &\leq \sup_{g \in \mathcal{F}^*, \|g\|_n \leq u} \left| \frac{\sigma}{n} \sum_i (w_i - w'_i) g(x_i) \right| \\ &\leq \frac{\sigma}{n} \|w - w'\|_2 \sup \left(\sum_i g(x_i)^2 \right)^{1/2} \\ &= \frac{\sigma}{\sqrt{n}} \|w - w'\|_2 \sup \|g\|_n \\ &\leq \frac{\sigma u}{\sqrt{n}} \|w - w'\|_2 \mathbb{P}(Z_n(u) \geq \mathbb{E}Z_n(u) + \delta) \\ &\leq \exp\left(-\frac{n\delta^2}{2\sigma^2 u^2}\right) \end{aligned}$$

where $\mathbb{E}Z_n(u) = \sigma G_n(u; \mathcal{F}^*)$ and here $\delta > 0$.

$u \geq \delta_n$ since δ_n is valid. Then (idk where this came from ngl)

$$\frac{G_n(u; \mathcal{F}^*)}{u} \leq \frac{G_n(\delta_n; \mathcal{F}^*)}{\delta_n} \leq \frac{\delta_n}{2\sigma}$$

$$\mathbb{E}Z_n(u) \leq \frac{u\delta_n}{2} \leq \frac{u^2}{2}$$

Then

$$\mathbb{P}(A(u)) \leq \mathbb{P}(Z_n(u) \geq 2u^2) \leq \mathbb{P}(Z_n(u) \geq \mathbb{E}Z_n(u) + \frac{\delta}{2}u^2) \leq \exp\left(-\frac{nu^2}{s\sigma^2}\right)$$

Fix $t \geq \delta_n$ and $u = \sqrt{t\delta_n}$. We have

$$\mathbb{P}(A(u)) \leq \exp\left(-\frac{nt\delta_n}{2\sigma^2}\right)$$

Two cases to consider 1: $\|\hat{\Delta}\|_n < u$ then $\|\hat{\Delta}\|_n^2 < u^2 \leq 16u^2$ 2: $\|\hat{\Delta}\|_n > u$ then $\hat{\Delta} \in \mathcal{F}^*$. Then

$$\left| \frac{\sigma}{n} \sum_i w_i \hat{\Delta}(x_i) \right| < 2u \|\hat{\Delta}\|_n$$

and

$$\begin{aligned} \frac{1}{2} \|\hat{\Delta}\|_n^2 &\leq \left| \frac{\sigma}{n} \sum_i w_i \hat{\Delta}(x_i) \right| < 2u \|\hat{\Delta}\|_n \\ \hat{\Delta}_n &< 4u \end{aligned}$$

■

Corollary 8.6. turn the last theorem into a $1 - \delta$ high probability statement:

AI Rewrite:

9 Nonparametric Least Squares and Localization

9.1 Setup

We study the fixed-design nonparametric regression model. Let $X_1, \dots, X_n \in \mathcal{X}$ be deterministic design points, and suppose we observe

$$Y_i = f^*(X_i) + \sigma w_i, \quad w_1, \dots, w_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1),$$

where $f^* \in \mathcal{F}$ for some function class $\mathcal{F} \subseteq \{f : \mathcal{X} \rightarrow \mathbb{R}\}$. In the nonparametric setting, $\mathcal{F}$ is typically very rich (infinite-dimensional), so classical parametric rates no longer apply.

For any $g : \mathcal{X} \rightarrow \mathbb{R}$, define the empirical L^2 seminorm

$$\|g\|_n^2 = \frac{1}{n} \sum_{i=1}^n g(X_i)^2,$$

and for a candidate $f : \mathcal{X} \rightarrow \mathbb{R}$, the empirical risk is

$$\|Y - f\|_n^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2.$$

The least squares estimator (LSE) is any minimizer

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \|Y - f\|_n^2.$$

Our goal is to bound the in-sample error $\|\hat{f} - f^*\|_n^2$.

9.2 The Basic Inequality (Symmetrization Step)

Lemma 9.1 (Basic inequality). The least square's estimator satisfies

$$\frac{1}{2} \left\| \hat{f} - f^* \right\|_n^2 \leq \frac{\sigma}{n} \sum_{i=1}^n w_i \left(\hat{f}(X_i) - f^*(X_i) \right).$$

Proof. By optimality of $\hat{f}$ and since $f^* \in \mathcal{F}$,

$$\left\| Y - \hat{f} \right\|_n^2 \leq \left\| Y - f^* \right\|_n^2.$$

Since $Y_i = f^*(X_i) + \sigma w_i$, we have pointwise

$$Y_i - \hat{f}(X_i) = \left(f^*(X_i) - \hat{f}(X_i) \right) + \sigma w_i$$

and

$$Y_i - f^*(X_i) = \sigma w_i$$

Substituting these two lines into the previous expression we have,

$$\left\| \left(f^* - \hat{f} \right) + \sigma w \right\|_n^2 \leq \left\| \sigma w \right\|_n^2.$$

Expanding the left side via $\|a + b\|_n^2 = \|a\|_n^2 + \frac{2}{n} \sum_i a(X_i)b(X_i) + \|b\|_n^2$,

$$\left\| f^* - \hat{f} \right\|_n^2 + \frac{2\sigma}{n} \sum_{i=1}^n w_i \left(f^*(X_i) - \hat{f}(X_i) \right) + \left\| \sigma w \right\|_n^2 \leq \left\| \sigma w \right\|_n^2.$$

$$\left\| \hat{f} - f^* \right\|_n^2 \leq \frac{2\sigma}{n} \sum_{i=1}^n w_i \left(\hat{f}(X_i) - f^*(X_i) \right).$$

Canceling $\left\| \sigma w \right\|_n^2$ and rearranging gives,

$$\left\| \hat{f} - f^* \right\|_n^2 \leq \frac{2\sigma}{n} \sum_{i=1}^n w_i \left(\hat{f}(X_i) - f^*(X_i) \right).$$

Dividing by 2 gives the claim. ■

Remark 9.2 (Intuition). The basic inequality is the workhorse of empirical risk minimization analysis. The left side is what we want to control (how well $\hat{f}$ estimates f^* in the empirical seminorm). The right side is a *noise-signal interaction*: it measures how well the noise w correlates with the error direction $\hat{f} - f^*$. Because $\hat{f}$ is random and depends on w , we cannot just take expectations — we need to control this quantity uniformly over all plausible error directions $g = \hat{f} - f^*$. This is what motivates introducing a supremum over a localized class.

9.3 Star-Shaped Classes and Localization

Definition 9.3 (Star-shaped class). $\mathcal{F}$ is *star-shaped around* f^* if

$$f^* + \alpha(f - f^*) \in \mathcal{F} \quad \text{for all } f \in \mathcal{F}, \alpha \in [0, 1].$$

Equivalently, setting $\mathcal{F}^* := \mathcal{F} - \{f^*\} = \{f - f^* : f \in \mathcal{F}\}$, the class $\mathcal{F}$ is star-shaped around f^* iff $\mathcal{F}^*$ is star-shaped around 0, i.e. $\alpha g \in \mathcal{F}^*$ for all $g \in \mathcal{F}^*$ and $\alpha \in [0, 1]$.

Remark 9.4 (Why star-shaped?). Convex classes are star-shaped around every point, so this is a strictly weaker assumption than convexity. Star-shapedness is exactly what we need to “shrink” elements toward 0: given any $g \in \mathcal{F}^*$, we can rescale it to smaller empirical norm and stay in the class. This rescaling is the core trick that makes the localization argument work.

Definition 9.5 (Localized class). For $\delta > 0$, define the *localized class*

$$T_n(\delta) := \{g \in \mathcal{F}^* : \|g\|_n \leq \delta\}.$$

Definition 9.6 (Localized Gaussian complexity). For $\delta > 0$, the *localized Gaussian complexity* of $\mathcal{F}^*$ is

$$G_n(\delta; \mathcal{F}^*) = \mathbb{E}_w \left(\sup_{g \in T_n(\delta)} \frac{1}{n} \sum_{i=1}^n w_i g(X_i) \right).$$

April 21 update: You can also work with

$$G_n(\delta; \mathcal{F}^*) = \mathbb{E}_w \left(\sup_{g \in T_n(\delta)} \left| \frac{1}{n} \sum_{i=1}^n w_i g(X_i) \right| \right).$$

and the arguments do not change that much.

This measures the richness of the class $\mathcal{F}^\star$ *restricted to a small empirical ball* of radius δ . The reason we localize: only functions g close to 0 (i.e. functions close to $f^\star$) matter for bounding $\hat{f} - f^\star$, since a good estimator will produce $\hat{f}$ close to $f^\star$.

Lemma 9.7 (Sub-linearity of localized complexity). If $\mathcal{F}^\star$ is star-shaped around 0, then the map

$$\delta \longmapsto \frac{G_n(\delta; \mathcal{F}^\star)}{\delta}$$

is *non-increasing* on $(0, \infty)$. Consequently, the set

$$\left\{ \delta > 0 : \frac{G_n(\delta; \mathcal{F}^\star)}{\delta} \leq \frac{\delta}{2\sigma} \right\}$$

is non-empty (and is of the form $[\delta_n, \infty)$).

Proof. Fix $0 < \delta \leq t$. We will show $G_n(\delta; \mathcal{F}^\star)/\delta \geq G_n(t; \mathcal{F}^\star)/t$. The argument has three steps: (i) rescale each function in the t -ball into the δ -ball; (ii) pass to the supremum over the t -ball; (iii) take expectation.

Step (i): Rescaling. Fix any $g \in \mathcal{F}^\star$ with $\|g\|_n \leq t$. Let $\alpha = \delta/t \in (0, 1]$ and define $\tilde{g} = \alpha g$. Since $\mathcal{F}^\star$ is star-shaped around 0, we have $\tilde{g} \in \mathcal{F}^\star$, and

$$\|\tilde{g}\|_n = \alpha \|g\|_n \leq \alpha t = \delta,$$

so $\tilde{g}$ lies in the δ -ball of $\mathcal{F}^\star$. By linearity,

$$\frac{1}{n} \sum_{i=1}^n w_i \tilde{g}(X_i) = \alpha \cdot \frac{1}{n} \sum_{i=1}^n w_i g(X_i) = \frac{\delta}{t} \cdot \frac{1}{n} \sum_{i=1}^n w_i g(X_i). \quad (\star)$$

Step (ii): Passing to the supremum. Because $\tilde{g}$ is a feasible candidate for the supremum over the δ -ball,

$$\frac{1}{n} \sum_{i=1}^n w_i \tilde{g}(X_i) \leq \sup_{\substack{h \in \mathcal{F}^\star \\ \|h\|_n \leq \delta}} \frac{1}{n} \sum_{i=1}^n w_i h(X_i).$$

Combining with $(\star)$,

$$\frac{\delta}{t} \cdot \frac{1}{n} \sum_{i=1}^n w_i g(X_i) \leq \sup_{\substack{h \in \mathcal{F}^\star \\ \|h\|_n \leq \delta}} \frac{1}{n} \sum_{i=1}^n w_i h(X_i). \quad (\star\star)$$

Now the key observation: the right-hand side of $(\star\star)$ *does not depend on g* . So $(\star\star)$ is a bound that holds uniformly for every g in the t -ball, with a fixed right-hand side. We may therefore take the supremum over all such g on the left:

$$\sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq t}} \left(\frac{\delta}{t} \cdot \frac{1}{n} \sum_{i=1}^n w_i g(X_i) \right) \leq \sup_{\substack{h \in \mathcal{F}^\star \\ \|h\|_n \leq \delta}} \frac{1}{n} \sum_{i=1}^n w_i h(X_i).$$

Since $\delta/t > 0$ is a positive scalar not depending on g , it commutes with the supremum:

$$\frac{\delta}{t} \cdot \sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq t}} \frac{1}{n} \sum_{i=1}^n w_i g(X_i) \leq \sup_{\substack{h \in \mathcal{F}^\star \\ \|h\|_n \leq \delta}} \frac{1}{n} \sum_{i=1}^n w_i h(X_i).$$

Step (iii): Expectation. This inequality is a pointwise inequality in $w = (w_1, \dots, w_n)$. Taking $\mathbb{E}_w$ on both sides (expectation preserves the inequality) and using the definition of the localized Gaussian complexity,

$$\frac{\delta}{t} \cdot G_n(t; \mathcal{F}^\star) \leq G_n(\delta; \mathcal{F}^\star).$$

Dividing through by $\delta > 0$,

$$\frac{G_n(t; \mathcal{F}^\star)}{t} \leq \frac{G_n(\delta; \mathcal{F}^\star)}{\delta},$$

which is precisely the statement that $\delta \mapsto G_n(\delta; \mathcal{F}^\star)/\delta$ is non-increasing.

Non-emptiness of the set. The function $\delta \mapsto G_n(\delta; \mathcal{F}^\star)/\delta$ is non-increasing by the above, while $\delta \mapsto \delta/(2\sigma)$ is strictly increasing from 0. For small δ the left side can be large, but as $\delta \rightarrow \infty$ we have $G_n(\delta; \mathcal{F}^\star)/\delta$ remains bounded by the non-increasing property (under mild boundedness of $\mathcal{F}^\star$ – an interesting comment for the LLM to add but I'm not sure exactly the purpose?? Is there a case where $G(\delta, \mathcal{F})$ is infinite for all δ ?) while $\delta/(2\sigma) \rightarrow \infty$. The two curves must therefore cross, and the crossing defines the smallest valid δ_n . The set of valid radii is a half-line $[\delta_n, \infty)$. ■

Remark 9.8 (Signal vs. Noise and the Critical Radius). Return to the basic inequality evaluated at the error $\hat{\Delta} = \hat{f} - f^\star$:

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \frac{\sigma}{n} \sum_{i=1}^n w_i \hat{\Delta}(X_i).$$

The left side, $\|\hat{\Delta}\|_n^2/2$, acts as the *signal term*. It is a purely quadratic penalty measuring how far our estimator is from the truth; in the noiseless case ($w = 0$), it forces $\hat{f} = f^\star$ exactly.

The right side is the *noise term*, representing the alignment of the error direction with the random Gaussian noise. If we restrict our attention to a sphere where $\|\hat{\Delta}\|_n \leq \delta$, the worst-case realized noise is bounded by the supremum $\sigma \sup_{g \in T_n(\delta)} \frac{1}{n} \sum_{i=1}^n w_i g(X_i)$. Taking the expectation of this supremum over the noise w gives precisely the localized Gaussian complexity, $\sigma G_n(\delta; \mathcal{F}^*)$.

This reveals a fundamental race between signal and noise as the radius δ grows. The signal penalty grows quadratically ($\delta^2/2$). The expected worst-case noise $\sigma G_n(\delta; \mathcal{F}^*)$, however, grows at most linearly (because $\delta \mapsto G_n(\delta)/\delta$ is non-increasing by Lemma 9.7). For very small δ , the expected noise dominates. But because quadratic growth eventually overtakes linear growth, there must be a crossing point where the signal definitively overpowers the expected noise:

$$\sigma G_n(\delta_n; \mathcal{F}^*) \leq \frac{\delta_n^2}{2}.$$

The *critical radius* δ_n is the smallest δ where this holds. Because the basic inequality *must* hold, this crossing point dictates the maximum possible size of our error. If the actual error $\|\hat{\Delta}\|_n$ were strictly greater than δ_n , the quadratic signal term would exceed the maximum available noise, creating a contradiction. Thus, the basic inequality forces the actual error to be bounded by δ_n (with high probability, once we account for the concentration of the realized noise around its expectation). The *critical radius* δ_n is the smallest δ where this holds. Beyond δ_n , the expected noise is too weak to satisfy the basic inequality. Through concentration of measure, one can then show that the realized noise tightly concentrates around its expectation, forcing the actual error $\|\hat{\Delta}\|_n$ to be bounded by δ_n with high probability.

This is just the vibes of the proof / strategy we will use. Not the full proof.

Definition 9.9 (Valid radius). We call $\delta_n > 0$ a *valid radius* (or *critical radius*) if

$$\frac{G_n(\delta_n; \mathcal{F}^*)}{\delta_n} \leq \frac{\delta_n}{2\sigma}, \quad \text{equivalently} \quad G_n(\delta_n; \mathcal{F}^*) \leq \frac{\delta_n^2}{2\sigma}.$$

By Lemma 9.7, if δ_n is valid then every $\delta \geq \delta_n$ is also valid.

9.4 Main Theorem: High-Probability Rate

Theorem 9.10 (Rate for the LSE). Assume $\mathcal{F}$ is star-shaped around f^* and δ_n is a valid radius. Then for every $t \geq 1$,

$$\mathbb{P} \left(\left\| \hat{f} - f^* \right\|_n^2 \geq 16 t \delta_n^2 \right) \leq \exp \left(-\frac{n t \delta_n^2}{2\sigma^2} \right).$$

Remark 9.11 (Statement fixes). The original statement had $16t\delta_n$ inside the probability and $nt\delta_n/(2\sigma^2)$ in the exponent. The correct dimensions require δ_n^2 in both places: $\|\hat{f} - f^\star\|_n^2$ has the units of δ_n^2 , and the exponent must be dimensionless once multiplied by n/σ^2 . The fix is consistent with the proof below, where we choose $u = \sqrt{t}\delta_n$ and the deviation scales as $u^2 = t\delta_n^2$.

Proof. Let $\hat{\Delta} := \hat{f} - f^\star$. Since $\mathcal{F}$ is star-shaped around $f^\star$, we have $\hat{\Delta} \in \mathcal{F}^\star$. By the basic inequality (Lemma 9.1),

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \frac{\sigma}{n} \sum_{i=1}^n w_i \hat{\Delta}(X_i). \quad (9.12)$$

Step 1: Bad event and supremum process. For $u \geq \delta_n$, define the “bad event”

$$A(u) = \left\{ \exists g \in \mathcal{F}^\star : \|g\|_n \geq u \text{ and } \left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(X_i) \right| \geq 2u \|g\|_n \right\},$$

and the localized supremum

$$Z_n(u) = \sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq u}} \left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(X_i) \right|.$$

Step 2: Reduction $A(u) \subseteq \{Z_n(u) \geq 2u^2\}$. Suppose $A(u)$ occurs, witnessed by some $g \in \mathcal{F}^\star$ with $\|g\|_n \geq u$ and $|\frac{\sigma}{n} \sum_i w_i g(X_i)| \geq 2u \|g\|_n$. Define the rescaled function

$$\tilde{g} = \frac{u}{\|g\|_n} g.$$

Since $u/\|g\|_n \in (0, 1]$ and $\mathcal{F}^\star$ is star-shaped around 0, we have $\tilde{g} \in \mathcal{F}^\star$, with $\|\tilde{g}\|_n = u$. Therefore $\tilde{g}$ is a valid candidate in the supremum defining $Z_n(u)$, and

$$Z_n(u) \geq \left| \frac{\sigma}{n} \sum_i w_i \tilde{g}(X_i) \right| = \frac{u}{\|g\|_n} \left| \frac{\sigma}{n} \sum_i w_i g(X_i) \right| \geq \frac{u}{\|g\|_n} \cdot 2u \|g\|_n = 2u^2.$$

Hence $A(u) \subseteq \{Z_n(u) \geq 2u^2\}$.

Step 3: Concentration of $Z_n(u)$. We view $Z_n(u) = Z_n(u; w)$ as a function of the Gaussian vector $w = (w_1, \dots, w_n)$. For any $w, w' \in \mathbb{R}^n$,

$$\begin{aligned}
|Z_n(u; w) - Z_n(u; w')| &\leq \sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq u}} \left| \frac{\sigma}{n} \sum_i (w_i - w'_i) g(X_i) \right| \\
&\stackrel{(a)}{\leq} \frac{\sigma}{n} \|w - w'\|_2 \cdot \sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq u}} \left(\sum_i g(X_i)^2 \right)^{1/2} \quad (\text{Cauchy-Schwarz}) \\
&= \frac{\sigma}{\sqrt{n}} \|w - w'\|_2 \cdot \sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq u}} \|g\|_n \quad (\text{defn. of } \|\cdot\|_n) \\
&\leq \frac{\sigma u}{\sqrt{n}} \|w - w'\|_2.
\end{aligned}$$

So $w \mapsto Z_n(u; w)$ is Lipschitz with constant $L = \sigma u / \sqrt{n}$. By Gaussian concentration for Lipschitz functions,

$$\mathbb{P}(Z_n(u) \geq \mathbb{E}Z_n(u) + r) \leq \exp\left(-\frac{r^2}{2L^2}\right) = \exp\left(-\frac{nr^2}{2\sigma^2 u^2}\right), \quad r > 0.$$

Step 4: Bounding $\mathbb{E}Z_n(u)$ via the valid radius. By definition, $\mathbb{E}Z_n(u) = \sigma G_n(u; \mathcal{F}^\star)$. Since $u \geq \delta_n$ and $\delta \mapsto G_n(\delta)/\delta$ is non-increasing (Lemma 9.7),

$$\frac{G_n(u; \mathcal{F}^\star)}{u} \leq \frac{G_n(\delta_n; \mathcal{F}^\star)}{\delta_n} \leq \frac{\delta_n}{2\sigma}.$$

Therefore

$$\mathbb{E}Z_n(u) = \sigma G_n(u; \mathcal{F}^\star) \leq \sigma \cdot u \cdot \frac{\delta_n}{2\sigma} = \frac{u \delta_n}{2} \leq \frac{u^2}{2},$$

where the last inequality uses $\delta_n \leq u$.

Step 5: Combining. Using Steps 2–4 with deviation $r = 3u^2/2$,

$$\begin{aligned}
\mathbb{P}(A(u)) &\leq \mathbb{P}(Z_n(u) \geq 2u^2) \\
&\leq \mathbb{P}(Z_n(u) \geq \mathbb{E}Z_n(u) + \tfrac{3}{2}u^2) \\
&\leq \exp\left(-\frac{n(3u^2/2)^2}{2\sigma^2 u^2}\right) = \exp\left(-\frac{9nu^2}{8\sigma^2}\right) \leq \exp\left(-\frac{nu^2}{2\sigma^2}\right).
\end{aligned}$$

Step 6: Plugging in $u = \sqrt{t} \delta_n$.

Fix $t \geq 1$ and set $u = \sqrt{t} \delta_n \geq \delta_n$. If you go back this is ok since our only assumption on u was that $u \geq \delta_n$. Then $u^2 = t \delta_n^2$, so

$$\mathbb{P}(A(u)) \leq \exp\left(-\frac{nt\delta_n^2}{2\sigma^2}\right).$$

Step 7: From $A(u)^c$ to the error bound. On $A(u)^c$, consider two cases:

Case 1. $\|\hat{\Delta}\|_n < u$. Then $\|\hat{\Delta}\|_n^2 < u^2 = t \delta_n^2 \leq 16 t \delta_n^2$.

Case 2. $\|\hat{\Delta}\|_n \geq u$. Then since $\hat{\Delta} \in \mathcal{F}^\star$ and $A(u)$ did not occur,

$$\left| \frac{\sigma}{n} \sum_i w_i \hat{\Delta}(X_i) \right| < 2u \|\hat{\Delta}\|_n.$$

Combined with the basic inequality (9.12),

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \left| \frac{\sigma}{n} \sum_i w_i \hat{\Delta}(X_i) \right| < 2u \|\hat{\Delta}\|_n.$$

Dividing by $\|\hat{\Delta}\|_n > 0$ gives $\|\hat{\Delta}\|_n < 4u$, so $\|\hat{\Delta}\|_n^2 < 16u^2 = 16 t \delta_n^2$.

In both cases, $\|\hat{\Delta}\|_n^2 < 16 t \delta_n^2$. Therefore

$$\mathbb{P} \left(\|\hat{f} - f^\star\|_n^2 \geq 16 t \delta_n^2 \right) \leq \mathbb{P}(A(u)) \leq \exp \left(-\frac{n t \delta_n^2}{2\sigma^2} \right). \quad \blacksquare$$

Remark 9.13 (What just happened?). The proof has three conceptual pieces:

1. *Basic inequality*: turns a global optimization property (LSE minimizes empirical risk) into a pointwise inequality relating $\|\hat{\Delta}\|_n^2$ to a noise-signal inner product.
2. *Peeling / rescaling*: we don't know a priori how big $\|\hat{\Delta}\|_n$ is, but star-shapedness lets us rescale any g to sit exactly on the boundary of a ball of radius u . This converts a statement about “all g in $\mathcal{F}^\star$ with $\|g\|_n \geq u$ ” into a statement about $Z_n(u)$ over the u -ball.
3. *Gaussian concentration*: the supremum process $Z_n(u)$ concentrates around its mean $\sigma G_n(u; \mathcal{F}^\star)$, and the valid-radius condition is exactly what makes this mean smaller than the deviation we want to detect ($2u^2$).

The valid radius δ_n balances the “signal” u^2 (from the basic inequality) against the “noise” $\sigma G_n(u)$ (from Gaussian concentration); the smallest u where signal beats noise is the achievable rate.

9.5 High-Probability Corollary

Corollary 9.14 ($1 - \delta$ statement). Under the assumptions of Theorem 9.10, for any $\delta \in (0, 1)$,

$$\left\| \hat{f} - f^* \right\|_n^2 \leq \frac{32 \sigma^2 \log(1/\delta)}{n} \vee 16 \delta_n^2$$

with probability at least $1 - \delta$. Equivalently, with probability at least $1 - \delta$,

$$\left\| \hat{f} - f^* \right\|_n^2 \leq 16 \delta_n^2 \cdot \max \left(1, \frac{2 \sigma^2 \log(1/\delta)}{n \delta_n^2} \right).$$

Proof. Set the failure probability to δ : we want $\exp(-nt\delta_n^2/(2\sigma^2)) = \delta$, i.e.

$$t = \frac{2\sigma^2 \log(1/\delta)}{n \delta_n^2}.$$

Applying Theorem 9.10 requires $t \geq 1$. If $t \geq 1$, we get

$$\left\| \hat{f} - f^* \right\|_n^2 \leq 16 t \delta_n^2 = \frac{32 \sigma^2 \log(1/\delta)}{n}$$

w.p. $\geq 1 - \delta$. If $t < 1$ (i.e. $\log(1/\delta) < n\delta_n^2/(2\sigma^2)$), apply the theorem at $t = 1$ to get $\left\| \hat{f} - f^* \right\|_n^2 \leq 16\delta_n^2$ w.p. $\geq 1 - \exp(-n\delta_n^2/(2\sigma^2)) \geq 1 - \delta$. Taking the maximum of the two bounds covers both cases. $\blacksquare$

Remark 9.15 (Interpretation of the rate). The typical regime has $\log(1/\alpha)$ of constant order, in which case

$$\left\| \hat{f} - f^* \right\|_n^2 \lesssim \delta_n^2$$

with probability at least $1 - \alpha$. The fundamental nonparametric rate is δ_n^2 , where δ_n is the critical radius. For parametric classes of dimension d , $\delta_n^2 \asymp \sigma^2 d/n$, recovering the usual parametric rate. For richer classes (e.g. Lipschitz functions on $[0, 1]$), δ_n^2 is slower, reflecting the harder estimation problem.

9.6 next section

For $\delta > 0$ let

$$T_n(\delta) = \{g \in \mathcal{F}^* : \|g\|_n \leq \delta\}$$

Proposition 9.16.

$$G_n(\delta; \mathcal{F}^*) \leq \frac{C}{\sqrt{n}} \int_0^\delta \sqrt{\log \mathcal{N}(T_n(\delta), \|\cdot\|_n, \epsilon)} d\epsilon$$

Proof. For $g \in T_n(\delta)$ let $X_g = \frac{1}{n} \sum_i (g(x_i) - h(x_i))^2 = \frac{1}{n} \dots$ he erased the rest of the proof immediately after writing it. ■

Corollary 9.17. All he wrote was

$$\frac{\text{upperbound}}{\delta_n} \leq \frac{\delta_n}{2\sigma}$$

Remark 9.18. If $\mathcal{F}$ is star shaped around f^* and $A > 0$, $p \in (0, 2)$ then for all $\epsilon > 0$

$$\log(\mathcal{N}(\mathcal{F}, \|\cdot\|_n, \epsilon)) \leq A\epsilon^{-p}$$

Then

$$\mathcal{F}^* \subseteq \mathcal{F}^*$$

Then

$$\log(\mathcal{N}(\mathcal{F}, \|\cdot\|_n, \epsilon)) \leq A\epsilon^{-p}$$

for all $0 < \epsilon < \delta$. We apply the previous integral and see that the local gaussian complexity is bounded above by

$$\frac{C}{\sqrt{n}} \int_0^\delta \sqrt{A\epsilon^{-p}} d\epsilon \leq \frac{C_{A,p}}{\sqrt{n}} \delta^{1-p/2}$$

and

$$\frac{C_{A,p}}{\sqrt{n}} \delta^{1-p/2} \leq \frac{\delta}{2\sigma} \iff \delta_n \leq C_{A,p} \left(\frac{\sigma^2}{n} \right)^{\frac{1}{p+2}}$$

Ai Clean up of next section:

9.7 Bounding the Localized Gaussian Complexity via Metric Entropy

Theorem 9.10 reduces the statistical rate to controlling the valid radius δ_n , which in turn requires an upper bound on the localized Gaussian complexity $G_n(\delta; \mathcal{F}^*)$. We now give such a bound using the *metric entropy* (logarithm of covering numbers) of the localized class.

For $\delta > 0$, recall the localization

$$T_n(\delta) := \{g \in \mathcal{F}^* : \|g\|_n \leq \delta\}.$$

Proposition 9.19 (Dudley’s entropy integral, localized form). There is a universal constant $C > 0$ such that, for every $\delta > 0$,

$$G_n(\delta; \mathcal{F}^*) \leq \frac{C}{\sqrt{n}} \int_0^\delta \sqrt{\log \mathcal{N}(T_n(\delta), \|\cdot\|_n, \epsilon)} d\epsilon.$$

Proof. For each $g \in T_n(\delta)$, define the centered Gaussian random variable

$$X_g = \frac{1}{n} \sum_{i=1}^n w_i g(X_i).$$

Then $\{X_g : g \in T_n(\delta)\}$ is a Gaussian process, and the L^2 increments are

$$\mathbb{E}(X_g - X_h)^2 = \frac{1}{n^2} \sum_{i=1}^n (g(X_i) - h(X_i))^2 = \frac{1}{n} \|g - h\|_n^2.$$

So the natural (“canonical”) metric on $T_n(\delta)$ induced by this process is

$$d(g, h) := \sqrt{\mathbb{E}(X_g - X_h)^2} = \frac{1}{\sqrt{n}} \|g - h\|_n.$$

Because $\mathcal{F}^*$ is star-shaped around 0, we have $0 \in T_n(\delta)$ and $X_0 = 0$. Thus $G_n(\delta; \mathcal{F}^*) = \mathbb{E} \sup_{g \in T_n(\delta)} X_g$ is the expected supremum of a centered Gaussian process over a set containing its origin.

By Dudley’s entropy integral bound for Gaussian processes,

$$\mathbb{E} \sup_{g \in T_n(\delta)} X_g \leq C \int_0^{\text{diam}_d(T_n(\delta))} \sqrt{\log \mathcal{N}(T_n(\delta), d, u)} du,$$

for a universal constant C . Two simplifications: $\text{diam}_d(T_n(\delta)) \leq 2\delta/\sqrt{n}$ (since functions in $T_n(\delta)$ have $\|\cdot\|_n \leq \delta$, so d -distances are at most $2\delta/\sqrt{n}$); and

$$\mathcal{N}(T_n(\delta), d, u) = \mathcal{N}(T_n(\delta), \|\cdot\|_n, u\sqrt{n}),$$

because the two metrics differ by the factor $\sqrt{n}$. Changing variables $\epsilon = u\sqrt{n}$,

$$\begin{aligned} \mathbb{E} \sup_{g \in T_n(\delta)} X_g &\leq C \int_0^{2\delta/\sqrt{n}} \sqrt{\log \mathcal{N}(T_n(\delta), \|\cdot\|_n, u\sqrt{n})} du \\ &= \frac{C}{\sqrt{n}} \int_0^{2\delta} \sqrt{\log \mathcal{N}(T_n(\delta), \|\cdot\|_n, \epsilon)} d\epsilon. \end{aligned}$$

Absorbing the factor of 2 in the upper limit into the constant C gives the stated bound. ■

Remark 9.20 (Chaining intuition). The proof of Dudley’s bound itself (which we quoted) is by the *chaining* argument: build a sequence of finer and finer ϵ -nets of $T_n(\delta)$, decomposing any X_g as a telescoping sum over the nets. Gaussian tail bounds and a union bound at each scale yield the $\sqrt{\log \mathcal{N}(\cdot, \cdot, \epsilon)}$ factor; integrating across scales gives the integral. Thus Dudley’s bound is essentially “local Gaussian deviations, summed over all resolutions.”

9.8 The Critical Radius via Dudley’s Bound

Combining Proposition 9.19 with the valid-radius condition gives an explicit route for computing δ_n .

Corollary 9.21 (Critical radius from metric entropy). Any $\delta_n > 0$ satisfying

$$\frac{C}{\sqrt{n} \delta_n} \int_0^{\delta_n} \sqrt{\log \mathcal{N}(T_n(\delta_n), \|\cdot\|_n, \epsilon)} d\epsilon \leq \frac{\delta_n}{2\sigma}$$

is a valid radius in the sense of Lemma 9.7. Equivalently,

$$\frac{C}{\sqrt{n}} \int_0^{\delta_n} \sqrt{\log \mathcal{N}(T_n(\delta_n), \|\cdot\|_n, \epsilon)} d\epsilon \leq \frac{\delta_n^2}{2\sigma}.$$

Proof. Proposition 9.19 bounds $G_n(\delta_n; \mathcal{F}^*)$ by the integral on the left. The valid-radius condition is $G_n(\delta_n; \mathcal{F}^*)/\delta_n \leq \delta_n/(2\sigma)$; substituting the upper bound suffices. ■

Remark 9.22 (Interpretation as a fixed point). The critical radius is the smallest scale δ_n at which the Dudley integral — a measure of cumulative class richness below δ_n — is dominated by $\delta_n^2/(2\sigma)$, a pure signal term. Below δ_n , noise overwhelms signal and the LSE cannot reliably resolve functions; above δ_n , the LSE’s error is driven by signal and converges. Finding δ_n is therefore solving a scalar fixed-point equation.

9.9 Polynomial Metric Entropy: Explicit Rates

To make Corollary 9.21 concrete, we specialize to classes whose metric entropy grows polynomially in $1/\epsilon$.

Theorem 9.23 (Rates under polynomial entropy). Assume $\mathcal{F}$ is star-shaped around f^* , and that there exist $A > 0$ and $p \in (0, 2)$ such that

$$\log \mathcal{N}(\mathcal{F}, \|\cdot\|_n, \epsilon) \leq A \epsilon^{-p} \quad \text{for all } \epsilon > 0.$$

Then there is a constant $C_{A,p} > 0$ (depending only on A, p and universal constants) such that

$$\delta_n := C_{A,p} \left(\frac{\sigma^2}{n} \right)^{\frac{1}{p+2}}$$

is a valid radius. Consequently, by Theorem 9.10,

$$\left\| \hat{f} - f^* \right\|_n^2 \lesssim \left(\frac{\sigma^2}{n} \right)^{\frac{2}{p+2}}$$

with high probability.

Proof. Step 1: Transfer the covering bound to $T_n(\delta)$. The empirical seminorm $\|\cdot\|_n$ is translation-invariant, so

$$\mathcal{N}(\mathcal{F}^*, \|\cdot\|_n, \epsilon) = \mathcal{N}(\mathcal{F} - f^*, \|\cdot\|_n, \epsilon) = \mathcal{N}(\mathcal{F}, \|\cdot\|_n, \epsilon).$$

Since $T_n(\delta) \subseteq \mathcal{F}^*$, covering numbers are monotone:

$$\log \mathcal{N}(T_n(\delta), \|\cdot\|_n, \epsilon) \leq \log \mathcal{N}(\mathcal{F}^*, \|\cdot\|_n, \epsilon) \leq A \epsilon^{-p}.$$

Step 2: Evaluate the Dudley integral. For any $\delta > 0$,

$$\int_0^\delta \sqrt{A \epsilon^{-p}} d\epsilon = \sqrt{A} \int_0^\delta \epsilon^{-p/2} d\epsilon = \frac{\sqrt{A}}{1-p/2} \delta^{1-p/2},$$

where the integral converges precisely because $p < 2$. By Proposition 9.19,

$$G_n(\delta; \mathcal{F}^*) \leq \frac{C\sqrt{A}}{(1-p/2)\sqrt{n}} \delta^{1-p/2} = \frac{C'_{A,p}}{\sqrt{n}} \delta^{1-p/2}.$$

Step 3: Solve the valid-radius condition. We seek the smallest δ_n with

$$\frac{C'_{A,p}}{\sqrt{n}} \delta_n^{1-p/2} \leq \frac{\delta_n^2}{2\sigma}.$$

Rearranging (and using $\delta_n > 0$),

$$\delta_n^{2-(1-p/2)} = \delta_n^{1+p/2} \geq \frac{2\sigma C'_{A,p}}{\sqrt{n}}.$$

Raising both sides to the power $2/(2+p)$,

$$\delta_n \geq \left(\frac{2\sigma C'_{A,p}}{\sqrt{n}} \right)^{\frac{2}{p+2}} = (2C'_{A,p})^{2/(p+2)} \cdot \left(\frac{\sigma^2}{n} \right)^{\frac{1}{p+2}}.$$

Setting $C_{A,p} := (2C'_{A,p})^{2/(p+2)}$, the choice $\delta_n = C_{A,p}(\sigma^2/n)^{1/(p+2)}$ is a valid radius.

Step 4: Final rate. Theorem 9.10 and Corollary 9.14 then give

$$\|\hat{f} - f^*\|_n^2 \lesssim \delta_n^2 = C_{A,p}^2 \left(\frac{\sigma^2}{n} \right)^{\frac{2}{p+2}}$$

with high probability. ■

Remark 9.24 (Sanity checks and examples). The rate $(\sigma^2/n)^{2/(p+2)}$ interpolates between familiar regimes:

- $p \rightarrow 0^+$: rate $\rightarrow \sigma^2/n$, the parametric rate. This matches finite-dimensional classes, whose covering numbers are polynomial in $1/\epsilon$ with exponent arbitrarily close to 0 (up to log factors).
- $p = 1$: rate $(\sigma^2/n)^{2/3}$. This is the classical minimax rate for Lipschitz functions on $[0, 1]$, whose $\|\cdot\|_\infty$ metric entropy satisfies $\log \mathcal{N} \asymp \epsilon^{-1}$.

- $p \rightarrow 2^-$: rate $(\sigma^2/n)^{1/2}$, the slowest polynomial rate Dudley can handle. At $p = 2$, the entropy integral diverges at 0 and one needs sharper tools (e.g. bracketing entropy, generic chaining).
- More generally, for Hölder balls of smoothness s on $[0, 1]^d$, one has $\log \mathcal{N} \asymp \epsilon^{-d/s}$, giving $p = d/s$ and the minimax rate $n^{-2s/(2s+d)}$ the classical nonparametric rate.

Remark 9.25 (Why $p < 2$ is sharp for Dudley). The Dudley integral $\int_0^\delta \epsilon^{-p/2} d\epsilon$ converges iff $p/2 < 1$, i.e. $p < 2$. For $p \geq 2$ the metric entropy grows too fast near $\epsilon = 0$ and the chaining argument no longer gives a useful bound. This is a genuine limitation of the technique, not of the problem: classes with $p \geq 2$ (such as certain ball-shaped Sobolev classes at low smoothness) may still admit rates, but the analysis requires bracketing entropy, Talagrand's γ_2 functional, or generic chaining.

9.10 Random Design non-parametric least squares

In the past example, the $X_1, \dots, X_n$ were fixed and the randomness was coming in only from the w_i s (gaussians).

Now assume $(X_i, Y_i)_{i=1}^n$ are iid from some unknown distribution.

If $\mathcal{F}$ is a class of functions such that

1. for all $f \in \mathcal{F}$, $|f| \leq m$; $|Y| \leq m$ (a.s.) and;
2. $\mathcal{F}$ is convex.

We DO NOT assume that $Y = f^*(X) + \xi$; with $\xi \perp X$.

The excess risk is

$$R(\hat{f}) - \inf_{f \in \mathcal{F}} R(f) = \mathbb{E}(\hat{f} - Y)^2 - \inf_{f \in \mathcal{F}} \mathbb{E}(f(X) - Y)^2$$

If we do assume that

$$Y = f^*(X) + \xi; \text{ with } \xi \perp X$$

and $\mathbb{E}\xi = 0$ then

$$R(f) - R(f^*) = \mathbb{E}(f - f^*)^2 + 2\mathbb{E}(f - f^*)\xi$$

and the by independence:

$$\mathbb{E}(f - f^\star)\xi = 0.$$

We have

$$\hat{f} := \arg \min_{f \in \mathcal{F}} \sum_{i=1}^n (f(X_i) - Y_i)^2$$

be the least squares estimator. Let

$$f^\star = \arg \min_{f \in \mathcal{F}} \mathbb{E}[f - Y]^2$$

Note that

$$\frac{1}{n} \left\langle Y - \hat{f}(X), Y - f^\star(X) \right\rangle \leq 0$$

where we are treating $Y = (Y_1, \dots, Y_n)$ and $X = (X_1, \dots, X_n)$. Then

$$\begin{aligned} R_n(f^\star) - R_n(\hat{f}) &= \frac{1}{n} \left(\sum_{i=1}^n (f^\star(X_i))^2 + Y_i^2 - 2f^\star(X_i)Y_i \right) - \frac{1}{n} \left(\sum_{i=1}^n (\hat{f}(X_i))^2 + Y_i^2 - 2\hat{f}(X_i)Y_i \right) \\ &= \frac{1}{n} \left(\sum_{i=1}^n (f^\star - \hat{f})^2 - (f^\star - \hat{f})^2 - 2f^\star Y_i + 2\hat{f}(X_i)Y_i + (f^\star(X_i))^2 - (\hat{f}(X_i))^2 \right) \\ &\geq \frac{1}{n} \left(\sum_{i=1}^n (f^\star - \hat{f})^2 \right) \end{aligned}$$

Now, to condense notation put

$$P(\hat{f} - Y) := \mathbb{E}[\hat{f}(X) - Y]^2$$

and

$$P_n(\hat{f} - Y) := \frac{1}{n} \sum_{i=1}^n [\hat{f}(X_i) - Y_i]^2$$

We define $\mathcal{G} = f^\star - \mathcal{F}$.

Can show

$$\begin{aligned}
& \mathbb{E}[\hat{f}(X) - Y]^2 - \mathbb{E}[f^* - Y]^2 \\
&= (P - P_n) \left((\hat{f})^2 - 2\hat{f}Y + 2f^*Y - (f^*)^2 \right) - P_n(\hat{f} - f^*)^2 \\
&= (P - P_n) \left((\hat{f} - f^*)^2 - (\hat{f} - (f^*)^2 + \hat{f})^2 - 2\hat{f}Y + 2f^*Y - (f^*)^2 \right) - P_n(\hat{f} - f^*)^2 \\
&= (P - P_n)(\hat{f} - f^*)^2 - P_n(\hat{f} - f^*)^2 + (P - P_n)(2\xi(f^* - \hat{f})) \\
&= P(\hat{f} - f^*)^2 - \frac{3}{2}P_n(\hat{f} - f^*)^2 + (P - P_n)(2\xi(f^* - \hat{f})) - \frac{1}{2}P_n(\hat{f} - f^*)^2 + \frac{1}{4}P(\hat{f} - f^*)^2 - \frac{1}{4}P(\hat{f} - f^*)^2 \\
&= \frac{9}{8}P(\hat{f} - f^*)^2 - \frac{3}{2}P_n(\hat{f} - f^*)^2 + (P - P_n)(2\xi(f^* - \hat{f})) - \frac{1}{2}P_n(\hat{f} - f^*)^2 - \frac{1}{8}P(\hat{f} - f^*)^2 \\
&\leq \sup_{g \in \mathcal{G}} \left(\frac{9}{8}P(g)^2 - \frac{3}{2}P_n(g^2) \right) + \sup_{g \in \mathcal{G}} \left((P - P_n)(2\xi g) - \frac{1}{4}P(g^2) - \frac{1}{4}P_n(g^2) \right)
\end{aligned}$$

For the above inequality, take the expectation wrt $(X_i, Y_i)_{i=1}^n$ being randomly drawn. The second term on the is

$$\begin{aligned}
& \mathbb{E} \sup_{g \in \mathcal{G}} \left((P - P_n)(2\xi g) - \frac{1}{4}P(g^2) - \frac{1}{4}P_n(g^2) \right) \\
&= \mathbb{E} \sup_{g \in \mathcal{G}} \left((\mathbb{E}' P'_n - P_n)(2\xi g) - \frac{1}{4}\mathbb{E}' P'_n(g^2) - \frac{1}{4}P_n(g^2) \right) \\
&\leq \mathbb{E} \sup_{g \in \mathcal{G}} \left((P'_n - P_n)(2\xi g) - \frac{1}{4}P'_n(g^2) - \frac{1}{4}P_n(g^2) \right) \\
&= \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{1}{n} \sum_{i=1}^n 2(\xi'_i g(X'_i) \epsilon_i - \xi_i g(X_i) \epsilon_i) - \frac{1}{4n} \sum_{i=1}^n (g(X_i)^2) - \frac{1}{4n} \sum_{i=1}^n (g(X'_i)^2) \right) \\
&\leq 2\mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{1}{n} \sum_{i=1}^n 2(\xi_i g(X_i) \epsilon_i) - \frac{1}{4n} \sum_{i=1}^n (g(X_i)^2) \right) \\
&\leq 2\mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{1}{n} \sum_{i=1}^n 2(2mg(X_i) \epsilon_i) - \frac{1}{4n} \sum_{i=1}^n (g(X_i)^2) \right) \\
&= \frac{8m}{n} \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{1}{n} \sum_{i=1}^n g(X_i) \cdot \epsilon_i - \frac{1}{16m} \sum_{i=1}^n (g(X_i)^2) \right)
\end{aligned}$$

Note that

$$(P'_n - P_n)(2\xi g) = \frac{1}{n} \sum_{i=1}^n 2(\xi'_i g(X'_i) - \xi_i g(X_i))$$

We can also add in iid Rademacher random variables in a standard symmetrization move.

We used the fact that P can be written as an expected value of P'_n with respect to a ghost sample $(X'_i, Y'_i)_{i=1}^n$.

We also used that

$$|\xi_i| = |Y_i - f^*(x_i)| \leq 2m$$

The contraction step where we use that multiplication of ξ_i is $2m$ -Lipschitz relied crucially on the presence of the Rademacher random variables.

Next, go back to the first term on the RHS. We apply similar logic to the last block:

$$\begin{aligned} & \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{9}{8} P(g)^2 - \frac{3}{2} P_n(g^2) \right) \\ &= \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{10}{8} P(g)^2 - \frac{10}{8} P_n(g^2) - \frac{1}{8} P g^2 - \frac{2}{8} P_n g^2 \right) \\ &\leq \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{10}{8} (P - P_n) - \frac{1}{8} P g^2 - \frac{2}{8} P_n g^2 \right) \\ &\leq \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{10}{8} (P'_n - P_n) g^2 - \frac{1}{8} P'_n g^2 - \frac{2}{8} P_n g^2 \right) \\ &\leq 2 \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{10}{8} \sum_{i=1}^n \epsilon_i [g(X_i)]^2 - \frac{1}{4} \sum_{i=1}^n [g(x_i)]^2 \right) \\ &\leq 2 \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{10}{8} 4m \sum_{i=1}^n \epsilon_i g(X_i) - \frac{1}{4} \sum_{i=1}^n [g(x_i)]^2 \right) \\ &= \frac{10m}{n} \mathbb{E} \sup_{g \in \mathcal{G}} \left(\sum_{i=1}^n \epsilon_i g(X_i) - \frac{1}{40m} \sum_{i=1}^n [g(x_i)]^2 \right) \end{aligned}$$

Note that $x \mapsto [g(x)]^2$ is locally $4m$ -lipschitz since the thing we are squaring is bounded by $2m$. So we can apply that contraction Lipschitz result again.

Now fix $\gamma \geq 0$ and consider the γ covering of $\mathcal{G}$ at scale γ using $L_2(P_n)$ -distance

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (g(X_i) - f(X_i))^2}$$

Given $g \in \mathcal{G}$ let $\pi(g)$ be the closest point in $\mathcal{G}$ in the net.

$$\begin{aligned} & \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i (g(X_i) - \pi(g)(X_i)) \right) + \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i \pi[g](X_i) - \frac{C}{4} \sum_{i=1}^n (\pi[g](X_i))^2 \right) \\ & + \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{C}{4} \sum_{i=1}^n (\pi[g](X_i))^2 - C \sum_{i=1}^n [g(X_i)]^2 \right) \end{aligned}$$

If $\|g\|_{L_2(P_n)} \leq \gamma$ then $\pi(g) = 0$.

If $\|g\|_{L_2(P_n)} > \gamma$ then

$$\|\pi[g]\|_{L_2(P_n)} \leq \|g\|_{L_2(P_n)} + \gamma$$

Based on this geometric setup, the third term

$$\frac{C}{4} \sum_{i=1}^n (\pi[g](X_i))^2 - C \sum_{i=1}^n [g(X_i)]^2 \leq \frac{C}{4} \left((\|g\|_{L_2(P_n)} + \gamma)^2 - 4 \|g\|_{L_2(P_n)}^2 \right) \leq 0$$

You can see this by expanding $(\|g\|_{L_2(P_n)} + \gamma)^2$ and the simplifying in the $\|g\| \geq \gamma$ case.

Note that we just assume 0 is in the net for this argument to go through. This could involve adding 0 to the minimal net if needed.

For the first term, we have

$$\begin{aligned} & \mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i (g(X_i) - \pi(g)(X_i)) \right) \\ & \leq C \left(\alpha + \frac{1}{\sqrt{n}} \int_{\alpha}^{\gamma} \sqrt{\log(\mathcal{N}(\mathcal{G}, L_2(P_n), \epsilon))} d\epsilon \right) \end{aligned}$$

by Dudley integral bound. We have $\alpha \leq \gamma$. I'm a bit confused on this term.

For the second term, let $W = \pi(\mathcal{G})$ and since W is finite $|W| = \mathcal{N}(\mathcal{G}, L_2(P_n), \gamma)$:

Use and MGF method: choose $\lambda = C/2$ then

$$\begin{aligned} & \mathbb{E} \left(\exp \left(\lambda \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i g(X_i) - \frac{C}{4} \sum_{i=1}^n (g(X_i))^2 \right) \right) \right) \\ & \leq \exp \left(\sum_{i=1}^n \frac{\lambda^2 [g(X_i)]^2}{2} - \frac{\lambda C}{4} \sum_{i=1}^n [g(X_i)]^2 \right) \end{aligned}$$

$$\begin{aligned}
\mathbb{E} \sup_{g \in \mathcal{G}} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i \pi[g](X_i) - \frac{C}{4} \sum_{i=1}^n (\pi[g](X_i))^2 \right) &= \mathbb{E} \max_{g \in W} \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i g(X_i) - \frac{C}{4} \sum_{i=1}^n (g(X_i))^2 \right) \\
&= \frac{1}{n} \frac{1}{\lambda} \mathbb{E} \log \left(\exp \left(\max_{g \in W} \lambda \left(\sum_{i=1}^n \left(\epsilon_i g(X_i) - \frac{C}{4} [g(X_i)]^2 \right) \right) \right) \right) \\
&\leq \frac{1}{n} \frac{1}{\lambda} \log |W| \max_{g \in W} \mathbb{E} \exp \left(\lambda \left(\sum_{i=1}^n \left(\epsilon_i g(X_i) - \frac{C}{4} [g(X_i)]^2 \right) \right) \right) \\
&\leq \frac{1}{n} \frac{2}{C} \log |W|
\end{aligned}$$

Final

$$\mathbb{E}(R(\hat{f}) - R(f^*)) \leq Cm \left(\alpha + \frac{1}{\sqrt{n}} \int_{\alpha}^{\gamma} \sqrt{\log(\mathcal{N}(\mathcal{G}, L_2(P_n), \epsilon))} d\epsilon + \frac{m}{n} \log(\mathcal{N}(\mathcal{G}, L_2(P_n), \gamma)) \right)$$

Theorem 9.26 (Random Design Nonparametric LSE Risk Bound). Let $(X_i, Y_i)_{i=1}^n$ be i.i.d. from an unknown distribution. Let $\mathcal{F}$ be a convex class of functions with $|f| \leq m$ for all $f \in \mathcal{F}$ and $|Y| \leq m$ a.s. Define the least squares estimator and its population counterpart

$$\begin{aligned}
\hat{f} &:= \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f(X_i) - Y_i)^2, \\
f^* &:= \arg \min_{f \in \mathcal{F}} \mathbb{E} (f(X) - Y)^2.
\end{aligned}$$

Set $\mathcal{G} := f^* - \mathcal{F}$. Then for any $0 \leq \alpha \leq \gamma$,

$$\mathbb{E} \left(R(\hat{f}) - R(f^*) \right) \leq Cm \left(\alpha + \frac{1}{\sqrt{n}} \int_{\alpha}^{\gamma} \sqrt{\log \mathcal{N}(\mathcal{G}, L_2(P_n), \varepsilon)} d\varepsilon + \frac{m}{n} \log \mathcal{N}(\mathcal{G}, L_2(P_n), \gamma) \right)$$

for a universal constant $C > 0$, where $R(f) := \mathbb{E}(f(X) - Y)^2$ denotes the population risk.

10 Minimax Lower Bounds

Let Θ be a space of parameters and let $d(\theta, \theta')$ be a pseudometric (triangle inequality and non-negativity). Let $\mathcal{P} = \{P_{\theta} : \theta \in \Theta\}$ be a family of distributions. Let $\hat{\theta}$ be

an estimator based on observations and θ be the true parameter. Fix $q > 0$. We are interested in

$$\left(d(\hat{\theta}, \theta)\right)^q$$

Example: least squares estimator.. do you minimized

$$\mathbb{E} \left\| \hat{\theta} - \theta \right\|_2^2 \text{ or } \mathbb{E} \left\| \hat{\theta} - \theta \right\|_2$$

The minimax rate is

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{P_\theta} d^q(\hat{\theta}, \theta)$$

What is the smallest possible error and what estimator achieves the estimator achieves the smallest possible expected error when θ true value is chosen in the worst case scenario??

Example: if $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2 I_d)$ the minimax rate is

$$\inf_{\hat{\theta}} \sup_{\theta \in R^d} \left\| \hat{\theta} - \theta \right\|_2^2$$

10.1 Fano's Inequality

Proposition 10.1 (Fano's Inequality). Suppose we have some space $\mathcal{X}$ and $P_1, \dots, P_n$ are distributions over $\mathcal{X}$ which are absolutely measurable with respect to some measure Q . Then

$$\inf_{\hat{j}: \mathcal{X} \rightarrow [N]} \frac{1}{N} \sum_{j=1}^N P_j(\hat{J}(\text{Obs}) \neq j) \geq 1 - \frac{\frac{1}{N} \sum_{j=1}^N \text{KL}(P_j || Q)}{\log(N)}$$

where Obs is an observation generated according to P_j .

Proof.

$$\begin{aligned}
\sum_{i=1}^N P_j(\hat{J}(Obs) = j) &= \sum_{j=1}^n \mathbb{E}_{P_j} \mathbf{1}(\hat{J}(Obs) = j) \\
&= \sum_{j=1}^n \int_{\mathcal{X}} \mathbf{1}(\hat{J}(Obs) = j) \frac{dP_j(x)}{dQ(x)} dQ(x) \\
&\leq \int_{\mathcal{X}} \max_{j \in [N]} \frac{dP_j(x)}{dQ(x)} \left(\sum_{j=1}^n \mathbf{1}(\hat{J}(Obs) = j) \right) dQ(x) \\
&= \int_{\mathcal{X}} \max_{j \in [N]} \frac{dP_j(x)}{dQ(x)} dQ(x) \\
&= \mathbb{E}_Q \max_{j \in [N]} \frac{dP_j(x)}{dQ(x)} := M
\end{aligned}$$

What is good about this M ? We know $M \leq \mathbb{E}_Q \sum_{j=1}^N \frac{dP_j(x)}{dQ(x)} = N$. Define $\phi(x) = x \log x - x + 1$. Properties: $\phi \geq 0$ and is convex. Then

$$\phi(M) = \phi[\mathbb{E}_Q \max_{j \in [N]} \frac{dP_j(x)}{dQ(x)}] \leq \mathbb{E}_Q \phi(\max_{j \in [N]} \frac{dP_j(x)}{dQ(x)}) \leq \sum_{j=1}^N \mathbb{E}_Q \phi(\frac{dP_j(x)}{dQ(x)}) = \sum_{i=1}^N \text{KL}(P_i, Q)$$

This is an upper bound. For the lower bound, since $M \leq N$ we have

$$\phi(M) \geq M \log N - N$$

(follows from plugging in M/N to ϕ and using non-negativity).

Combining these two result we have

$$M \log(N) - N \leq \sum_{j=1}^N \text{KL}(P_j, Q)$$

which implies

$$\frac{1}{N} M \leq \frac{1 + \frac{1}{N} \sum_{j=1}^N \text{KL}(P_j, Q)}{\log N}$$

which implies

$$\frac{1}{N} \sum_{j=1}^N P_j(\hat{J}(Obs) = j) \leq \frac{1 + \frac{1}{N} \sum_{j=1}^N \text{KL}(P_j, Q)}{\log N}$$

TODO go from this to the desired result in one step :)

■

10.2 more lower bound stuff

Suppose Θ is a parameter space; $P_{\theta_1}, \dots, P_{\theta_N}$ where $\theta_i \in \Theta$ for all i (finite number of em). Assume Q is a measure such that $P_j \ll Q$ for all j . We require that

$$d(\theta_i, \theta_j) \geq \Delta > 0$$

for all $i \neq j$. Then

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d^q(\hat{\theta}, \theta) \geq \left(\frac{\Delta}{2}\right)^q \left(1 - \frac{1 + \frac{1}{N} \sum_{j=1}^N \text{KL}(P_j, Q)}{\log(N)}\right)$$

Proof. Let $\hat{\theta}$ be an estimator (map from observation space $\mathcal{X}$ to Θ)

$$\hat{J}(x) = \arg \min_{j \in [N]} d(\hat{\theta}(x), \theta_j)$$

(Estimator to testing reduction)

If $\hat{J}(\text{Obs}) \neq j$ with respect to the true distribution

Then

$$\Delta \leq d(\theta_j, \theta_j) \leq d(\theta_j, \hat{\theta}) + d(\theta_j, \hat{\theta}) \leq 2d(\hat{\theta}, \theta_j)$$

so

$$d(\hat{\theta}, \theta_j) \geq \left(\frac{\Delta}{2}\right)^q$$

Then

$$\sup_{\theta \in \Theta} \mathbb{E}_{\theta} [d^q(\hat{\theta}, \theta)] \geq \frac{1}{N} \sum_j \mathbb{E}_{\theta_j} [d^q(\hat{\theta}, \theta_j) \mathbf{1}(\hat{J} \neq j)]$$

By Fano's inequality

$$\frac{1}{N} \sum_{j=1}^N P_j(\hat{J}(\text{Obs}) \neq j) \geq 1 - \frac{\frac{1}{N} \sum_{j=1}^N \text{KL}(P_j || Q)}{\log(N)}$$

Then combine to conclude :)

■

Example 10.2. If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2 I_d)$ and $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$; $\theta \sim \mathcal{N}(\theta, \sigma^2/n I_d)$. Then

$$\mathbb{E}[\|\hat{\theta} - \theta\|_2^2] = \mathbb{E}\|Y\|_2^2 = \text{Tr}(\sigma^2/n I_d) = d\sigma^2/n$$

where $Y \sim \mathcal{N}(0, \sigma^2/n I_d)$.

for the lower bound, let $\mathbb{Q}^\Theta \sim N(0, \sigma^2 I_d)$. Choose an $\epsilon > 0$ separated set in the ball of radius $R > 0$ (with respect to euclidean metric): $\theta_1, \dots, \theta_N$ is an ϵ -packing and a volumetric argument implies $N \geq (R/\epsilon)^d$.

Recall in the lower bound, everything was in terms of a single shot observation X not an iid sample. Thus we define

$$P_{\theta_j} = (\mathcal{N}(\theta_j, \sigma^2 I_d))^{\textcircled{n}}$$

The n is circled idk why this just what he wrote haha to show its a power.

Then

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_\theta \|\hat{\theta} - \theta\| \geq \left(\frac{\epsilon}{2}\right)^2 \left(1 - \frac{nR^2/(2\sigma^2)}{\log(R/\epsilon)^d}\right)$$

Choosing $\epsilon = a\sigma\sqrt{d/n}$ and $R = b\sigma\sqrt{d/n}$ gives

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_\theta \|\hat{\theta} - \theta\| \geq \frac{a^2}{4} \sigma^2 d/n \left(1 - \frac{1 + b^2 d/2}{d \log(b/a)}\right)$$

Choose b and a such that $\left(1 - \frac{1+b^2 d/2}{d \log(b/a)}\right) \geq \frac{1}{2}$ to attain the desired rate:

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_\theta \|\hat{\theta} - \theta\| \geq \frac{a^2}{8} \sigma^2 d/n$$

Why is the $\sigma^2 d/n$ the correct rate to aim for?

The rate $\frac{\sigma^2 d}{n}$ is the correct target because it is the **minimax optimal rate** for this estimation problem. We establish this by matching the upper and lower bounds up to absolute constant factors:

- **Upper Bound (Achievability):** As shown earlier, the standard sample mean $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$ has a known distribution of $\mathcal{N}(\theta, \frac{\sigma^2}{n} I_d)$. Taking the trace of the covariance matrix gives an expected squared error exactly equal to $\frac{d\sigma^2}{n}$. This proves an error rate of $\mathcal{O}(\frac{\sigma^2 d}{n})$ is strictly achievable.

- **Lower Bound (Fundamental Limit):** The entire purpose of Fano's inequality and the ϵ -packing argument is to prove that *no estimator* can do fundamentally better. By deliberately setting the packing separation $\epsilon = a\sigma\sqrt{d/n}$ and bounding radius $R = b\sigma\sqrt{d/n}$, the lower bound resolves to $\Omega(\frac{\sigma^2 d}{n})$.

Because the lower bound matches the upper bound, the “minimax sandwich” is closed. This proves that $\frac{\sigma^2 d}{n}$ is not just an arbitrary rate, but the fundamental information-theoretic limit for estimating the mean of this Gaussian.

An alternative to Fano's inequality is called Birge's Lemma.

Lemma 10.3 (Birge's Lemma). If I have $P_1, \dots, P_N$ are distributions over $\mathcal{X}$ and $A_1, \dots, A_n$ are disjoint events then

$$\min_{i \in [N]} P_i(A_i) \leq \max \left\{ \frac{2e}{2e+1}, \frac{\max_{i,j} \{ \text{KL}(P_i, P_j) \}}{\log N} \right\}$$

Compare this to what Fano's gives us:

$$\inf_{\hat{J}: \mathcal{X} \rightarrow [N]} \frac{1}{N} \sum_{j=1}^N P_j(\hat{J}(\text{Obs}) \neq j) \geq 1 - \frac{\frac{1}{N} \sum_{j=1}^N \text{KL}(P_j || Q)}{\log(N)}$$

where Obs is an observation generated according to P_j .

To make Birge's lemma look more comparable to Fano's, we let $\hat{J}$ be any selector $\mathcal{X} \rightarrow [N]$ and

$$A_j : \{ \hat{J} = j \}$$

Then the $A_1, \dots, A_j$ are disjoint by construction and

$$\min_{i \in [N]} P_i(A_i) = 1 - \max_{i \in [N]} P_i\{ \hat{J} \neq i \}$$

In the regime where

$$\frac{2e}{2e+1} > \frac{\max_{i,j} \{ \text{KL}(P_i, P_j) \}}{\log N}$$

then

$$\max_{i \in [N]} P_i(\hat{J} \neq i) \geq 1 - \frac{2e}{2e+1} = \frac{1}{2e+1}$$

and then

$$\max_{i \in [N]} P_i(\hat{J} \neq i) \geq \frac{1}{2e+1}$$

then we take the inf over all possible $\hat{J}$:

$$\inf_{\hat{J}} \max_{i \in [N]} P_i(\hat{J} \neq i) \geq \frac{1}{2e + 1}$$

Example 10.4 (Two points lower bound). Let Θ be a parameter space, and $\theta_1, \theta_2 \in \Theta$ with $d(\theta_1, \theta_2) \geq \Delta$.

Then we have

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d^q(\hat{\theta}, \theta) \geq \inf_{\hat{\theta}} \max_{j \in \{1,2\}} \mathbb{E}_{\theta} d^q(\hat{\theta}, \theta) \geq \left(\frac{\Delta}{2}\right)^q \max_{j \in \{1,2\}} P_j(\hat{J} \neq j)$$

The $\hat{J}$ we have here, recall, is defined for a specific estimator $\hat{\theta}$ by

$$\hat{J} = \arg \min_{j \in \{1,2\}} d(\hat{\theta}, \theta_j)$$

We can further reduce by taking the infimum over *all possible* testing rules $\hat{J}$ on the RHS:

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d^q(\hat{\theta}, \theta) \geq \left(\frac{\Delta}{2}\right)^q \inf_{\hat{J}} \max_{j \in \{1,2\}} P_j(\hat{J} \neq j)$$

Why deliberately weaken the bound with this last step? Mathematically, taking the infimum does give away some tightness. However, we make this sacrifice for **computability**:

- **Decoupling from estimators:** If we keep $\hat{J}$ tied to $\hat{\theta}$, we have to analyze the behavior of every conceivable estimator $\hat{\theta}$ over the entire parameter space, which is usually impossible to compute directly.
- **Reduction to pure testing:** By taking the infimum over all $\hat{J}$, we sever the link to $\hat{\theta}$. We stop asking “what is the error of the best estimator?” and start asking “what is the fundamental limit of distinguishing P_1 from P_2 with a binary classifier?”
- **Unlocking Total Variation:** This reduction to hypothesis testing unlocks Le Cam’s Lemma, which bounds the testing error using the Total Variation (TV) distance between the two distributions:

$$\inf_{\hat{J}} \max_{j \in \{1,2\}} P_j(\hat{J} \neq j) \geq \frac{1}{2}(1 - \text{TV}(P_1, P_2))$$

Unlike arbitrary estimator risk, we *do* know how to calculate or bound the distance between two specific probability distributions (e.g., using KL-divergence and Pinsker’s inequality).

Example 10.5. Suppose we have n i.i.d. observations $X_1, \dots, X_n \in \{-1, 1\}$. We want to estimate their true mean $\mathbb{E}[X_1] = \mu$ using the squared error risk $\mathbb{E}[(\hat{\mu} - \mu)^2]$.

1. The Upper Bound (Achievability)

If we use the standard sample mean $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$, the error is exactly the variance of the sample mean. Since $X_i \in \{-1, 1\}$, the variance of a single observation is at most 1. Therefore:

$$\mathbb{E} \left[\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right)^2 \right] \leq \frac{1}{n}$$

2. Constructing the “Hard” Sub-problem (Lower Bound)

To prove we cannot do fundamentally better than $1/n$, we construct two difficult-to-distinguish distributions to act as our worst-case scenario. Let $\delta \in (0, 1/2)$ be a small shift. We define two biased coin distributions:

$$\begin{aligned} P_+(X_1 = 1) &= \frac{1+\delta}{2}, & P_+(X_1 = -1) &= \frac{1-\delta}{2} \implies \mathbb{E}_+[X_1] = \delta \\ P_-(X_1 = 1) &= \frac{1-\delta}{2}, & P_-(X_1 = -1) &= \frac{1+\delta}{2} \implies \mathbb{E}_-[X_1] = -\delta \end{aligned}$$

The parameter distance between these two scenarios is:

$$\Delta = d(\mathbb{E}_+, \mathbb{E}_-) = |\delta - (-\delta)| = 2\delta$$

3. Calculating the Information Overlap (KL Divergence)

We calculate the Kullback-Leibler (KL) divergence for a single observation to see how hard they are to distinguish:

$$\text{KL}(P_+ || P_-) = \frac{1+\delta}{2} \log \left(\frac{1+\delta}{1-\delta} \right) + \frac{1-\delta}{2} \log \left(\frac{1-\delta}{1+\delta} \right) = \delta \log \left(\frac{1+\delta}{1-\delta} \right)$$

Using the Taylor expansion for the logarithm, for a sufficiently small $\delta \in (0, \frac{1}{4})$, we know $\log(\frac{1+\delta}{1-\delta}) \approx 2\delta$. Thus we can upper bound it:

$$\text{KL}(P_+ || P_-) \leq 3\delta^2$$

By symmetry, $\text{KL}(P_- || P_+) \leq 3\delta^2$ as well.

4. Tensorization (Moving to n samples)

For n independent samples, the KL divergence tensorizes (it simply adds up). For the joint distributions P_+^n and P_-^n :

$$\text{KL}(P_+^n || P_-^n) = n \cdot \text{KL}(P_+ || P_-) \leq 3n\delta^2$$

and similarly $\text{KL}(P_-^n || P_+^n) \leq 3n\delta^2$.

5. Applying Birge's Lemma to Finish

Birge's Lemma guarantees a strictly positive testing error if the KL divergence is bounded by a specific constant. For $N = 2$ hypotheses, the condition is:

$$\frac{\max_{i,j} \text{KL}(P_i^n, P_j^n)}{\log 2} \leq \frac{2e}{2e+1}$$

Substituting our KL bound into this requirement:

$$\frac{3n\delta^2}{\log 2} \leq \frac{2e}{2e+1}$$

Solving this inequality for δ tells us exactly how to choose our separation:

$$\delta \leq \frac{c}{\sqrt{n}}, \quad \text{where } c = \sqrt{\frac{2e \log 2}{3(2e+1)}}$$

Finally, we plug $\delta = \frac{c}{\sqrt{n}}$ into our generic two-point lower bound formula. Since $\frac{\Delta}{2} = \delta = \frac{c}{\sqrt{n}}$, and Birge's Lemma guarantees the testing error is at least $\frac{1}{2e+1}$, we conclude:

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d^2(\hat{\theta}, \theta) \geq \left(\frac{\Delta}{2}\right)^2 \inf_j \max_{j \in \{1,2\}} P_j(\hat{J} \neq j) \geq \left(\frac{c}{\sqrt{n}}\right)^2 \frac{1}{2e+1} = c' \frac{1}{n}$$

Because this $\Omega(1/n)$ lower bound matches our $\mathcal{O}(1/n)$ upper bound, we have proved that $1/n$ is the optimal minimax rate.

Example 10.6 (Yang-Barron Approach). Let Θ be a space of parameters. We have an assumption: I can find a cover of Θ of size $N = N_{\epsilon}$,

$$\theta_1, \dots, \theta_N$$

: For all $\theta \in \Theta$ there exists $j \in N$ such that

$$\text{KL}(P_{\theta}, P_{\theta_j}) \leq n\epsilon^2$$

Why ϵ^2 ? Motivate this by

$$\text{KL}(\mathcal{N}(\theta, I_d), \mathcal{N}(\nu, I_d)) = \frac{\|\nu - \theta\|_2^2}{2}$$

This holds iff

$$nd^2(\theta, \theta_j) \leq n\epsilon^2$$

since $\delta(\theta, \theta_j) \leq \epsilon$?

and the same time

$$d(\theta_i, \theta_j) \geq \Delta$$

(assuming $\Delta \approx \epsilon$).

Then

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d(\hat{\theta}, \theta) \geq \left(\frac{\Delta}{2} \right) \left(1 - \frac{1 + \frac{1}{N} \sum_{i=1}^N \text{KL}(P_i, Q)}{\log N} \right)$$

Take $Q = \frac{1}{N} \sum_{i=1}^N P_{\theta_i}$.

Then

$$\text{KL}(P_{\theta}, Q) = \int_{\mathcal{X}} p_{\theta}(x) \log \frac{p_{\theta}(x)}{q(x)} dx$$

then for each θ , there is a closet element in the set, θ_j . Then $q(x) \geq p_{\theta_j}(x)$.

Then

$$\int_{\mathcal{X}} p_{\theta}(x) \log \frac{p_{\theta}(x)}{q(x)} dx \leq \int_{\mathcal{X}} p_{\theta}(x) \log \frac{p_{\theta}(x)}{\frac{1}{N} p_{\theta_j}(x)} dx = \log(N) \text{KL}(\theta, \theta_j)$$

High level principal: once we construct the set (cover) we can choose a measure which is an average over the entire net then when we compute the KL divergence it will be bounded by the so called critical equation which involved the $\log N$ term. The nice balance on how to choose everything in Fano's lemma is you want the KL term to scale

like $\log N$ so that the term $\frac{1 + \frac{1}{N} \sum_{i=1}^N \text{KL}(P_i, Q)}{\log N}$ can be approximately constant. Choose a solution for ϵ so that $\epsilon^2 \approx \log(N)/n$. See how this affects the upper bound of

$$\log(N) + n\epsilon^2$$

you want to avoid

$$\frac{1 + \frac{1}{N} \sum_{i=1}^N \text{KL}(P_i, Q)}{\log N} \geq 1$$

Make your cover consist of $N(\epsilon)$ points and the set you are taking the supremum over have $10 \cdot N(\epsilon)$ points and this somehow helps.

This makes it so any θ can be approximated well by Q which is defined in terms of the net. This makes the term of interest become something like

$$\frac{1 + \frac{1}{N} \sum_{i=1}^N \text{KL}(P_i, Q)}{\log(10 \cdot N)} < 1$$

Then for the final rate we can focus on the $\Delta/2$ term which gives us the rate we are trying to achieve.

Final expression should look like

$$\left(\frac{\Delta}{2}\right) \left(1 - \frac{1 + n\epsilon^2 + \log N_\epsilon}{\log(N'_\epsilon)}\right)$$

where N_ϵ is the size of the ϵ -net used to construct Q and N'_ϵ is the size of the set we are taking the supremum over which has pairwise distances at least Δ .

Example 10.7 (Yang-Barron Approach). 1. The Goal and the Two Sets

To use Fano's inequality in a large parameter space Θ , we actually need to construct *two* different sets of points:

- **The Packing (The Test Hypotheses):** A set of $M = N'_\epsilon$ points $\theta_1, \dots, \theta_M \in \Theta$ that are sufficiently far apart: $d(\theta_i, \theta_j) \geq \Delta$. This is the set we take the supremum over to bound our error.
- **The Cover (The Net for Q):** A set of $N = N_\epsilon$ points $\phi_1, \dots, \phi_N$ that blanket the entire space. For *any* $\theta \in \Theta$, there exists some ϕ_j in the cover such that the KL divergence is bounded:

$$\text{KL}(P_\theta^n || P_{\phi_j}^n) \leq n\epsilon^2$$

Motivation for ϵ^2 : For Gaussians $\mathcal{N}(\theta, I_d)$ and $\mathcal{N}(\nu, I_d)$, the KL divergence is exactly $\frac{n}{2} \|\theta - \nu\|_2^2$. Bounding the parameter distance $d(\theta, \nu) \leq \epsilon$ naturally bounds the KL divergence by $n\epsilon^2$.

2. The Clever Choice of Reference Measure Q

Instead of picking a single distribution for Fano's reference measure, Yang and Barron define Q as a uniform mixture over the entire ϵ -cover:

$$q(x) = \frac{1}{N} \sum_{k=1}^N p_{\phi_k}(x)$$

Because all probabilities are positive, we can lower-bound $q(x)$ using just the single closest point in the cover. For any target θ , let ϕ^* be its closest point in the cover. Then:

$$q(x) \geq \frac{1}{N} p_{\phi^*}(x)$$

3. Bounding the KL Divergence to Q (The Math Trick)

We need to bound $\text{KL}(P_\theta||Q)$. Using the lower bound for $q(x)$ we just found:

$$\begin{aligned}\text{KL}(P_\theta||Q) &= \int_{\mathcal{X}} p_\theta(x) \log \frac{p_\theta(x)}{q(x)} dx \\ &\leq \int_{\mathcal{X}} p_\theta(x) \log \frac{p_\theta(x)}{\frac{1}{N} p_{\phi^*}(x)} dx \\ &= \int_{\mathcal{X}} p_\theta(x) \left(\log N + \log \frac{p_\theta(x)}{p_{\phi^*}(x)} \right) dx \\ &= \log N + \text{KL}(P_\theta||P_{\phi^*})\end{aligned}$$

(Note: The log properties split the term into addition, not multiplication!)

Because ϕ^* is from our ϵ -cover, we know $\text{KL}(P_\theta||P_{\phi^*}) \leq n\epsilon^2$. Therefore:

$$\text{KL}(P_\theta||Q) \leq \log N_\epsilon + n\epsilon^2$$

4. The Final Fano Bound and the Critical Equation

We plug this bounded KL divergence into the standard Fano's inequality over our packing of size N'_ϵ :

$$\begin{aligned}\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_\theta d(\hat{\theta}, \theta) &\geq \left(\frac{\Delta}{2} \right) \left(1 - \frac{1 + \max_i \text{KL}(P_{\theta_i}||Q)}{\log N'_\epsilon} \right) \\ &\geq \left(\frac{\Delta}{2} \right) \left(1 - \frac{1 + n\epsilon^2 + \log N_\epsilon}{\log N'_\epsilon} \right)\end{aligned}$$

High Level Principle (The Critical Equation): Fano's bound is only useful if the multiplier is strictly positive (e.g., $\geq 1/2$). To ensure this, we must prevent the fraction from exceeding 1. We need the denominator (the log size of the packing) to be larger than the numerator (the metric entropy + the approximation error):

$$\log N'_\epsilon \gtrsim n\epsilon^2 + \log N_\epsilon$$

We balance this by finding the critical radius ϵ . We choose an ϵ where the statistical error $n\epsilon^2$ roughly balances the complexity of the space $\log N_\epsilon$. By constructing the packing to be slightly larger than the cover (e.g., using a slightly smaller radius for the packing so N'_ϵ is larger), we force the fraction strictly below 1, leaving us with our optimal lower bound rate of $\frac{\Delta}{2}$.

